

Permutations with exactly one occurrence of 1234, 1243 or 12345

OEIS A217057, A224179 and A224248

Half decompositions, leading amplitudes, rational diagonals
and asymptotic expansions to every order

Six research reports of one external research session
(bundle Reports 134, 135, 136, 137, 138 and 140), merged into one ProveIt report

Parts I–VI: 2 October 2026 merged: 5 October 2026

Abstract

A permutation contains exactly one classical occurrence of a pattern when exactly one of its subsequences, adjacent or not, is order-isomorphic to the pattern. This report prints six manuscripts on three such counts: u_n , for exactly one 1234 (OEIS A217057); b_n , for exactly one 1243 (A224179); and U_n , for exactly one 12345 (A224248). They are normalized by the avoidance counts: $A_n = |\text{Av}_n(1234)| = |\text{Av}_n(1243)|$ (A005802) for the first two, and $|\text{Av}_n(12345)|$ (A047889) for the third. Part I proves that $u_n/A_n \rightarrow R$, where R is an explicit series of positive terms, so that $u_n \sim (81\sqrt{3}/(16\pi))R9^n n^{-4}$; its finite rational certificate $R > 0.50009$ refutes Conway and Guttmann’s conjecture $R = 1/2$, and an exact partial sum computed in the write raises the bound to $R > 0.50153$. Part II proves that $\sum u_n z^n$ is the diagonal of a rational power series, hence D-finite, gives an explicit rational constant term, and proves $u_n/A_n = R + S/n + S_2/n^2 + (43020\sqrt{3}/\pi) \log n/n^3 + O(n^{-3})$. Part III extends this expansion to every fixed order, with coefficients affine in $\log n$; the next logarithmic coefficient is $-5041845\sqrt{3}/\pi$, and every logarithmic coefficient lies in $(\sqrt{3}/\pi)\mathbb{Q}$. Part IV proves $b_n/A_n \rightarrow R_{1243}$, with R_{1243} an explicit positive series, rational diagonality and non-algebraicity, and $b_n/A_n = R_{1243} + o(n^{-1/2})$; the value $149/160$ suggested by Conway and Guttmann remains open. Part V proves an exact four-binomial gluing and a six-term formula for one 12345, rational diagonality, and $U_n/|\text{Av}_n(12345)| \rightarrow R_5 > 0$, so that $U_n \sim (1536R_5/\pi^{3/2})16^n n^{-15/2}$. Part VI proves an expansion of that ratio in powers of $1/n$ to every fixed order, with no logarithms. The same method thus has opposite outcomes for 1234 (logarithms from relative order three) and for 12345 (none). Each Part also proves inverse-threshold statements with the rounding uncertainty kept inside the ceilings. The amplitudes and the non-logarithmic correction coefficients are given as convergent series, not in closed form, and none is certified numerically beyond lower bounds. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The six Parts are six manuscripts of the session bundle of Reports 1–243 (arrival commit `60f54ea06`), placed in this directory by commit `e85586b7c` (batch 105). They form two chains and one offshoot on one spine, the decomposition of a permutation with a unique occurrence into two halves glued along that occurrence:

Part	Report	Delivered title	Sequence	Labels
I	134	Permutations with exactly one increasing subsequence of length four	A217057	<code>upo:amp:</code>
II	135	Rational structure and logarithmic asymptotics for A217057	A217057	<code>upo:rat:</code>
III	137 (base)	All orders logarithmic asymptotics for A217057	A217057	<code>upo:all:</code>
IV	136	Permutations with exactly one classical 1243 occurrence	A224179	<code>upo:1243:</code>
V	138	Exact formulas and leading asymptotics for unique 12345	A224248	<code>upo:ex5:</code>
VI	140	All orders asymptotics and inverse thresholds for unique 12345	A224248	<code>upo:all5:</code>

Arrangement. The Parts are printed in order of dependence. Reports 134, 135 and 137 are one chain on one 1234 occurrence: each cites the one before and was delivered with it as an unchanged companion. The base of the merge is Report 137, the most general of the three; its `Report137.tex` was staged as this `article.tex`. It is printed as Part III, after Reports 134 and 135, because it uses their exact gluing identity, global bounds, unitary integral and first expansions as inputs and restates none of their proofs. Report 136 (one 1243) depends on Reports 134 and 135: it reproves Report 134’s tableau formula and fixed-boundary lemma word for word, and adapts Report 135’s unitary-integral method; it follows the 1234 chain as Part IV. Reports 138 and 140 are the chain for one 12345, with the same architecture at length five; Report 140 builds on Report 138. They are Parts V and VI. Report 139 of the same batch (one 1432, an unrelated method) is the separate report `a224182-unique-1432-order`.

Numbering. Sections are numbered continuously through the six Parts, and statements and equations within sections. The offsets of the section numbers are 0, 10, 22, 29, 43 and 54 for Parts I–VI; Section 66, after Part VI, is the write’s synthesis. Reports 135, 137, 138 and 140 number statements within sections, so their Theorem $k.j$ is Theorem $(k + \text{offset}).j$ here: Report 135’s Theorem 9.1 is Theorem 19.1, Report 137’s Theorem 1.1 is Theorem 23.1, Report 138’s Theorem 8.1 is Theorem 51.1, Report 140’s Theorem 7.1 is Theorem 61.1. Reports 134 and 136 number their statements consecutively through the document; here they are renumbered within sections. Report 134’s Theorem 1, Lemmas 2, 3, Proposition 4, Lemma 5, Lemmas 6, 7 and Corollary 8 are Theorem 1.1, Lemmas 2.1 and 3.1, Proposition 4.1, Lemma 5.1, Lemmas 7.1 and 7.2 and Corollary 9.1. Report 136’s Theorems 1, 2, Lemmas 3–5, Lemmas 8–10, Proposition 11, Corollary 12, Lemmas 13, 14 and Theorems 15, 16 are Theorems 30.1 and 31.1, Lemmas 31.2, 32.1, 32.2, Lemmas 35.1, 35.2, 35.3, Proposition 36.1, Corollary 39.1, Lemmas 40.1, 40.2 and Theorems 40.3, 41.1; its Proposition 6 and Lemma 7 are Report 134’s Proposition 4 and Lemma 5 word for word and are printed once, as Proposition 4.1 and Lemma 5.1. Report 136’s Appendix A is Section 43. The manuscripts number equations consecutively through each document; here equations are numbered within sections, so a delivered equation number is not kept, but every delivered equation keeps its label name, with its Part’s prefix. Statements added in the write (Proposition 6.1, Remarks 6.2,

13.2 and 66.1) are the last numbered statements of their sections, so no delivered statement number moved.

Notes. Notes added when the manuscripts were merged are set in small type and tagged **[write]**, with their date. They give cross-references between the Parts, mark statements that a later Part supersedes, pin citations that a manuscript left implicit, and mark results proved twice. Superseded statements are printed as delivered, with a dated note pointing forward. No delivered statement was changed and no delivered symbol was renamed; where a symbol means different things in different Parts, the notation section below and the table at the head of each Part say which meaning applies. Each Part has, after its own questions, a subsection “Further questions and research”, which, under Vladimir’s standing rule of 4 October 2026, records which of that Part’s questions a later Part answers and every claim of the Part that is stated without proof; Section 66.1 gathers the open problems of the whole report.

What is proved, and where

Statement	Where
<i>One</i> 1234 (<i>A217057</i>); $A_n = \text{Av}_n(1234) $, $C_0 = 81\sqrt{3}/(16\pi)$ Twelve allowed cells; exact two-half gluing $u_n = \sum H_s H_t \binom{i+k}{i} \binom{j+l}{j}$; halves as differences $H_s(i, j) = F_{s+2}(i+1, j+1) - F_{s+1}(i, j)$; three-row tableau formula for $F_m(p, q)$ $u_n/A_n \rightarrow R$, R a series of positive terms, $u_n \sim C_0 R 9^n n^{-4}$ $\liminf u_n/A_n \geq L_{40,10} > 0.50009$ without the existence proof; Conway–Guttmann’s $R = 1/2$ is false $\liminf u_n/A_n \geq \Pi_{44} > 0.50153$ $\sum u_n z^n$ is a rational diagonal over $\mathbb{Q}$, D-finite; (u_n) P-recursive; a 42-variable rational constant term	Part I, Lemmas 2.1, 3.1, Proposition 4.1
Not algebraic	[write] Proposition 6.1 Part II, Theorems 11.1 and 14.1; [write] Remark 13.2 checks the hypotheses of the cited closure theorem Cited: Bóna–Burstein [9, Theorem 6.2], see Section 21
$A_m = C_0 9^m m^{-4} (1 - \frac{11}{2m} + \frac{20}{m^2} + O(m^{-3}))$; uniform boundary expansions $u_n/A_n = R + S/n + S_2/n^2 + \kappa_3 \log n/n^3 + O(n^{-3})$, $\kappa_3 = 43020\sqrt{3}/\pi$ Every fixed order: $u_n/A_n = \sum_{j \leq J} (r_j + \kappa_j \log n) n^{-j} + O_J((1 + \log n) n^{-J-1})$; $\kappa_4 = -5041845\sqrt{3}/\pi$; r_3 exists	Part II, Lemma 15.1, (19.9), Proposition 16.1, (19.11) Part II, Theorems 11.2 and 19.1
A coefficient-level convolution theorem in any Banach space (integer exponent 4, logarithmic models) <i>One</i> 1243 (<i>A224179</i>); <i>the same</i> A_n Marked-avoider bijection; shared decreasing spine; $b_n = \sum J_k(s) J_k(t)$ $b_n/A_n \rightarrow R_{1243}$, a positive series; rational diagonal, D-finite, not algebraic	Part III, Theorem 23.1 Part III, Theorem 25.3
$b_n/A_n = R_{1243} + O(n^{-1/2})$, then $+o(n^{-1/2})$ (square-root cancellation); diffusive spine profile $e^{-u^2/2}$ <i>One</i> 12345 (<i>A224248</i>); <i>avoiders</i> <i>A047889</i> , $C_A = 1536/\pi^{3/2}$ Twenty allowed cells, four-binomial gluing, nested boundary bijections, six-term inverse-Cauchy formula	Part IV, Theorems 31.1, 30.1, Lemmas 32.1, 32.2 Part IV, Theorem 30.1, Sections 37–39 Part IV, Theorems 40.3 and 41.1 Part V, Theorems 45.3, 46.2, 47.3

Statement	Where
Rational diagonal, P-recursive	Part V, Theorem 48.1; the hypothesis check of Remark 13.2 applies
$U_n/ \text{Av}_n(12345) \rightarrow R_5 \geq 15/32768$, $U_n \sim C_A R_5 16^n n^{-15/2}$	Part V, Theorems 44.1 and 51.1
Every fixed order in powers of $1/n$, $r_1 = S_5$ explicit; no logarithms	Part VI, Theorem 55.1
A coefficient-level convolution theorem for half-integer exponent $15/2$	Part VI, Theorem 61.1
<i>All three</i>	
Inverse thresholds with the rounding error inside the ceilings	Corollaries 9.1, 20.1, 27.1, 39.1; Theorems 40.3, 41.1; Proposition 52.1; Theorem 63.1
The same final step, opposite outcomes for 1234 and 12345	[write] Remark 66.1
<i>Open in all Parts.</i> Closed forms and certified numerical enclosures of R , S , S_2 , r_3 , R_{1243} , R_5 , S_5 ; whether $R_{1243} = 149/160$; effective constants and onsets; usable differential equations or recurrences; the corrections for one 1243 beyond the vanishing square-root term; algebraicity for one 12345; longer patterns and several occurrences. See Section 66.1.	

Finite computations in all six Parts (exact counts, rational certificates, symbolic identities, mutation tests) are regression checks of formulas and implementations; no asymptotic statement rests on them. The lower bounds $L_{40,10}$, Π_{44} , L_{35} and $15/32768$ are exact rational evaluations of finitely many terms of proved series of nonnegative terms; they are rigorous given the proofs. The inverse statements are instances of the transseries volume’s generic inverse theorems (the front-matter section Relation to neighbouring reports); no new inversion theory is claimed.

Notation across the six Parts

The three counts are u_n (one 1234, Parts I–III), b_n (one 1243, Part IV) and U_n (one 12345, Parts V–VI). In the write’s own text the two avoidance sequences are $A_n^{(4)} = |\text{Av}_n(1234)|$ and $A_n^{(5)} = |\text{Av}_n(12345)|$, and the three amplitudes are $R = R_{1234}$, R_{1243} and R_5 ; the Parts themselves write A_n and R in each case. Common to Parts I–IV: $F_m(p, q)$, the half counts $H_s(i, j)$, $d_p = \binom{p+2}{2} 3^{-p}$, $C(i, j) = 81d_{i+1}d_{j+1} - 9d_id_j$, $C_0 = 81\sqrt{3}/(16\pi)$, $\lambda = \log 9$. Common to Parts V–VI: the boundary vector $v = (i_2, i_3; j_2, j_3)$, the kernel $K(v, w)$ and pairing $\langle \cdot, \cdot \rangle_K$, $G_{P,R}$, the set $\mathcal{T}$ of six partitions, $h(v)$, $C_A = 1536/\pi^{3/2}$, $p = 15/2$, $\lambda = \log 16$. The symbols below change meaning between the Parts or inside one; each Part’s text fixes its meaning there.

Symbol	Meanings
A	Parts I–IV: $A_n = A_n^{(4)}$ (A005802; equal to $ \text{Av}_n(1243) $ by West’s bijection, Part IV). Parts V–VI: $A_n = A_n^{(5)}$ (A047889). <i>Tempting false reading:</i> A_n in Part V is not Part I’s A_n ; R_5 is normalized by $A_n^{(5)}$. Also integration variables A, B, C in the constant terms (Section 14, Section 43, which says they are “unrelated to the earlier sequence”), and alphabets A, B, C, D in Section 47.
U	Parts V–VI: U_n is the count A224248. Part II: $U(z) = \sum u_n z^n$ and $U \in U(3)$ a unitary matrix. Part III: U_j the Banach coefficient arrays of the half expansion. Part VI: $\mathbf{u}_j$ (bold) the corresponding arrays at length five; $U \in U(4)$. Parts I and IV: U is a word in $U1V2W$; Part IV’s appendix: $U_r = c(q_r)$.

Symbol	Meanings
R	Parts I–III: the amplitude $R = \lim u_n/A_n$. Part IV: $R = \lim b_n/A_n$, written R_{1243} by the write, and $R_{m,p,q}$, R_s , R_j , $R_k, \dots$ (weights in the appendix). Parts V–VI: $R = i_3 + 1$ a boundary parameter, while the amplitude is R_5 . Part II: twelve weights $R_s, R_i, \dots$ (Section 14). Part III: $R_K(z)$ the remainder generating function. <i>False reading</i> : Part V's $G_{P,R}$ does not involve an amplitude.
H	Parts I–III: half counts $H_s(i, j)$; Part IV: the same object written $H_h(i, j)$. Parts V–VI: four-index half counts $H_s(v)$, a different object. Part II: $H = c(b)$ a constant-term variable (also Part IV's appendix). Part III: harmonic numbers H_m (it warns). <i>False reading</i> : Part III's H_6 in (26.4) is a harmonic number.
P, Q	Part I: polynomials $P_k = k^2 + 15k + 38$, $Q_k = k^2 + 11k + 18$. Part II: P, Q shuffle weights in Section 14 (it warns), $P(U) = \text{tr } U$, $Q = \sum \theta_j^2$, $Q_{\alpha,\gamma}$, $P_2(t; i, j)$. Part III: $P = \text{tr } U$, $Q = \sum \theta_i^2$, $P_K(z)$ a Taylor polynomial, Q_j inverse polynomials. Part IV: $P = \text{tr } U$, $Q = \sum \theta_i^2$, $Q_{\alpha,\beta}$. Parts V–VI: $P = i_2 + i_3 + 1$, $Q = j_2 + j_3 + 1$ boundary parameters; Part VI also $Q_k = \sum \theta_a^k$ power sums, Q a decay exponent (Lemma 61.2), P_j inverse polynomials. The write's partial sums are Π_T .
B	Part I: $B_k^{(I)}$ truncated transforms, $\mathcal{B}$ a majorant sum. Parts II–III: $B(i, j)$ the first boundary correction array; Part II: $B(a, b)$ the binomial coefficient and B a constant-term variable; Part III: $\mathcal{B}$ a bilinear form. Part IV: $B(z) = \sum b_n z^n$, $B_p = \binom{p+2}{2}$ (Part I's c_p), $\mathcal{B}(a, b)$ the binomial coefficient, B_{mid} . Part V: $b_{P,R}^T(\nu)$. Part VI: $B(v)$ the first half correction, B_k Bernoulli polynomials.
T	Thresholds T in $\tau(T)$ (Parts I–II) and $N(T)$ (Part III); truncation index in $L_{T,I}$ (Part I) and L_T (Part IV); T_j regularized coefficients (Part III); $T_3 = \sum \theta_j^3$ and $T_r = c(p_r)$ (Part II); T_p an incidence matrix (Part IV); Parts V–VI: $T = j_2 + 1$ a boundary parameter, $\mathcal{T}$ the six partitions, T_j matched arrays (Part VI).
L	Part I: $L = \log T$, the certificate $L_{T,I}$. Part II: $L = \log T$, the cutoff L_n , transforms L_k , a tilt L . Part III: polynomials L_j , $L = p + q$, L_{2h} . Part IV: $L = \log x$, the lower bound L_T , a tilt L . Part V: $L = \log x$ and a cutoff L . Part VI: $L = \log X$, a tilt L , an order L .
r	Part II: $r_p(\theta)$ the character ratio. Part III: r the Banach radius, r_j the coefficients, $r_n^{(K)}$, r_p . Part IV: r the spine length, r_p , $r(x)$ a root. Part V: $r = \lfloor E/2 \rfloor$, $r = M - k$. Part VI: r_j the coefficients, r_n , $r_n^{[L]}$ remainders, $r = s + 2 - t$. <i>False reading</i> : Part III's r_3 is a coefficient, not a character ratio.
τ	Parts I–II: $\tau(T)$ the inverse threshold (Part III: $N(T)$; Part IV: $N(x)$; Part V: $n_*(x)$; Part VI: $N(X)$). Parts V–VI: $\tau \in \mathcal{T}$ a partition. Part II: τ a triple of summation indices.
p	Parts I–IV: a boundary parameter, as in $F_m(p, q)$. Parts V–VI: $p = 15/2$ the exponent; but in Part V's Section 46, $p = i_2 + i_3 + 2$. Part II: p_r constant-term variables; Part VI: p_η probabilities.
w	Part II: $w_i = (i + 1)^6 3^{-i}$, $\tilde{w}_i$, and the inverse coefficient w of (20.4). Part III: the density factor $w(\theta)$. Part IV: $w(t, k) = 9^{-t} \gamma_k J_k(t)$. Part V: w a second boundary vector and $w(v)$ the domination weight (both in (49.9)). Part VI: w a formal variable; $\omega(v)$ the weight.
d	Parts I–IV: $d_p = \binom{p+2}{2} 3^{-p}$. Part IV: also d_i counts of allowed values (Section 31). Part III: $d = \log M_0 + 4 \log \lambda$. Part V: d, e degrees; d_i in the determinant expansion. Part VI: $d = P - t$ a degree, and in Section 63 the constant $d = \log M + p \log \lambda$ (it warns).
κ	Part II: $\kappa = 43020\sqrt{3}/\pi$. Part III: κ_j the logarithmic coefficients (κ_3 is Part II's κ). Part IV: $\kappa = \log 9$ in Corollary 39.1. Part VI: κ_ν a Casimir value.
α	Part II: $\alpha = -11/2$, and $\alpha \in \{0, 1\}$ in (13.4). Parts III and VI: α_j avoidance coefficients ($\alpha_1 = -11/2$ for 1234, $-135/8$ for 12345). Part V: partitions α . Part VI: $\alpha = p - 1 = 13/2$.
S	Parts II–III: $S = r_1$, $S_2 = r_2$ for 1234. Part VI: $S_5 = r_1$ for 12345. Part I: $S_k^{(I)}$ truncated transforms. Part III: $S_K(z)$. Part VI: S_n a model sum.

Symbol	Meanings
E, D, J	E_k, D_k, J_k, K_k binomial transforms (Parts I–II); $\mathcal{D}(x), \mathcal{D}_2(i, j), \mathcal{E}$ (Part II). Part III: E_r the Banach space, $E_j(s; p, q)$. Part IV: $J_k(s)$ the spine half sums (not Part II's J_k), $D(x)$ a dimension, D a constant and a mismatch. Part V: $D_M(P, R; Q, T)$, $E = 2M - P - Q$. Part VI: $E = E_{x,y}$ the Banach space, $D(\theta)$ the density factor, $D(F)$ a dimension.
C	Parts I–IV: $C(i, j)$; Part IV: $C = C_0 R$ in Corollary 39.1. Part V: $C_5 = C_A R_5$. Constants C, c, K vary within proofs.
N	Thresholds (above); Part I: $N = n - 4$ and N_*, D_* in the certificate; Part V: $N = n - 5$; Part III: $N_j(p, q)$.

Provenance and merge decisions

Six manuscripts of the session bundle of Reports 1–243, all dated 2 October 2026. Their title blocks read “Report 134”, . . . , “Report 140”; none names an author, a tool or an addressee (each sets an empty PDF author field), and none carries “prepared for private review” wording. None pins a ProveIt commit or names a repository path. They arrived in commit 60f54ea06 and were placed by e85586b7c (batch 105).

- **Part I**, Report 134, archive A217057_Unique_1234_Asymptotics_and_Conjecture_Refutation_Source.zip (Report134.tex, 791 lines, 12 pages). Contributes the exact two-half decomposition, the tableau formula, the fixed-boundary limits, the finite certificate refuting $R = 1/2$, the uniform domination, the series for R and the first inverse theorem.
- **Part II**, Report 135, archive A217057_Rational_Structure_Logarithmic_Asymptotics_and_Inversion_Source.zip (Report135.tex, 1455 lines, 22 pages). Its archive carries Report 134’s eight files under companion134/ (byte copies). Contributes rational diagonality and D-finiteness, the explicit constant term, the unitary-integral expansions, S, S_2 , the first logarithm and the sharper inverse.
- **Part III**, Report 137, archive A217057_All_Orders_Logarithmic_Asymptotics_and_Inversion_Source.zip (Report137.tex, 812 lines, 13 pages). The base. Its archive carries Report 135’s twenty files, Report 134’s included, under companion135/ (byte copies). Contributes the expansion to every order, κ_4 , the existence of r_3 with a regularized-series recipe, a general convolution theorem, and the inverse to every order.
- **Part IV**, Report 136, archive A224179_Leading_Amplitude_Diffusive_Cancellation_and_Inversion_Source.zip (Report136.tex, 1737 lines, 24 pages). Cites Reports 134 and 135 and reproves two statements of Report 134 word for word. Contributes the marked-avoider bijection, the spine decomposition, R_{1243} , rational diagonality, non-algebraicity, the rate and the square-root cancellation, and a 40-variable constant term.
- **Part V**, Report 138, archive A224248_Exact_Formulas_Leading_Asymptotics_and_Inversion_Source.zip (Report138.tex, 1159 lines, 17 pages). Cites none of Reports 134–137. Contributes the length-five gluing, the boundary bijections, the six-term formula, rational diagonality and the leading ratio R_5 .
- **Part VI**, Report 140, archive A224248_All_Orders_Asymptotics_and_Inverse_Thesis_holds_Source.zip (Report140.tex, 1122 lines, 18 pages). Its archive carries the whole Report 138 archive as dependencies/Report138_reproducible.zip, byte-identical to the standalone delivery (SHA-256 checked at placement). Contributes the expansion to every

order without logarithms, S_5 , a half-integer convolution theorem and the inverse to every order.

Where the merge had to choose:

1. *Order and base.* Dependency order, with the base (Report 137) third; the 1243 Part after the 1234 chain because it depends on it; the 12345 chain last. Later Parts improve earlier statements; the earlier statements stay, with dated notes.
2. *Verbatim repetition.* Report 136's Sections 4 and 5 (Proposition 6 and Lemma 7 with their proofs) are Report 134's Sections 4 and 5 character for character, except that Report 136 writes B_p for Report 134's c_p and closes Section 4 with a paragraph of its own (checked by the merge script); they are printed once, in Part I, and Sections 33 and 34 keep their titles, Report 136's own closing paragraph and a pointer. Report 135's Section 2 and Report 137's Section 1 restate inputs from the earlier reports in their own words, and Report 140's Section 2 restates inputs from Report 138; they are printed in full, because they fix the statements each Part uses, with notes naming the proved statements they summarize.
3. *Second routes.* Part II's Proposition 16.1 and Part IV's Lemma 40.1 prove boundary expansions by the same unitary-integral method (Part IV's is weaker and global; Part IV says it is adapted from Part II). Part IV's Gaussian strip bounds (Lemmas 35.1–35.3) refine Part I's Cauchy–Schwarz bound (7.4) by a mismatch factor. Part VI's Proposition 60.1 reproves $r_1 = S_5$ of its own Theorem 55.1 by an elementary route. All are printed.
4. *Front matter, appendix and bibliography.* The six title blocks and abstracts are printed at the heads of their Parts; one table of contents replaces those of Reports 138 and 140. Report 136's Appendix A is printed as Section 43, the last section of Part IV. The six bibliographies are merged into one, keeping every detail of every entry; entries for Reports 134, 135 and 138 point to Parts I, II and V.
5. *Write additions.* Proposition 6.1 with its exact rational, Remarks 6.2, 13.2 and 66.1, the further-questions subsections and Section 66, the dated notes, the reading tables and this front matter. Everything else is delivered text.

Relation to neighbouring reports

All neighbours named here are in `SetTheory/Cardinals/docs/reports/`. No other report treats A217057, A224179 or A224248; the only other report on a prescribed number of pattern occurrences is the sibling below (searched 5 October 2026).

The sibling for 1432. `generating-functions-and-asymptotics/oeis-sequence-asymptotics/a224182-unique-1432-order`, placed in the same batch from Report 139, proves that exactly one 1432 is $\Theta(9^n n^{-3})$, normalized by the same $A_n = |\text{Av}_n(1432)| = |\text{Av}_n(1234)|$, by a minima-skeleton argument that shares nothing with the half decompositions here; Conway and Guttmann treat 1234, 1243 and 1432 in consecutive sections. Neither manuscript cites the other.

Longest increasing subsequences. `generating-functions-and-asymptotics/oeis-sequence-asymptotics/a047874-long-increasing-subsequences` studies the counts of permutations by the length of a longest increasing subsequence; $A_n^{(4)}$ and $A_n^{(5)}$ are its counts for length at most 3 and 4, and both reports use RSK and strip tableaux. It studies a different regime (length near $N/2$); a cross-reference only.

Inverse thresholds. Every inverse statement here is an instance of the generic theorems of the transseries volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:thm:staircase` (parts (1) and (2): the integer threshold is the ceiling of any admissible interpolated inverse, and an approximation within η fixes it when the two ceilings agree) and, for Part I's W_{-1} formula, `p0:thm:lambert-core`. The Parts prove their inverse statements directly; the notes at each say which part of these theorems it instantiates.

Formal status. No statement of this report is formalized: no Lean or Rocq file in the repository concerns permutation patterns, A217057, A224179 or A224248 (searched 5 October 2026). Placement in the `SetTheory/Cardinals` report collection confers no formal status.

Part I

Permutations with exactly one increasing subsequence of length four

Bundle Report 134, archive A217057_Unique_1234_Asymptotics_and_Conjecture_Refutation_Source.zip, delivered as Report134.tex (12 pages), dated 2 October 2026; title block “Permutations with exactly one increasing subsequence of length four”, author line “Report 134”. Printed as delivered, with [write] notes. Report 134 numbers its statements consecutively; here they are numbered within sections (its Theorem 1 is Theorem 1.1, its Corollary 8 is Corollary 9.1). Proposition 6.1, Remark 6.2 and Section 10.1 are added in the write.

Reading conventions of Part I	Elsewhere in the report
u_n : exactly one 1234 (A217057); $A_n = \text{Av}_n(1234) $ $R = \lim u_n/A_n$, the amplitude	Part IV: b_n (one 1243); Parts V–VI: U_n (one 12345), and $A_n = \text{Av}_n(12345) $ Part IV: $R = R_{1243}$; Parts V–VI: R a boundary parameter, amplitude R_5
$H_s(i, j)$ half counts; $c_p = \binom{p+2}{2}$	Part IV: $H_h(i, j)$, B_p ; Parts V–VI: $H_s(v)$, another object
P_k, Q_k polynomials (1.3); $B_k^{(I)}, \mathcal{B}$ $b_i = (i+1)^4 3^{-i}$ (Lemma 7.1); e_t (7.8)	Parts II–IV: $P = \text{tr } U$, $Q = \sum \theta^2$; Parts II–III: $B(i, j)$ Part III: models $b_k(n)$; Part IV: counts b_n and its own e_t
$\tau(T)$, $L = \log T$, $x(T)$, $y(T)$, $K_0 = C_0 R$	Part III: $N(T)$ with $y = \log T$

Abstract of this Part (as delivered). Let u_n count permutations of $[n]$ with exactly one classical occurrence of 1234, and let A_n count 1234-avoiders. We prove $u_n \sim R A_n \sim (81\sqrt{3}/(16\pi)) R 9^n n^{-4}$, where R is an explicit convergent positive series of exactly computable tableau counts. A finite rational lower bound gives $R > 0.50009$, contradicting the conjecture $R = 1/2$ stated by Conway and Guttmann. The argument separates an exact two-half decomposition, a finite lower-bound certificate, and the uniform estimates needed for existence of the limit. It also gives the inverse threshold consequence justified by a relative equivalent. No closed form, certified decimal approximation to R , or higher-order expansion is claimed.

[write] **5 October 2026.** Later Parts go beyond the limits this Part states: Part II proves rational diagonality, the first two corrections and the first logarithmic term (Theorems 11.1, 11.2 and 19.1), and Part III the expansion to every fixed order (Theorem 23.1). No closed form and no certified decimal of R is known anywhere in the report; Proposition 6.1 sharpens the lower bound.

1 The problem and the main result

An occurrence of 1234 in $\pi \in S_n$ is a choice of positions $r_1 < r_2 < r_3 < r_4$ with $\pi_{r_1} < \pi_{r_2} < \pi_{r_3} < \pi_{r_4}$. Positions need not be consecutive. We count permutations, not marked occurrences or equivalence classes. Set $u_n = 0$ for $0 \leq n < 4$ and $A_0 = 1$. The sequence u_n is OEIS A217057 [4]; for $n = 4, \dots, 15$ it is

1, 12, 102, 770, 5545, 39220, 276144, 1948212,
13817680, 98679990, 710108396, 5150076076.

Regev's strip-sum theorem gives

$$A_n \sim C_0 \frac{9^n}{n^4}, \quad C_0 = \frac{81\sqrt{3}}{16\pi}. \quad (1.1)$$

The specialization is $d = 1$, three rows and exponent $\alpha = 2$ in [15, Section 3.1.4]: its constant is $(2\pi)^{-1}2^{-4}3^{9/2}2! = C_0$. That paper confirms this ordinary strip case of the 1981 result [14]; the correction it discusses concerns $d > 1$.

Here are the finite quantities defining our answer. Let $F_m(p, q)$ count 1234-avoiders of size m whose final p entries are decreasing and whose q smallest values occur in decreasing order. We use $0 \leq p, q \leq m$; parameters zero or one impose no restriction. For $s, i, j \in \mathbb{N}_0$, define

$$H_s(i, j) = F_{s+2}(i+1, j+1) - F_{s+1}(i, j) \quad (i+j \leq s), \quad (1.2)$$

and set $H_s(i, j) = 0$ outside that triangle. Section 4 gives a finite integer formula for every $F_m(p, q)$. Put

$$P_k = k^2 + 15k + 38, \quad Q_k = k^2 + 11k + 18. \quad (1.3)$$

Theorem 1.1. *The limit $R = \lim_{n \rightarrow \infty} u_n/A_n$ exists, is finite and positive, and is given by the absolutely convergent series*

$$R = \frac{1}{10368} \sum_{t=0}^{\infty} 9^{-t} \sum_{\substack{k, l \geq 0 \\ k+l \leq t}} H_t(k, l) \left(\frac{3}{2}\right)^{k+l} (P_k P_l - Q_k Q_l). \quad (1.4)$$

Consequently

$$u_n \sim C_0 R \frac{9^n}{n^4}. \quad (1.5)$$

Moreover $R \geq L_{40,10} > 50009/100000 > 1/2$, where

$$L_{40,10} = \frac{6958884420052382006187220194732220723168380665062}{13915193059764305937984450503671774362956903094027}. \quad (1.6)$$

Every coefficient $H_t(k, l)$ is nonnegative, and the factor in parentheses in (1.4) is strictly positive. Thus every nonzero summand is positive. The dimensionless ratio R must be distinguished from the leading coefficient $C_0 R$.

[write] The refutation on record, 5 October 2026. The conjecture is Conway and Guttmann's [8]. In arXiv:2306.12682v2, Section 5.1 (printed page 19 of the journal version), they estimate the limit of the ratio from about 200 known terms, find it "plausibly 0.5", and conjecture that the ratio of one-occurrence to avoiding counts is $1/2$ for 1234, equivalently $u_n \sim 81\sqrt{3}9^n/(32\pi n^4)$ (read 5 October 2026). Theorem 1.1 refutes it. The refutation uses only Lemmas 2.1, 3.1 and 5.1 and the nonnegativity of every summand: Section 6 proves $\liminf u_n/A_n \geq L_{40,10}$ without assuming that the limit exists, and (6.2) certifies $L_{40,10} > 1/2$ in exact rational arithmetic. The proof is printed in full in Sections 2–6. Proposition 6.1 raises the bound to 0.50153, and Remark 6.2 explains why extrapolation from the data pointed to $1/2$. No text in the repository repeated the conjecture before this report, so none needed correction.

What was known and what is established here. Bóna and Burstein's inequalities already imply $u_n = \Theta(9^n/n^4)$ by symmetry and Regev's theorem [9, Section 6, equation (6.1)]. They also prove nonalgebraicity of the ordinary generating function in this case. Catalytic functional-equation algorithms of Nakamura and Zeilberger enumerate prescribed small occurrence counts and include this sequence [11]. Conway and Guttmann give further enumeration and asymptotic analysis [8]. In

the published journal version, Section 5.1, printed page 19, they explicitly conjecture $u_n/A_n \rightarrow 1/2$, equivalently leading coefficient $81\sqrt{3}/(32\pi)$. Theorem 1.1 disproves that specific amplitude and establishes a leading equivalent. The existence proof and the finite disproof are given separately below. This report makes no worldwide priority claim and no claim about their other pattern or occurrence-count conjectures.

[write] **Citations pinned, 5 October 2026.** Checked against arXiv:2101.00332v3, the preprint of the published [9]: the Θ statement is their equation (6.1); the nonalgebraicity is their Theorem 6.2, stated for even $k > 2$ and proved by their Lemma 6.3; their Theorem 5.3 proves nonrationality for every $k \geq 3$. Report 134 cites the arXiv version, Reports 135, 136 and 138 the published one; the merged bibliography gives both.

2 The twelve allowed cells and exact gluing

Fix the unique increasing four-chain, with entries $c_1 < c_2 < c_3 < c_4$ in their position order. For a nonchain entry, let a be the number of chain positions before it, and b the number of chain values below it. Thus $(a, b) \in \{0, 1, 2, 3, 4\}^2$. The longest increasing chain using this entry and only distinguished-chain entries has length

$$\min(a, b) + 1 + 4 - \max(a, b) = 5 - |a - b|.$$

If $|a - b| \leq 1$, it contains an increasing four-chain different from the distinguished one. All extra entries therefore lie in

$$\begin{aligned}\mathcal{N} &= \{(0, 2), (0, 3), (0, 4), (1, 3), (1, 4), (2, 4)\}, \\ \mathcal{S} &= \{(2, 0), (3, 0), (3, 1), (4, 0), (4, 1), (4, 2)\}.\end{aligned}$$

The chain together with its $\mathcal{N}$ entries is a *northern half*; the analogous restriction is a *southern half*. Each half contains exactly its distinguished four-chain. Reverse-complement, sending a point to its reflection in both coordinates, exchanges the half types.

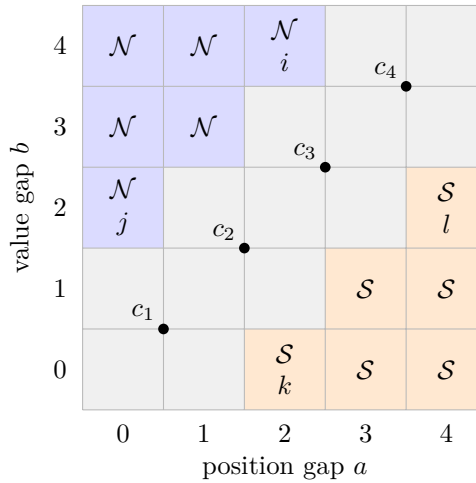


Figure 1: Only the twelve colored cells with $|a - b| \geq 2$ can contain extra entries. The four black dots are the distinguished chain. The marked boundary cells have sizes i, j, k, l . A half must itself have exactly one four-chain; the cells are not independent unrestricted avoider classes.

Let a northern half have s extra points, i of them in $(2, 4)$ and j in $(0, 2)$. For a southern half with t extra points, define k, l by reflection: they count $(2, 0)$ and $(4, 2)$, respectively.

Lemma 2.1. *Writing $H_s(i, j)$ for the number of northern halves with those parameters, the number of full permutations is*

$$u_n = \sum_{s+t=n-4} \sum_{i,j,k,l \geq 0} H_s(i, j) H_t(k, l) \binom{i+k}{i} \binom{j+l}{j}. \quad (2.1)$$

Proof. The two halves share only position gap $a = 2$ and value gap $b = 2$. In the former, merge the i northern and k southern positions while preserving each half's order. There are $\binom{i+k}{i}$ choices. In the latter, merge the j northern and l southern values, giving $\binom{j+l}{j}$ choices. These involve four distinct cells and are independent; all other cross-half comparisons are forced. Total position and value orders determine one permutation, and both shuffles can be recovered from it.

It remains to check uniqueness of the four-chain after gluing. Each of the four marked cells in Figure 1 is decreasing: an increasing pair there would combine with two distinguished points to make another four-chain in its half. An increasing pair from $\mathcal{N}$ to $\mathcal{S}$ can only run from $(0, 2)$ to $(4, 2)$. An increasing pair in the other direction can only run from $(2, 0)$ to $(2, 4)$. In either case no further point of either half can extend the mixed pair in an increasing chain: any additional opposite-half pair is constrained to the same two cells, and within either cell there is no increasing pair. Thus a mixed increasing chain has at most two extra entries.

Nor can a distinguished point join such a mixed pair. In the first case both values are between c_2, c_3 , while the two positions are before all and after all chain positions. In the second case both positions are between those of c_2, c_3 , while their values are below all and above all chain values. These comparisons exclude a distinguished point before, between, or after the pair. Hence every four-chain is contained in a half, and its distinguished one is the only possibility.

Finally the unique chain is determined by the full permutation, so restriction recovers the two halves uniquely. There is no marking multiplicity or division factor. Summing the shuffle counts proves the formula. $\square$

3 A half is a difference of boundary restricted avoiders

Lemma 3.1. *The half count in Lemma 2.1 is exactly (1.2). In particular it is nonnegative and vanishes unless $i + j \leq s$.*

Proof. Standardize a northern half of total size $s + 4$. Its first two chain values are 1, 2, and its last two chain positions have values c, d . The j points in $(0, 2)$ have values $3, \dots, j + 2$; hence $c = j + 3$. Its form is

$$U \, 1 \, V \, 2 \, W \, c \, d. \quad (3.1)$$

Here W is decreasing of length i , all its values exceed d , the values $3, \dots, j + 2$ occur decreasingly in U , and every value in V exceeds c . These facts follow from the cell geometry and the decreasingness of the two marked northern cells.

Delete $2, c$ and standardize to a permutation β of size $s + 2$. It avoids 1234, since the only original four-chain has been broken. Its last $i + 1$ entries, corresponding to W, d , are decreasing; its $j + 1$ smallest values occur decreasingly; and its final value is greater than $j + 1$.

Conversely let β be an avoider with these three properties. Its decreasing suffix of length $i + 1$ contains none of its $j + 1$ smallest values, because its last and therefore smallest suffix value exceeds $j + 1$. In particular its value 1 is before that suffix. Insert a new value 2 immediately before the specified suffix, then a new value $c = j + 3$ immediately before the last entry, increasing old values as needed to preserve their relative order. The result has form (3.1). The low values $3, \dots, j + 2$ lie

decreasingly before 1; all entries between 1 and the inserted 2 exceed c ; and W is decreasing above $d > c$.

A four-chain using neither inserted value would have existed in β . A chain using the inserted 2 has at most the single smaller predecessor 1, and the only increasing pair to its right is c, d . A chain using the inserted c has at most the single successor d , and the only increasing pair of smaller entries to its left is 1, 2. Thus the new permutation has exactly the chain 1, 2, c, d . The two operations are inverse, including when either boundary condition holds for a longer block than required.

It remains to exclude the avoiders whose final value is at most $j + 1$. Since their $j + 1$ smallest values occur decreasingly, such a final value must be 1. Deleting this final minimum changes the suffix condition from $i + 1$ to i and the small-value condition from $j + 1$ to j . Appending a new minimum reverses this operation and never makes a new increasing four-chain. The excluded number is therefore $F_{s+1}(i, j)$, giving (1.2).

In the retained class the last $i + 1$ entries and the $j + 1$ smallest values are disjoint. Their combined size $i + j + 2$ is at most $s + 2$, proving triangular support. In particular $H_0(0, 0) = 1$ and $H_s(0, 0) = A_{s+2} - A_{s+1}$. $\square$

4 The exact tableau formula

For a partition λ , let f^λ denote the number of standard Young tableaux of its diagram. A vertical strip has at most one box per row; a horizontal strip has at most one box per column. Define

$$V_p(\lambda) = \sum_{\substack{\mu \subseteq \lambda \\ \lambda/\mu \text{ a vertical } p\text{-strip}}} f^\mu.$$

Proposition 4.1. *For $0 \leq p, q \leq m$,*

$$F_m(p, q) = \sum_{\substack{\lambda \vdash m \\ \lambda_1 \leq 3}} V_p(\lambda) V_q(\lambda). \quad (4.1)$$

After transposition, use three-row shapes $\rho = (a, b, c)$, with $a \geq b \geq c \geq 0$ and $a + b + c = m$. Set

$$W_p(\rho) = \sum_{\substack{a \geq u \geq b \geq v \geq c \geq w \geq 0 \\ u+v+w=m-p}} f^{(u,v,w)}. \quad (4.2)$$

Then $F_m(p, q) = \sum_{\rho \vdash m, \ell(\rho) \leq 3} W_p(\rho) W_q(\rho)$, where

$$f^{(u,v,w)} = \frac{(u+v+w)!(u-v+1)(u-w+2)(v-w+1)}{(u+2)!(v+1)!w!}. \quad (4.3)$$

Proof. Ordinary row-insertion RSK is a bijection from permutations to pairs (P, Q) of standard tableaux of the same shape λ . The longest increasing subsequence has length λ_1 , so avoidance is precisely $\lambda_1 \leq 3$. The descent correspondences are $\text{Des}(\pi) = \text{Des}(Q)$ and $\text{Des}(\pi^{-1}) = \text{Des}(P)$, where a tableau descent means that the next label is in a strictly lower row.

A decreasing suffix of length p requires the last p labels of Q to occupy strictly increasing row numbers. Their deletion leaves a vertical p -strip. Conversely, any such strip has exactly one admissible filling by these labels, in row order from top to bottom. It is standard: any retained box has a smaller label, and the retained Young diagram has no box to the right of a removed box in its row or below it in its column. Comparisons between removed boxes in a column follow the row order. Hence there are $V_p(\lambda)$ choices for Q .

The q smallest values of π occurring decreasingly means that every index $1, \dots, q-1$ is a descent of P , so these q labels occupy its first column. The remaining number is the skew-tableau dimension $f^{\lambda/(1^q)}$. For completeness, the vertical Pieri rule says that the coefficient of s_λ in $s_\mu e_q$ is one if λ/μ is a vertical q -strip and zero otherwise. Taking the adjoint gives $s_{\lambda/(1^q)} = \sum_{\lambda/\mu \text{ vertical } q} s_\mu$. In homogeneous degree $m-q$, the exponential specialization sends s_ν to $f^\nu/(m-q)!$, and a skew Schur function to its skew standard-tableau count divided by the same factorial. It follows that $f^{\lambda/(1^q)} = V_q(\lambda)$. The cases $p=0$ or $q=0$ have empty strips and the same interpretation.

Conditional on the common shape, P, Q are independent choices. This proves (4.1). Transposition changes vertical strips to horizontal ones and preserves tableau dimensions. For three-row shapes, the horizontal-strip condition is exactly the interlacing in (4.2). Finally (4.3) is the three-row hook-length formula, valid also with zero parts and for the empty shape. $\square$

These formulas are finite and exact. They also show $A_m = F_m(0,0) = \sum_{\ell(\rho) \leq 3} (f^\rho)^2$. The remaining argument requires more than coefficient agreement: first a fixed-boundary limit, then a bound uniform in every boundary parameter.

5 Fixed boundary asymptotics

Put

$$c_p = \binom{p+2}{2}, \quad d_p = c_p 3^{-p}. \quad (5.1)$$

Lemma 5.1. *For every fixed $p, q \in \mathbb{N}_0$,*

$$\frac{F_m(p, q)}{A_m} \longrightarrow d_p d_q. \quad (5.2)$$

Consequently, for fixed i, j ,

$$\frac{H_s(i, j)}{A_s} \longrightarrow C(i, j) := 81d_{i+1}d_{j+1} - 9d_i d_j > 0. \quad (5.3)$$

Proof. Give the three-row partitions $\rho = (a, b, c)$ of m probability $(f^\rho)^2/A_m$. Fix $\eta = 1/20$. We show that with probability tending to one,

$$\max\{|a - m/3|, |b - m/3|, |c - m/3|\} \leq m^{1/2+\eta}, \quad a - b > m^\eta, \quad b - c > m^\eta. \quad (5.4)$$

Assigning the row of every tableau label gives $f^\rho \leq \binom{m}{a, b, c}$. For any triple outside the balance window, the multinomial coefficient is at most $3^m \exp(-c_* m^{2\eta})$ for some fixed $c_* > 0$ and sufficiently large m . One elementary justification is to regard $3^{-m} \binom{m}{a, b, c}$ as a multinomial probability; some coordinate is outside its $m/3$ mean by more than $m^{1/2+\eta}$, and the binomial Chernoff bound applies. There are $O(m^2)$ triples. Their squared dimensions have total $O(m^{29} m^{-2c_* m^{2\eta}}) = o(A_m)$ by (1.1).

Inside the window, all row lengths are of order m and the multinomial coefficient is uniformly $O(3^m/m)$, for example by the factorial bounds in Stirling's formula. Rewrite (4.3) as

$$f^{(a, b, c)} = \binom{m}{a, b, c} \frac{(a-b+1)(a-c+2)(b-c+1)}{(a+2)(a+1)(b+1)}.$$

If either adjacent gap is at most m^η , the squared product of the three numerator factors is $O(m^{2+6\eta})$. There are $O(m^{1/2+2\eta})$ such shapes: choose the small gap in $O(m^\eta)$ ways and one row in a window of size $O(m^{1/2+\eta})$, with the third row fixed by the total. Their total squared weight is therefore

$$O\left(g^m m^{-2} m^{-6} m^{2+6\eta} m^{1/2+2\eta}\right) = O(g^m m^{-11/2+8\eta}) = o(A_m).$$

This proves (5.4).

For fixed p , every weak composition $\delta_1 + \delta_2 + \delta_3 = p$ gives a legal horizontal strip removal $\rho/(\rho - \delta)$ on (5.4), once m is large. The adjacent gaps and last row eventually exceed p , which proves all interlacing inequalities. These are all possible removals, numbering c_p . The exact ratio of dimensions is

$$\frac{f^{\rho-\delta}}{f^\rho} = \frac{(m-p)!}{m!} (a+2)_{\underline{\delta_1}} (b+1)_{\underline{\delta_2} c_{\delta_3}} \frac{\Delta(\rho-\delta)}{\Delta(\rho)}, \quad (5.5)$$

where $\Delta(a, b, c) = (a-b+1)(a-c+2)(b-c+1)$ and underlining denotes a falling factorial. Uniformly on the good shapes, these three factors are respectively asymptotic to m^{-p} , $(m/3)^p$, and 1. Thus $W_p(\rho)/f^\rho \rightarrow d_p$ uniformly there.

For all shapes, $0 \leq W_p(\rho)/f^\rho \leq 1$, since the untransposed strip sum counts a subset of standard tableaux with a specified suffix descent condition. Express (4.1) divided by A_m as the expectation of the product of the two ratios. Its difference from $d_p d_q$ is bounded by a good-event supremum error tending to zero plus a constant times the probability of the bad event. This proves (5.2) without assuming a fixed probability space.

Finally $A_{s+h}/A_s \rightarrow 9^h$ for fixed h by (1.1), which proves (5.3). The strict positivity follows from

$$C(i, j) = 9d_i d_j \left(\frac{(i+3)(j+3)}{(i+1)(j+1)} - 1 \right) > 0. \quad (5.6)$$

□

6 A finite certificate already rules out one half

This section does not assume existence of R . Fix nonnegative integers T, I . In (2.1), retain only terms with one half size $t \leq T$ and the other half's two boundary parameters at most I . For $n > 2T + 4$ the two endpoint ranges are disjoint. They contribute equally by exchanging the halves. All terms are nonnegative. For each retained fixed t, k, l, i, j , Lemma 5.1 gives

$$\frac{H_{n-4-t}(i, j)}{A_n} \rightarrow 9^{-t-4} C(i, j).$$

Taking a limit in this finite sum proves

$$\liminf_{n \rightarrow \infty} \frac{u_n}{A_n} \geq L_{T,I} := 2 \sum_{t=0}^T 9^{-t-4} \sum_{k+l \leq t} H_t(k, l) \sum_{i,j=0}^I C(i, j) \binom{i+k}{i} \binom{j+l}{j}. \quad (6.1)$$

No infinite interchange or boundary uniformity is needed for this step.

For exact computation define

$$S_k^{(I)} = \sum_{i=0}^I \binom{i+k}{i} d_i, \quad B_k^{(I)} = \sum_{i=0}^I \binom{i+k}{i} d_{i+1}.$$

The innermost sum in (6.1) is exactly $81B_k^{(I)}B_l^{(I)} - 9S_k^{(I)}S_l^{(I)}$. Integer dimensions and rational arithmetic now give (1.6) at $T = 40, I = 10$. Write its numerator and denominator as N_*, D_* . Then

$$L_{40,10} - \frac{1}{2} = \frac{2575780340458074389989885792667083379858236097}{27830386119528611875968901007343548725913806188054} > 0. \quad (6.2)$$

The stronger displayed decimal inequality means the exact rational comparison $100000N_* > 50009D_*$, not a floating-point comparison. This proves the finite disproof independently of the next section.

Independent reconstruction. The accompanying code computes the complete certificate in two ways. One uses (4.2)–(4.3). The other generates tableau dimensions by Young-lattice branching, starting with dimension one for the empty shape and adding one legal corner at a time. For each contained inner and outer three-row shape, it explicitly lists removed-box columns and accepts the skew diagram exactly when no column is repeated. Thus the second method uses neither the hook product nor the interlacing loop to construct the needed F and H arrays. Both paths evaluate the entire finite rational sum, rather than checking a supplied decimal or only its final inequality. Reproduction details are in Section 10.

Proposition 6.1 ([write]). *For an integer $T \geq 0$ let Π_T be the partial sum of the series (1.4) over $0 \leq t \leq T$:*

$$\Pi_T = \frac{1}{10368} \sum_{t=0}^T 9^{-t} \sum_{k+l \leq t} H_t(k, l) \left(\frac{3}{2}\right)^{k+l} (P_k P_l - Q_k Q_l).$$

Then $\liminf_{n \rightarrow \infty} u_n/A_n \geq \Pi_T$ for every T ; this does not use Lemmas 7.1 or 7.2. In exact arithmetic,

$$\Pi_{44} = \frac{616064898260691226056814615415016451341424247360270300969}{1228351671029471449207812122938224643885861920422359990272} > \frac{50153}{100000}, \quad (6.3)$$

so $R \geq \Pi_{44} > 0.50153$.

Proof. Fix T . For every I , (6.1) gives $\liminf u_n/A_n \geq L_{T,I}$. In $L_{T,I}$ the outer sums over $t \leq T$ and $k+l \leq t$ are finite, and the coefficients $H_t(k, l)$ are nonnegative (Lemma 3.1). For fixed k, l the inner sum $\sum_{i,j \leq I} C(i, j) \binom{i+k}{i} \binom{j+l}{j}$ has positive terms by (5.6), so it increases with I ; it converges as $I \rightarrow \infty$ because $d_i \leq (i+1)^2 3^{-i}$ and $\binom{i+k}{i} \leq (i+1)^k$. Its limit is $81E_k E_l - 9D_k D_l$ with the absolutely convergent transforms D_k, E_k of Section 8, and by (8.1) this equals $\frac{81}{256} \left(\frac{3}{2}\right)^{k+l} (P_k P_l - Q_k Q_l)$. Hence, as $I \rightarrow \infty$,

$$L_{T,I} \longrightarrow 2 \sum_{t=0}^T 9^{-t-4} \sum_{k+l \leq t} H_t(k, l) \frac{81}{256} \left(\frac{3}{2}\right)^{k+l} (P_k P_l - Q_k Q_l) = \Pi_T,$$

since $2 \cdot 81/(256 \cdot 9^4) = 1/10368$. Taking the supremum over I gives $\liminf u_n/A_n \geq \Pi_T$. With Theorem 1.1, $R \geq \Pi_T$.

The value (6.3) was computed in the write (5 October 2026) by a program written for the purpose, independent of the delivered code: it evaluates $F_m(p, q)$ for $m \leq 46$ from Proposition 4.1 (hook formula (4.3) and the interlacing sum (4.2)), then $H_t(k, l)$ by (1.2), and sums Π_{44} in exact rational arithmetic. As checks, the same program reproduces $u_4, \dots, u_{15}$ of Section 1 through (2.1), gives $H_t(k, l) \geq 0$ for all $t \leq 44$, gives $\Pi_0 = 35/324$, and reproduces $L_{40,10}$ of (1.6) as the same fraction. The intermediate partial sums are $\Pi_{10} = 0.457116$, $\Pi_{20} = 0.490148$, $\Pi_{30} = 0.498089$, $\Pi_{40} = 0.500923$ and $\Pi_{44} = 0.501538$ (rounded); the inequality in (6.3) is the exact comparison $100000 N > 50153 D$ of numerator and denominator. $\square$

The bound is one-sided. The tail of (1.4) beyond T is $O((T+1)^{-3})$ with an unspecified constant (Section 7), so Π_{44} is not an approximation with a certified error, and no upper bound for R is known.

Remark 6.2 ([write]) Why the numerics pointed to $1/2$. Conway and Guttmann’s estimate rests on ratio and quadratic estimators plotted against $1/n^2$ [8, Section 5.1], which presuppose corrections in powers of $1/n$. Parts II and III prove that u_n/A_n has the term $\kappa_3 \log n/n^3$ with $\kappa_3 = 43020\sqrt{3}/\pi \approx 2.37 \times 10^4$, followed by $\kappa_4 \log n/n^4$ with $\kappa_4 = -5041845\sqrt{3}/\pi \approx -2.78 \times 10^6$ (Theorems 19.1 and 23.1). At $n = 200$ these two terms are about 0.0157 and -0.0092 ; at $n = 50$ they are about 0.74 and -1.74 . At the end of the available data they are therefore an order of magnitude larger than the

margin $R - 1/2 > 0.0015$ proved above, and below $n \approx 70$ the asymptotic series is not yet in its useful range. No extrapolation that omits these terms can decide whether R exceeds $1/2$. The exact ratios computed in the write with the program of Proposition 6.1 make the slow approach visible: $u_n/A_n = 0.70858, 0.73021, 0.71055, 0.69538, 0.69035$ at $n = 20, 30, 40, 46, 48$. The intake dossier for this batch reports least-squares fits of OEIS b-file ratios for $40 \leq n \leq 70$, with κ_3 included, that return $0.474\text{--}0.483$, below the proved lower bound. These fits are diagnostics, not estimates, and nothing in the report rests on them. The refutation of $R = 1/2$ rests on the proof of Sections 2–6 alone.

7 Uniform domination and existence of the amplitude

Fixed boundary limits alone do not justify (1.4). We give uniform control of the unbounded parameters and of the middle of the convolution.

Lemma 7.1. *There is an absolute constant K such that for every $s, i, j \in \mathbb{N}_0$,*

$$0 \leq \frac{H_s(i, j)}{A_s} \leq K b_i b_j, \quad b_i = \frac{(i+1)^4}{3^i}. \quad (7.1)$$

Also

$$\mathcal{B} := \sum_{i, k \geq 0} b_i b_k \binom{i+k}{i} < \infty. \quad (7.2)$$

Proof. The horizontal p -strip incidence matrix between three-row shapes of sizes $m-p$ and m has at most c_p ones in any row or any column: the numbers of removed or added boxes in the three rows form a weak composition of p . Cauchy–Schwarz first inside the strip sum and then between the two boundary vectors gives

$$F_m(p, p) \leq c_p \sum_{\rho} \sum_{\rho/\mu \text{ horizontal } p} (f^\mu)^2 \leq c_p^2 A_{m-p}, \quad (7.3)$$

$$F_m(p, q) \leq c_p c_q \sqrt{A_{m-p} A_{m-q}}. \quad (7.4)$$

All indices here satisfy $0 \leq p, q \leq m$. Equation (1.1), adjusted over finitely many small values, provides constants $0 < c < C < \infty$ with

$$c \frac{9^m}{(m+1)^4} \leq A_m \leq C \frac{9^m}{(m+1)^4} \quad (m \geq 0). \quad (7.5)$$

Consequently

$$\frac{F_m(p, q)}{A_m} \leq K_1 c_p c_q 3^{-p-q} \frac{(m+1)^4}{(m-p+1)^2 (m-q+1)^2}.$$

Since $(m+1)/(m-p+1) \leq p+1$ for $0 \leq p \leq m$ (the difference after cross multiplication is $p(m-p) \geq 0$), and $c_p \leq (p+1)^2$, this yields

$$\frac{F_m(p, q)}{A_m} \leq K_1 (p+1)^4 (q+1)^4 3^{-p-q}.$$

Use $H_s(i, j) \leq F_{s+2}(i+1, j+1)$, the globally bounded ratio A_{s+2}/A_s , and $(i+2)^4 \leq 16(i+1)^4$ to obtain (7.1). Outside the support the same assertion is immediate.

For the second assertion, set $r = i + k$. Then

$$\sum_{i+k=r} b_i b_k \binom{r}{i} \leq (r+1)^8 \left(\frac{2}{3}\right)^r.$$

The right side is summable in r . The complete four-index coupling therefore has summable majorant with total $K^2 \mathcal{B}^2$. $\square$

For clarity let the aggregate contribution of half sizes s, t be

$$G_{s,t} = \sum_{i,j,k,l \geq 0} H_s(i,j) H_t(k,l) \binom{i+k}{i} \binom{j+l}{j}.$$

It is symmetric in s, t and satisfies

$$0 \leq G_{s,t} \leq K^2 \mathcal{B}^2 A_s A_t. \quad (7.6)$$

Lemma 7.2. *There is a constant K_2 such that, with $N = n - 4$,*

$$\frac{1}{A_n} \sum_{\substack{s+t=N \\ s,t > T}} G_{s,t} \leq K_2 (T+1)^{-3} \quad (7.7)$$

whenever the sum is nonempty.

Proof. By (7.5) and (7.6), the left side is bounded by a constant times

$$9^{-4} (n+1)^4 \sum_{T < s < N-T} \frac{1}{(s+1)^4 (N-s+1)^4}.$$

For $s \leq N/2$, $(n+1)/(N-s+1) \leq 2(N+5)/(N+2) \leq 5$. Use the symmetric estimate on the other half of the range. The result is at most a constant times $\sum_{s > T} (s+1)^{-4} \leq (T+1)^{-3}/3$, with an inessential adjustment if needed at the endpoint. This is uniform in n . $\square$

Proof of existence and the endpoint representation. For fixed t , the boundary support of $H_t(k, l)$ is finite. For each fixed k, l , Lemma 7.1 dominates the sum over i, j by a constant times $b_i b_j \binom{i+k}{i} \binom{j+l}{j}$, which is summable by (7.2). The additional ratio A_{n-4-t}/A_n is uniformly bounded for sufficiently large n and has limit 9^{-4-t} . Dominated summation and Lemma 5.1 therefore give

$$\frac{G_{n-4-t,t}}{A_n} \longrightarrow e_t := 9^{-4-t} \sum_{k+l \leq t} H_t(k, l) \sum_{i,j \geq 0} C(i, j) \binom{i+k}{i} \binom{j+l}{j}. \quad (7.8)$$

All terms are nonnegative. Independently, taking the limit in (7.6) yields

$$0 \leq e_t \leq K^2 \mathcal{B}^2 9^{-4-t} A_t = O((t+1)^{-4}).$$

Thus $\sum_t e_t$ converges and its tail is $O((T+1)^{-3})$.

For fixed T and $n > 2T + 4$, the two endpoint ranges of the exact convolution are disjoint and sum to $2 \sum_{t=0}^T G_{n-4-t,t}$. The remaining middle is bounded by (7.7). Taking $n \rightarrow \infty$ at fixed T , then $T \rightarrow \infty$, proves

$$\frac{u_n}{A_n} \longrightarrow R = 2 \sum_{t \geq 0} e_t. \quad (7.9)$$

In particular there is neither unaccounted mass in large boundary parameters nor mass remaining in the middle half-size range. $\square$

8 Evaluation of the boundary sums

Define convergent sums

$$D_k = \sum_{i \geq 0} \binom{i+k}{i} d_i, \quad E_k = \sum_{i \geq 0} \binom{i+k}{i} d_{i+1}.$$

Since $\sum_{i \geq 0} d_i w^i = (1 - w/3)^{-3}$ and $\sum_{k \geq 0} \binom{i+k}{i} z^k = (1 - z)^{-i-1}$, summation gives, initially for sufficiently small $|z|$,

$$\begin{aligned} \sum_{k \geq 0} D_k z^k &= \frac{27(1-z)^2}{(2-3z)^3}, \\ \sum_{k \geq 0} E_k z^k &= \frac{27(1-z)^3}{(2-3z)^3} - 1. \end{aligned}$$

Extracting coefficients using $[z^k](1 - 3z/2)^{-3} = \binom{k+2}{2}(3/2)^k$ gives

$$D_k = \frac{3}{16} Q_k \left(\frac{3}{2}\right)^k, \quad E_k = \frac{1}{16} P_k \left(\frac{3}{2}\right)^k. \quad (8.1)$$

The subtraction of 1 in the second generating function is essential at $k = 0$: $D_0 = 27/8$ and $E_0 = 19/8$.

Substitution of (5.3) in (7.8) and (7.9) now yields

$$\begin{aligned} R &= 2 \sum_{t \geq 0} 9^{-4-t} \sum_{k+l \leq t} H_t(k, l) (81 E_k E_l - 9 D_k D_l) \\ &= \frac{2}{81} \sum_{t \geq 0} 9^{-t} \sum_{k+l \leq t} H_t(k, l) (E_k E_l - D_k D_l / 9). \end{aligned}$$

Equation (8.1) proves precisely (1.4), including its factor $1/10368$. Since $P_k - Q_k = 4k + 20 > 0$ and both polynomials are positive for $k \geq 0$, all nonzero summands are positive. The $t = 0$ summand is $35/324$. Existence, positivity, and convergence have already been proved, and (6.1) gives the lower bound for R . Together with (1.1), this completes the proof of Theorem 1.1.

The $O(T^{-3})$ tail statements here have unspecified constants. They prove convergence and passage to the limit; they do not certify digits of R . In particular, a positive partial sum of (1.4) is a lower bound, not a two-sided approximation with a certified decimal error.

[write] **5 October 2026.** The partial sums Π_T of Proposition 6.1 are such lower bounds. Effective tail constants remain open (Section 66.1, item 1).

9 What the equivalent implies for inverse thresholds

Let $\tau(T) = \min\{n \in \mathbb{N}_0 : u_n \geq T\}$ for real $T > 0$. Write

$$\lambda = \log 9, \quad L = \log T, \quad K_0 = C_0 R, \quad x(T) = \frac{L + 4 \log L - 4 \log \lambda - \log K_0}{\lambda}.$$

Here $x(T)$ is used only for sufficiently large T .

Corollary 9.1. *There is a positive function $\epsilon(T) \rightarrow 0$ such that for all sufficiently large T ,*

$$\lceil x(T) - \epsilon(T) \rceil \leq \tau(T) \leq \lceil x(T) + \epsilon(T) \rceil. \quad (9.1)$$

No effective rate for $\epsilon(T)$ is asserted.

Proof. Prepending a new largest value to a permutation cannot use that value in an increasing subsequence of length at least two: it has no predecessor and no larger successor. All old occurrences are unchanged. This is an injection from the permutations counted by u_n to those counted by u_{n+1} , so u_n is nondecreasing. It is unbounded by Theorem 1.1, hence $\tau(T)$ exists.

Put $h(y) = \lambda y - 4 \log y + \log K_0$. For large y , it is strictly increasing with $h'(y) \rightarrow \lambda$. The unique large solution $y(T)$ of $h(y) = L$ satisfies

$$y(T) = x(T) + O(\log L/L) = x(T) + o(1). \quad (9.2)$$

Indeed, inserting $x(T) = L/\lambda + O(\log L)$ gives $h(x(T)) - L = O(\log L/L)$, and h' is bounded away from zero.

The relative equivalent says $\log u_n = h(n) + r_n$ with $r_n \rightarrow 0$ for $n \geq 4$. Let $a(T) = \sup_{n \geq \lceil y(T)/2 \rceil} |r_n|$, which tends to zero. Choose $\delta(T) = 3a(T)/\lambda + 1/y(T)$, and take T large enough that $h'(v) \geq \lambda/2$ near and above $y(T)/2$. Then $h(n) + r_n < L$ for integers $y(T)/2 \leq n \leq y(T) - \delta(T)$, and $h(n) + r_n > L$ for integers $n \geq y(T) + \delta(T)$. For the first assertion one uses $(\lambda/2)\delta(T) > a(T)$; the same margin proves the second. Earlier integers also have $u_n < T$: for instance the equivalent gives $\log u_{\lceil y(T)/2 \rceil} - L \rightarrow -\infty$, and monotonicity applies.

Thus $\lceil y(T) - \delta(T) \rceil \leq \tau(T) \leq \lceil y(T) + \delta(T) \rceil$. Take $\epsilon(T) = |y(T) - x(T)| + \delta(T)$ and use (9.2) to obtain (9.1). $\square$

The rounding uncertainty must remain *inside* the ceilings. A relative $o(1)$ equivalent does not decide on which side of a nearby integer the threshold lies. It also does not supply first or higher correction terms. As an optional exact expression for the continuous model only,

$$y(T) = -\frac{4}{\lambda} W_{-1} \left(-\frac{\lambda}{4} (K_0/T)^{1/4} \right),$$

where W_{-1} is the real branch at negative arguments tending to zero. This is not an exact formula for the discrete threshold. Finite threshold checks must use exact u_n values from (2.1), not a partial sum substituted for R .

[write] Instances of the generic inverse theorems, 5 October 2026. The ceiling sandwich (9.1) is the form that parts (1) and (2) of `p0:thm:staircase` in the transseries volume (the front-matter section Relation to neighbouring reports) take once a real inverse is located within $\epsilon(T)$ of $x(T)$; the proof above establishes this directly. The W_{-1} expression for $y(T)$ is `p0:thm:lambert-core` for $aX + b \log X = L'$ with $a = \lambda$, $b = -4$ and $L' = \log T - \log K_0$, on the branch W_{-1} , whose values tend to $-\infty$ as its argument increases to 0, which gives the large root. Part II narrows the width to $O((\log T)^{-3})$ (Corollary 20.1) and Part III to every fixed order (Corollary 27.1).

10 Reproduction and limits of the computation

The accompanying reproducibility archive contains this TeX source and PDF, exact source fixtures, two certificate implementations, and a standard-library Python driver. Its README specifies the complete commands and dependency requirements for checking, deterministic PDF rebuilding, and deterministic archive creation. The programs use integer and rational arithmetic for every certificate decision.

The checks cover exact agreement with the A217057 fixture, direct permutation enumeration at manageable sizes, independently constructed tableau dimensions and horizontal strips, the complete $T = 40, I = 10$ certificate, and the strict rational inequalities above. Selected mutations test that failures are detected rather than silently ignored. The driver has explicit validation active under Python’s optimized mode. Its closed inventory rejects unexpected entries, symbolic links, and nonregular objects. Normal commands create outputs exclusively outside the bundle and reject symbolic-link ancestors before computation. Intentional maintainer sealing is a separately documented operation.

These tests address transcription, normalization, implementation, and reproducibility. They do not turn finite checks into a proof of the analytic theorem. The latter rests on the bijections, the fixed-boundary concentration argument, the uniform Cauchy–Schwarz majorant, and the convolution-tail estimate proved above.

Questions left open here. The series (1.4) is an exact answer for the leading amplitude, but no simpler closed form is obtained. Making the domination constants effective enough for efficient certified numerical enclosures would be useful. First and higher asymptotic corrections for u_n , and the corresponding effective or all-orders inverse expansions, remain open in this report. The two-half geometry suggests comparisons with other single-pattern classes, but does not by itself settle any of them.

10.1 Further questions and research

[write] **Status of the questions above, 5 October 2026.** (a) A closed form for R : open. (b) Effective domination constants and certified enclosures of R : open (Section 66.1, item 1); Proposition 6.1 gives only a lower bound. (c) First and higher corrections: *answered* by Part II (S, S_2, κ_3 ; Theorems 11.2 and 19.1) and Part III (every fixed order, Theorem 23.1). (d) Effective or all-orders inverse expansions: the all-orders inverse is Part III’s Corollary 27.1; effective constants remain open (item 4). (e) Other single-pattern classes: advanced by Part IV (one 1243), Parts V–VI (one 12345) and the sibling report for one 1432; Remark 66.1 compares the outcomes, and longer patterns remain open (item 9). This Part states no claim without proof: its open points are the questions above and the limitations it prints (unspecified tail constants, no rate for $\epsilon(T)$). Nothing in it was found to be wrong; the one wrong claim it addresses is external, Conway and Guttmann’s $R = 1/2$.

Part II

Rational structure and logarithmic asymptotics for A217057

Bundle Report 135, archive A217057_Rational_Structure_Logarithmic_Asymptotics_and_Inversion_Source.zip, delivered as Report135.tex (22 pages), dated 2 October 2026; title block “Rational structure and logarithmic asymptotics for A217057”, author line “Report 135”. Printed as delivered, with [write] notes; a section number is Report 135’s plus 10 (its Theorem $k.j$ is Theorem $(k + 10).j$), and equations are renumbered within sections. Remark 13.2 and Section 22.1 are added in the write.

Reading conventions of Part II	Elsewhere in the report
$u_n, A_n, R, H_s(i, j)$ as in Part I	
$U(z) = \sum u_n z^n; U \in U(3)$	Parts V–VI: U_n the 12345 count; Part III: arrays U_j
P, Q shuffle weights (Section 14); $P = \text{tr } U$;	Part I: polynomials P_k, Q_k ; Parts V–VI: boundary
$Q = \sum \theta_j^2$	parameters
$B(a, b) = [v^b](1 + v)^a; B(i, j)$ (11.4); B a	Part IV: $B(a, b), B(z), B_p$
constant-term variable	
$D_k, E_k, J_k, K_k, L_k, M_k$ transforms; $\mathcal{D}(x)$;	Part IV: $J_k(s)$ spine sums, $D(x)$
$\mathcal{D}_2(i, j)$	
$\alpha = -11/2; \alpha \in \{0, 1\}$ in (13.4)	Part VI: $\alpha = 13/2$
$\kappa = 43020\sqrt{3}/\pi$	Part III: κ_3 ; Part IV: $\kappa = \log 9$
$\tau(T), L = \log T, y(T), x_3(T), \beta, \delta, \eta, \mu$	Part III: $N(T), y = \log T$
z the generating variable, and $z = t + 4$ in	
Section 19	

Abstract of this Part (as delivered). Let u_n count permutations with exactly one classical increasing subsequence of length four, and A_n count the corresponding avoiders. Continuing Report 134, we prove that $\sum u_n z^n$ is a rational diagonal and hence D-finite, and give an explicit 42-variable rational constant term with convergent contours. A uniform unitary-integral expansion and weighted convolution analysis also prove

$$\frac{u_n}{A_n} = R + \frac{S}{n} + \frac{S_2}{n^2} + \frac{43020\sqrt{3}}{\pi} \frac{\log n}{n^3} + O(n^{-3}).$$

The constants S, S_2 are explicit absolutely convergent signed series; $R > 0$ is the amplitude from Report 134. The positive logarithmic term rules out a pure power expansion at the third relative correction. The full corrected model gives an eventual $O((\log T)^{-3})$ uncertainty inside the ceilings for inverse thresholds. No explicit recurrence, certified numerical values for the series constants, effective onset, finite third coefficient, or all-orders expansion is asserted.

[write] **5 October 2026.** In this report “Report 134” [1] is Part I; the companion it calls unchanged is Part I’s text and the 134-amplitude- files. The finite n^{-3} coefficient and the further logarithms that this Part does not reach are proved in Part III (Theorem 23.1: r_3 exists, $\kappa_4 = -5041845\sqrt{3}/\pi$).

11 Scope and the new conclusions

An occurrence of the classical pattern 1234 is a choice of four positions $r_1 < r_2 < r_3 < r_4$ with increasing values. The positions need not be consecutive. Let u_n be the number of permutations of $[n]$ with exactly one such choice, with $u_0 = u_1 = u_2 = u_3 = 0$. This is OEIS A217057 [4]. Let A_n count the 1234-avoiders and set $A_0 = 1$.

Report 134 [1] proves an exact two-half decomposition and

$$u_n \sim RA_n \sim C_0 R 9^n n^{-4}, \quad C_0 = \frac{81\sqrt{3}}{16\pi}, \quad R > 0.50009. \quad (11.1)$$

Its finite rational certificate disproves the particular conjecture $R = 1/2$ of Conway and Guttmann [8, Section 5.1]. The present report is a continuation, with that report included unchanged as a companion in the archive. Section 12 defines all the finite counts and records the exact identities and bounds used here. The proofs of those inherited combinatorial facts remain in the companion; all the new algebraic and asymptotic steps are proved below.

Theorem 11.1. *The ordinary generating function $U(z) = \sum_{n \geq 0} u_n z^n$ is the diagonal of a rational power series over $\mathbb{Q}$. In particular it is D -finite, and (u_n) is P -recursive. Section 14 constructs an explicit rational constant-term representation of U .*

For precision, a rational diagonal means that for some finite d and rational function Q regular at the origin,

$$U(z) = \text{Diag } Q(z) := \sum_{n \geq 0} [x_1^n \cdots x_d^n] Q(x_1, \dots, x_d) z^n.$$

P -recursiveness means existence of a nonzero polynomial-coefficient recurrence of finite order. D -finiteness means existence of a nonzero linear differential equation with polynomial coefficients. Neither an explicit recurrence nor an explicit origin-regular diagonal function Q is computed here. The rational constant term is a separate, fully displayed constructive representation with specified contours.

To state the refinement of (11.1), put

$$d_p = \binom{p+2}{2} 3^{-p}, \quad a_p = \frac{p(p-1)}{2}, \quad (11.2)$$

and define

$$C(i, j) = 81d_{i+1}d_{j+1} - 9d_i d_j, \quad (11.3)$$

$$B(i, j) = -81d_{i+1}d_{j+1}(8 + a_{i+1} + a_{j+1}) + 9d_i d_j(4 + a_i + a_j). \quad (11.4)$$

The half counts $H_t(k, l)$ are defined in (12.1).

Theorem 11.2. *The following signed series is absolutely convergent:*

$$S = 2 \sum_{t \geq 0} 9^{-t-4} \sum_{k+l \leq t} H_t(k, l) \sum_{i, j \geq 0} \binom{i+k}{i} \binom{j+l}{j} \cdot (4(t+4)C(i, j) + B(i, j)). \quad (11.5)$$

It satisfies

$$\frac{u_n}{A_n} = R + \frac{S}{n} + O(n^{-2}). \quad (11.6)$$

Consequently, with $M_0 = C_0 R$ and $\beta = S/R - 11/2$,

$$u_n = M_0 9^n n^{-4} \left(1 + \frac{\beta}{n} + O(n^{-2}) \right). \quad (11.7)$$

The infinite i, j sums in (11.5) have exact polynomial binomial transforms, given in Section 18.

The implied constants are not made effective. Absolute convergence specifies S exactly, but does not certify its decimal value, its sign, or its nonvanishing. The same qualifications apply to β . Section 19 goes farther: Theorem 19.1 adds an absolutely convergent second coefficient S_2 and evaluates the first logarithmic term, with coefficient $43020\sqrt{3}/\pi > 0$. Section 20 derives the resulting $O((\log T)^{-3})$ inside-ceiling uncertainty. The first-correction proof is retained as a useful intermediate result and supplies the endpoint estimate needed to identify the logarithm.

12 Exact reduction and inherited estimates

For $0 \leq p, q \leq m$, let $F_m(p, q)$ count the 1234-avoiders whose last p entries are decreasing and whose q smallest values occur in decreasing order. Conditions with parameter zero or one are vacuous. For $s, i, j \in \mathbb{N}_0$ put

$$H_s(i, j) = F_{s+2}(i+1, j+1) - F_{s+1}(i, j) \quad (i+j \leq s), \quad (12.1)$$

and set $H_s(i, j) = 0$ otherwise. These differences are nonnegative integers, not an inclusion-exclusion approximation.

[write] **A summary of Part I, 5 October 2026.** This section restates, in its own words and citing Part I, results proved there: (12.2) is Lemma 2.1; the interpretation of (12.1) is Lemma 3.1; (12.3)–(12.5) are Proposition 4.1; (12.6) is (7.5); (12.7) is Lemma 7.1; (12.8) is (7.9) with (7.8). It is printed in full because its displays are the ones this Part uses; it adds no claim.

12.1 Why the two halves and the two shuffles appear

Fix the unique increasing four-chain. For any extra point, let a count chain positions before it and b count chain values below it. A chain using that point and distinguished-chain points has maximal length $5 - |a - b|$. Thus $|a - b| \leq 1$ is forbidden. The twelve allowed cells split as

$$\begin{aligned} \mathcal{N} &= \{(0, 2), (0, 3), (0, 4), (1, 3), (1, 4), (2, 4)\}, \\ \mathcal{S} &= \{(2, 0), (3, 0), (3, 1), (4, 0), (4, 1), (4, 2)\}. \end{aligned}$$

The northern and southern halves share the four-chain. They have s, t extra points. In the northern half let i, j count the cells $(2, 4), (0, 2)$; in the southern half let k, l count $(2, 0), (4, 2)$. Each of these four boundary cells is decreasing, since an increasing pair would create another four-chain inside its half.

The only position-order ambiguity is the shuffle of the i and k points in position gap 2. Independently, the j and l points in value gap 2 may be shuffled in value order. Cross-half increasing pairs cannot extend to a four-chain: they can only connect the two marked cells in one of these shared gaps, and no distinguished point can extend such a pair. Thus gluing valid halves produces exactly one four-chain and recovers the two shuffles uniquely. The result is

$$u_n = \sum_{s+t=n-4} \sum_{i,j,k,l \geq 0} H_s(i, j) H_t(k, l) \binom{i+k}{i} \binom{j+l}{j}. \quad (12.2)$$

There is no marking multiplicity or division factor.

For the interpretation of (12.1), standardize a northern half. Its distinguished values are $1, 2, c, d$ with $c = j + 3$, and it has form $U1V2Wcd$, where W is decreasing of size i above d . Deleting $2, c$ gives an avoider counted by $F_{s+2}(i+1, j+1)$ with final value greater than $j+1$. The excluded avoiders end in their minimum; deleting that minimum gives $F_{s+1}(i, j)$. The two insertions reverse the construction. The suffix and low-value blocks of a retained avoider are disjoint, which also proves the triangular support $i+j \leq s$. The full inverse-bijection checks are in [1, Sections 2–3].

12.2 The fixed-row tableau formula

For a partition $\lambda = (\lambda_1, \lambda_2, \lambda_3)$, including zero parts, let $|\lambda|$ be its size and f^λ its standard-tableau dimension. If $\lambda \vdash m$ and $\ell(\lambda) \leq 3$, define

$$W_p(\lambda) = \sum_{\substack{\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \lambda_3 \geq \mu_3 \geq 0 \\ |\mu| = m-p}} f^\mu. \quad (12.3)$$

The interlacing condition says precisely that λ/μ is a horizontal strip. The exact formula is

$$F_m(p, q) = \sum_{\substack{\lambda \vdash m \\ \ell(\lambda) \leq 3}} W_p(\lambda) W_q(\lambda), \quad (12.4)$$

with

$$f^{(x_1, x_2, x_3)} = \frac{(x_1 + x_2 + x_3)!(x_1 - x_2 + 1)(x_1 - x_3 + 2)(x_2 - x_3 + 1)}{(x_1 + 2)!(x_2 + 1)!x_3!}. \quad (12.5)$$

In ordinary RSK, avoidance bounds the first row by three. The suffix condition removes a vertical strip in the recording tableau. The smallest-value condition fixes an initial first-column segment in the insertion tableau; the vertical Pieri rule gives the same strip sum. Transposition turns both conditions into (12.3) and preserves dimensions. This explains (12.4); details appear in [1, Section 4]. In particular $A_m = F_m(0, 0) = \sum_{\ell(\lambda) \leq 3} (f^\lambda)^2$.

12.3 Uniform bounds and the established amplitude

The ordinary three-row strip case of Regev's theorem gives $A_m \sim C_0 9^m m^{-4}$ [14, 15]. After adjusting constants at finitely many small m , there are $c_*, C_* > 0$ such that

$$c_* \frac{9^m}{(m+1)^4} \leq A_m \leq C_* \frac{9^m}{(m+1)^4}. \quad (12.6)$$

Report 134 proves, for an absolute constant K_* ,

$$0 \leq H_s(i, j) \leq K_* A_s b_i b_j, \quad b_i = (i+1)^4 3^{-i}. \quad (12.7)$$

One useful way to see the bound is that a horizontal p -strip incidence matrix has at most $c_p = \binom{p+2}{2}$ ones in each row and column. Cauchy–Schwarz in (12.4) gives $F_m(p, q) \leq c_p c_q \sqrt{A_{m-p} A_{m-q}}$. Combining this with (12.6), $(m+1)/(m-p+1) \leq p+1$, and $H_s(i, j) \leq F_{s+2}(i+1, j+1)$ proves (12.7).

For fixed i, j , $H_s(i, j)/A_s \rightarrow C(i, j) > 0$. The convolution in (12.2), controlled by (12.7), then yields

$$R = 2 \sum_{t \geq 0} 9^{-t-4} \sum_{k+l \leq t} H_t(k, l) \sum_{i, j \geq 0} C(i, j) \binom{i+k}{i} \binom{j+l}{j}. \quad (12.8)$$

All terms here are nonnegative and the series converges. Unlike S , its finite partial sums are rigorous lower bounds. The lower bound in (11.1) is inherited unchanged; its full rational certificate and independent reconstructions are in the companion archive.

13 A binomial-sum proof of rational diagonality

We use the binomial-sum algebra of Bostan, Lairez and Salvy (BLS) [16]. Their Definition 1.1 permits products, integral affine substitutions, and directed partial sums. Their Theorem 3.5 identifies nonnegative-index binomial sums with rational diagonals, and Corollary 3.6 implies P-recursiveness. These numbers refer to the authors' arXiv version 2, pages 6–7 and 17. The following explicit encoding checks the hypotheses, including all moving boundaries and negative arguments.

13.1 An everywhere-defined dimension replacement

Use the coefficient convention

$$B(a, b) = [v^b](1 + v)^a \quad (a, b \in \mathbb{Z}). \quad (13.1)$$

For $b < 0$ this is zero; for $b \geq 0$ it is the usual falling-factorial polynomial in a divided by $b!$. Negative upper arguments are allowed. Set $h(r) = B(r, r)$ and $\delta(r) = B(0, r)$. Then h is the indicator of $r \geq 0$, and δ is the indicator of $r = 0$.

For any integer triple x , put $M = x_1 + x_2 + x_3$ and, for $\sigma \in S_3$, let $e_i = x_i - i + \sigma(i)$. Define

$$\mathcal{D}(x) = \sum_{\sigma \in S_3} \text{sgn}(\sigma) B(M, e_1) B(M - e_1, e_2) B(e_3, e_3). \quad (13.2)$$

Every argument is integral affine in x .

Lemma 13.1. *For every three-row partition x , $\mathcal{D}(x) = f^x$.*

Proof. Since $e_1 + e_2 + e_3 = M$, the product for one permutation is $M!/(e_1!e_2!e_3!)$ if all e_i are nonnegative. If any e_i is negative, the corresponding first, second or third factor is zero. All factors are defined, so there is no implicit product of zero and an undefined factorial ratio. For a partition this gives

$$\mathcal{D}(x) = M! \det \left(\frac{1}{(x_i - i + j)!} \right)_{i,j=1}^3,$$

where negative reciprocal factorials in this auxiliary determinant are zero. Put $X_i = x_i + 3 - i$. Each entry is $(X_i)_{3-j}/X_i!$. The column degrees are 2, 1, 0, so the numerator determinant is $\prod_{i < j} (X_i - X_j)$. The falling factorials vanish correctly when $0 \leq X_i < 3 - j$. Thus this computation covers zero parts and the empty partition and is exactly (12.5). $\square$

13.2 A fixed-dimensional encoding of the entire sequence

For integer triples λ, μ , set

$$I(\lambda, \mu) = h(\lambda_1 - \mu_1) h(\mu_1 - \lambda_2) h(\lambda_2 - \mu_2) \\ \cdot h(\mu_2 - \lambda_3) h(\lambda_3 - \mu_3) h(\mu_3).$$

It is one exactly when the interlacing in (12.3) holds. Define

$$\mathcal{E}(m, p, q; \lambda, \mu, \nu) = \delta(|\lambda| - m) \delta(|\mu| - m + p) \delta(|\nu| - m + q) \\ \cdot I(\lambda, \mu) I(\lambda, \nu) \mathcal{D}(\mu) \mathcal{D}(\nu). \quad (13.3)$$

For $0 \leq p, q \leq m$ and any $N \geq m$, summing this over $\lambda, \mu, \nu \in \{0, \dots, N\}^3$ gives exactly $F_m(p, q)$. Indeed, the deltas impose the three sizes and each surviving coordinate is at most m .

Use the 24 summation indices

$$\mathbf{z} = (s, t, i, j, k, l, \lambda_1, \lambda_2, \lambda_3, \mu_1, \mu_2, \mu_3, \nu_1, \nu_2, \nu_3, \Lambda_1, \Lambda_2, \Lambda_3, \rho_1, \rho_2, \rho_3, \tau_1, \tau_2, \tau_3).$$

For $\alpha, \gamma \in \{0, 1\}$ put

$$\begin{aligned} \mathcal{Q}_{\alpha, \gamma}(n, \mathbf{z}) = & \delta(s + t - n + 4) h(s - i - j) h(t - k - l) B(i + k, i) B(j + l, j) \\ & \cdot \mathcal{E}(s + 2 - \alpha, i + 1 - \alpha, j + 1 - \alpha; \lambda, \mu, \nu) \\ & \cdot \mathcal{E}(t + 2 - \gamma, k + 1 - \gamma, l + 1 - \gamma; \Lambda, \rho, \tau). \end{aligned} \quad (13.4)$$

Then the following single fixed-dimensional formula is valid for every $n \geq 0$:

$$u_n = \sum_{\alpha, \gamma \in \{0, 1\}} (-1)^{\alpha + \gamma} \sum_{\mathbf{z} \in \{0, \dots, n+2\}^{24}} \mathcal{Q}_{\alpha, \gamma}(n, \mathbf{z}). \quad (13.5)$$

The first delta forces $s + t = n - 4$, the two h factors impose the triangular supports, and the binary choices expand the two differences in (12.1). The surviving half sizes are at most $n - 4$, and each outer tableau size is at most $n - 2$. Thus the common box is large enough. For $n < 4$ its delta has no nonnegative solutions and the result is zero. Expanding the four $\mathcal{D}$ factors gives at most $4 \cdot 6^4 = 5184$ signed products of the single primitive B evaluated at affine arguments; both this number and 24 are independent of n .

Proof of Theorem 11.1. Every summand in (13.5) belongs to the BLS algebra. Replace each ordinary box sum by the directed sum from zero to a fresh integer upper-bound parameter, then substitute $n + 2$ for that parameter. For $n \geq 0$ these are precisely the ordinary finite sums in (13.5). Multiplication by $h(n)$ extends the sequence by zero on negative n ; the directed construction is still defined everywhere. BLS Theorem 3.5 and Corollary 3.6 apply and give the rational diagonal and a nontrivial recurrence

$$p_0(n)u_n + \dots + p_r(n)u_{n+r} = 0, \quad p_j \in \mathbb{Q}[n], \quad (p_0, \dots, p_r) \neq 0.$$

The displayed form is an existence statement, not a computed recurrence. Clearing denominators gives integer coefficients. Summing a polynomial-coefficient recurrence against powers of z yields a linear differential equation, initially with a polynomial inhomogeneous term; sufficient differentiation makes it homogeneous. Hence U is D-finite. $\square$

The controlled bounds matter. Merely saying that a sum is finite or its summand is hypergeometric would not check the required closure theorem. For example $\sum_{k=1}^{2^{n^2}} 1 = 2^{n^2}$ is not P-recursive: in any alleged recurrence divide by the largest nonzero shift term, and all smaller-shift ratios tend to zero, a contradiction. That upper bound is not affine or rational-polyhedral. Similarly, summing a formal Wilf–Zeilberger certificate requires support and singularity checks [19]; the everywhere-defined binomial encoding avoids that issue.

Remark 13.2 ([**write**] The hypotheses of the closure theorem). Read 5 October 2026 in arXiv:1510.07487v2, the version whose numbering the text uses. There, for a field K of characteristic zero, Definition 1.1 (page 7) defines the algebra of binomial sums as the smallest subalgebra of the sequences $\mathbb{Z}^d \rightarrow K$ that contains (a) the Kronecker delta, (b) the geometric sequences $n \mapsto C^n$, (c) the binomial sequence $(n, k) \mapsto \binom{n}{k}$, and is closed under (d) composition with affine maps $\mathbb{Z}^d \rightarrow \mathbb{Z}^e$ and (e) the directed indefinite sum $(n, m) \mapsto \sum_{k=0}^m u_{n, k}$. By Section 1.1 there, $\binom{n}{k}$ is the coefficient of x^k in $(1 + x)^n$ for all integers n, k , and a directed sum with upper limit below its lower limit is defined by a sign convention, not as zero. Theorem 3.5 (page 17): a sequence $u : \mathbb{N} \rightarrow K$ is

a binomial sum if and only if $\sum_{n \geq 0} u_n t^n$ is the diagonal of a rational power series; “ $u : \mathbb{N} \rightarrow K$ is a binomial sum” means that some binomial sum $v : \mathbb{Z} \rightarrow K$ with support in $\mathbb{N}$ coincides with u on $\mathbb{N}$. Corollary 3.6 (page 17): every binomial sum $u : \mathbb{Z} \rightarrow K$ satisfies a recurrence with polynomial coefficients, not all zero, for all $n \in \mathbb{Z}$.

The encoding (13.5) meets these hypotheses, with $K = \mathbb{Q}$. The primitive B of (13.1) is the binomial sequence of (c), with the same convention at negative arguments. The factors $h(r) = B(r, r)$ and $\delta(r) = B(0, r)$ are (c) composed with affine maps, by (d). $\mathcal{D}$, $\mathcal{E}$ and $\mathcal{Q}_{\alpha, \gamma}$ are finite signed sums of products of such factors at integral affine arguments of the 24 indices and n . Each box sum over $\{0, \dots, n+2\}$ is the directed sum from 0 to the affine bound $n+2$, obtained by (e) with a fresh upper variable followed by the affine substitution of (d) (display (6) there). For $n \geq 0$ these directed sums are ordinary sums, so $v(n) = h(n) \cdot (\text{right side of (13.5)})$ is a binomial sum on $\mathbb{Z}$ with support in $\mathbb{N}$ and $v(n) = u_n$ for $n \geq 0$. Theorem 3.5 then gives the rational diagonal over $\mathbb{Q}$, and Corollary 3.6 the recurrence used in the proof of Theorem 11.1. The same check applies without change to Part IV’s 23-index encoding (38.5), and to Part V’s Theorem 48.1 with the reading noted there.

14 A compressed rational constant term

We now give a single rational integrand, rather than the 5184-term binomial expansion. Its constant term has specified contour semantics. It is not asserted to be analytic at the simultaneous origin of all integration variables.

14.1 The two blocks and the twelve weights

For a triple X , define

$$c(v) = \frac{1+v}{v}, \quad \Sigma(X) = X_1 + X_2 + X_3, \quad J(X) = \frac{(X_1 - X_2)(X_1 - X_3)(X_2 - X_3)}{X_1^2 X_2}.$$

One block has 20 internal variables

$$A, B, C; b; p_1, \dots, p_5; q_1, \dots, q_5; X_1, X_2, X_3; Y_1, Y_2, Y_3.$$

Write $H = c(b)$, $T_r = c(p_r)$ and $V_r = c(q_r)$. The other block has independent primed copies of all these variables. Two additional, shared variables ξ, η give

$$P = \frac{1+\xi}{\xi}, \quad Q = \frac{1+\eta}{\eta}, \quad P' = 1+\xi, \quad Q' = 1+\eta.$$

Here P, Q are shuffle weights, unrelated to the trace notation used later. Define the following twelve rational weights for one block:

$$\begin{aligned} R_s &= \frac{zH}{ABC}, & R_i &= \frac{BP}{H}, & R_j &= \frac{CQ}{H}, \\ R_{\lambda_1} &= AT_1 V_1, & R_{\lambda_2} &= \frac{AT_3 V_3}{T_2 V_2}, & R_{\lambda_3} &= \frac{AT_5 V_5}{T_4 V_4}, \\ R_{\mu_1} &= \frac{B\Sigma(X)T_2}{X_1 T_1}, & R_{\mu_2} &= \frac{B\Sigma(X)T_4}{X_2 T_3}, & R_{\mu_3} &= \frac{B\Sigma(X)}{X_3 T_5}, \\ R_{\nu_1} &= \frac{C\Sigma(Y)V_2}{Y_1 V_1}, & R_{\nu_2} &= \frac{C\Sigma(Y)V_4}{Y_2 V_3}, & R_{\nu_3} &= \frac{C\Sigma(Y)}{Y_3 V_5}. \end{aligned} \tag{14.1}$$

For this list $\mathcal{R}$ of twelve weights, put

$$\mathcal{K}(z; P, Q) = \frac{1-A}{A^2 BC} J(X) J(Y) \prod_{R \in \mathcal{R}} \frac{1}{1-R}. \quad (14.2)$$

The primed block $\mathcal{K}'(z; P', Q')$ has the identical formula in primed internal variables and the indicated primed shuffle weights.

Theorem 14.1. *With the expansion or contour prescription below,*

$$U(z) = z^4 \text{CT} [\mathcal{K}(z; P, Q) \mathcal{K}'(z; P', Q')], \quad (14.3)$$

where the constant term is over all $20 + 20 + 2 = 42$ internal and shuffle variables. The generating-function variable z is not integrated. The integrand contains 24 geometric factors of the displayed form $(1-R)^{-1}$.

Four Vandermonde factors compress the 6^4 dimension expansion, while $(1-A)(1-A')$ compresses the four choices in the two half differences. This is an explicit rational-period input for elimination; its size alone gives no useful bound on the order or practical cost of a recurrence.

14.2 Verification of all extractions

Every geometric factor is expanded in nonnegative powers of its displayed weight. For an integer d , on any positively oriented circle $|v| < 1$,

$$\text{CT}_v c(v)^d = [v^d](1+v)^d = h(d). \quad (14.4)$$

For negative d , $(1+v)^d$ is analytic inside the circle, and the other factor is v^{-d} of positive degree. Its constant term is zero. For a nonnegative triple μ of size M ,

$$\begin{aligned} \text{CT}_X J(X) \prod_{r=1}^3 \left(\frac{\Sigma(X)}{X_r} \right)^{\mu_r} \\ = [X_1^{\mu_1+2} X_2^{\mu_2+1} X_3^{\mu_3}] (X_1 - X_2)(X_1 - X_3)(X_2 - X_3) \Sigma(X)^M. \end{aligned} \quad (14.5)$$

For partitions this is f^μ , by the six-term determinant computation in Lemma 13.1. The strip indicators ensure that only partitions survive, so dimensions of nonpartitions need no interpretation.

Let the twelve nonnegative geometric exponents be $s, i, j, \lambda_1, \lambda_2, \lambda_3, \mu_1, \mu_2, \mu_3, \nu_1, \nu_2, \nu_3$. Before including the numerator in (14.2), their product is

$$\begin{aligned} z^s H^{s-i-j} A^{|\lambda|-s} B^{|\mu|-s+i} C^{|\nu|-s+j} P^i Q^j \\ \cdot \prod_{r=1}^3 \left(\frac{\Sigma(X)}{X_r} \right)^{\mu_r} \left(\frac{\Sigma(Y)}{Y_r} \right)^{\nu_r} \\ \cdot T_1^{\lambda_1-\mu_1} T_2^{\mu_1-\lambda_2} T_3^{\lambda_2-\mu_2} T_4^{\mu_2-\lambda_3} T_5^{\lambda_3-\mu_3} \\ \cdot V_1^{\lambda_1-\nu_1} V_2^{\nu_1-\lambda_2} V_3^{\lambda_2-\nu_2} V_4^{\nu_2-\lambda_3} V_5^{\lambda_3-\nu_3}. \end{aligned} \quad (14.6)$$

Extraction in b imposes $i+j \leq s$, and extraction in p_r, q_r imposes both interlacings. Lower-bound indicators for μ_3, ν_3 are redundant because the geometric indices are already nonnegative. The numerator $(1-A)/(A^2 BC)$ and extraction in A, B, C impose

$$|\lambda| = s + 2 - \alpha, \quad |\mu| = s + 1 - i, \quad |\nu| = s + 1 - j, \quad (14.7)$$

with $\alpha \in \{0, 1\}$ and sign $(-1)^\alpha$. These are precisely the size constraints in $F_{s+2}(i+1, j+1) - F_{s+1}(i, j)$. The inner sizes are independent of α , explaining why one factor $1 - A$ suffices. The primed block gives the analogous southern half with indices t, k, l . Finally

$$\text{CT}_\xi P^i(P')^k = \binom{i+k}{i}, \quad \text{CT}_\eta Q^j(Q')^l = \binom{j+l}{j}.$$

The powers of z and the leading z^4 give $n = s + t + 4$, so the result is (12.2).

14.3 A convergent contour prescription

The preceding expansion is unambiguous under the following explicit radii. Give every dimension variable X_r, Y_r, X'_r, Y'_r radius 1; every boundary, strip and shuffle variable $b, b', p_r, q_r, p'_r, q'_r, \xi, \eta$ radius $1/2$; and every equality variable A, B, C, A', B', C' radius $\varepsilon = 1/100$. Take $|z| < \varepsilon^3/6$. On $|v| = 1/2$, $1 \leq |c(v)| \leq 3$. On the dimension circles, $|\Sigma(X)/X_r| \leq 3$, and similarly for the other triples. Thus

$$\begin{aligned} |R_s| &\leq 3|z|\varepsilon^{-3} < 1/2, \\ |R_{\lambda_r}|, |R_{\mu_r}|, |R_{\nu_r}| &\leq 9\varepsilon < 1, \\ |R_i|, |R_j| &\leq 3\varepsilon < 1. \end{aligned}$$

Primed shuffle weights have modulus at most $3/2$, so their estimates are no worse. All 24 geometric series converge absolutely and uniformly on the full product torus. The remaining numerator is bounded and continuous there. Sum-integral interchange is therefore valid. The constant term is the resulting normalized integral with measure $\prod_v dv/(2\pi i v)$.

The integral is analytic in z near zero. After a coefficient z^n is fixed, the equalities in (14.7) and $s + t = n - 4$ leave only finitely many surviving indices. The proof above therefore establishes both analytic and formal-series equality. It does not license an arbitrary alternative Laurent expansion at a simultaneous origin.

15 A unitary integral and the avoidance correction

The new asymptotic argument starts from the same tableau formula, now using characters. Let $U \in U(3)$, write $P(U) = \text{tr } U$, and let $h_p(U) = s_{(p)}(U)$ be the complete homogeneous symmetric polynomial in its three eigenvalues, with $h_0 = 1$. Haar measure has total mass one. Schur–Weyl decomposition and the horizontal Pieri identity give

$$\begin{aligned} P(U)^{m-p} &= \sum_{\substack{\mu \vdash m-p \\ \ell(\mu) \leq 3}} f^\mu s_\mu(U), \\ P(U)^{m-p} h_p(U) &= \sum_{\substack{\lambda \vdash m \\ \ell(\lambda) \leq 3}} W_p(\lambda) s_\lambda(U). \end{aligned}$$

Orthogonality of irreducible unitary characters therefore yields

$$F_m(p, q) = \int_{U(3)} P(U)^{m-p} h_p(U) \overline{P(U)^{m-q} h_q(U)} dU. \quad (15.1)$$

In particular $A_m = \int |P(U)|^{2m} dU$. The trace-moment identity is also in Rains [17]; the displayed Pieri insertion supplies the boundary conditions here.

Under $U \mapsto zU$ with $|z| = 1$, each unconjugated factor in (15.1) has combined degree m . Thus the central phase cancels exactly, including when $p \neq q$. Weyl integration expresses the integral as $1/(6(2\pi)^3)$ times its three-angle integral with density

$$\prod_{a < b} |e^{i\theta_a} - e^{i\theta_b}|^2.$$

The maximum $|P| = 3$ occurs exactly when all eigenvalues coincide. The maxima form a center circle; modulo common phase there is a unique maximum on the quotient torus. A local centered chart has $\theta_1 + \theta_2 + \theta_3 = 0$ and constant Jacobian. We work on this quotient rather than count the three equivalent central points of an $SU(3)$ formulation separately.

Put $Q = \sum \theta_j^2$ and $\Delta(\theta) = \prod_{a < b} (\theta_a - \theta_b)$. Elementary Taylor expansions on the centered plane give

$$\begin{aligned} \frac{|P|^2}{9} &= 1 - \frac{Q}{3} + \frac{Q^2}{24} + O(Q^3), \\ \log \frac{|P|^2}{9} &= -\frac{Q}{3} - \frac{Q^2}{72} + O(Q^3), \end{aligned} \tag{15.2}$$

$$\prod_{a < b} |e^{i\theta_a} - e^{i\theta_b}|^2 = \Delta(\theta)^2 \left(1 - \frac{Q}{4} + O(Q^2) \right). \tag{15.3}$$

For the first calculation use $\sum \theta_j^4 = Q^2/2$ on that plane. For the density use $4 \sin^2(x/2) = x^2(1 - x^2/12 + O(x^4))$ and $\sum_{a < b} (\theta_a - \theta_b)^2 = 3Q$. The factored remainder remains uniform at $\Delta = 0$, because the sinc factors are analytic there.

Rescale $\theta = y/\sqrt{m}$. The leading measure is proportional to $e^{-Q_y/3} \Delta(y)^2 dy$ on a two-dimensional plane. Its radial factor is $r^7 dr$, since the Vandermonde square has degree six. Thus Q_y has Gamma radial moments of shape 4 and scale 3:

$$\mathbb{E}_0 Q_y = 12, \quad \mathbb{E}_0 Q_y^2 = 180. \tag{15.4}$$

Equations (15.2)–(15.4) give the relative first correction $-12/4 - 180/72 = -11/2$.

Lemma 15.1. *As $m \rightarrow \infty$,*

$$A_m = C_0 9^m m^{-4} \left(1 - \frac{11}{2m} + O(m^{-2}) \right). \tag{15.5}$$

Let $\mathbb{E}_m$ denote expectation for density $|P|^{2m}/A_m$ against Haar measure. In the fixed small centered chart,

$$\mathbb{E}_m[Q; \text{local}] = \frac{12}{m} + O(m^{-2}). \tag{15.6}$$

For any fixed $c_1 > 0$ and nonnegative integer r , there are $\varepsilon > 0, C_r < \infty$ such that

$$\mathbb{E}_m[Q^r e^{c_1 L Q}; \text{local}] \leq C_r m^{-r} \quad (0 \leq L \leq \varepsilon m). \tag{15.7}$$

Proof. Outside a fixed neighborhood of the center circle, compactness gives $|P|/3 \leq \rho < 1$. Inside a sufficiently small chart, $|P|^{2m}/9^m \leq e^{-cmQ}$ and the Weyl density is bounded by a constant times Δ^2 . After the two displayed Taylor corrections, the remaining errors can be bounded by constants times Q^2, mQ^3, m^2Q^4 with a slightly weaker Gaussian $e^{-c'mQ}$. Their integrated sizes are $O(m^{-2})$ relative to the leading integral. This follows either by Taylor's theorem with exponential absorption or by splitting at $Q = m^{-2/3}$ and using Gaussian decay on its complement. Regions away from the center give exponentially small contributions. The leading constant is the Regev constant already specified in (11.1); the moment calculation above proves (15.5). The same estimates with one extra Q give (15.6). For (15.7), choose ε so that $c_1 L Q$ is absorbed by a fixed part of the Gaussian. Radial scaling, together with the lower bound in (12.6), gives m^{-r} . $\square$

16 A boundary expansion uniform in the parameters

A fixed- p, q expansion alone is insufficient for the four-index convolution. The following remainder allows those indices to grow.

Proposition 16.1. *There are absolute $\varepsilon > 0$ and $C < \infty$ such that, uniformly for all sufficiently large m and $p, q \geq 0$ with $p + q \leq \varepsilon m$,*

$$\frac{F_m(p, q)}{A_m} = d_p d_q \left(1 - \frac{a_p + a_q}{m} + O\left(\frac{(1 + p + q)^4}{m^2}\right) \right). \quad (16.1)$$

Proof. Write $c_p = \binom{p+2}{2}$. In the local centered chart, put

$$r_p(\theta) = \frac{h_p(e^{i\theta})}{c_p(P(\theta)/3)^p}, \quad h_p/P^p = d_p r_p.$$

The c_p weak compositions $\delta_1 + \delta_2 + \delta_3 = p$ index the terms of h_p . Their mean has each coordinate $p/3$, so the centered linear term vanishes. Their second moment is

$$\frac{1}{c_p} \sum_{\delta} (\delta \cdot \theta)^2 = \frac{p(p+3)}{12} Q. \quad (16.2)$$

For example, differentiation of $\prod_j (1 - x_j z)^{-1}$ and extraction of z^p give $\mathbb{E}[\delta_1(\delta_1 - 1)] = p(p-1)/6$ and $\mathbb{E}[\delta_1 \delta_2] = p(p-1)/12$, from which (16.2) follows. Hence

$$\frac{h_p}{c_p} = 1 - \frac{p(p+3)}{24} Q + O_p(|\theta|^3), \quad (P/3)^{-p} = 1 + \frac{p}{6} Q + O_p(|\theta|^3),$$

and $r_p = 1 - p(p-1)Q/24 + O_p(|\theta|^3)$. The needed real product has quadratic term

$$\Re(r_p \overline{r_q}) = 1 - \frac{p(p-1) + q(q-1)}{24} Q + \text{remainder}. \quad (16.3)$$

We now make that remainder uniform instead of using the fixed-parameter O_p estimates.

Fix a centered unit vector v and put $\theta = rv$, so $Q = r^2$. Let $z(r) = P(rv)/3$ and $g_p(r) = h_p(e^{irv})/c_p$. For $0 \leq j \leq 4$, $|g_p^{(j)}(r)| \leq p^j$ (with the evident bound 1 at $j = 0$), because $|\delta \cdot v| \leq p$. On a fixed small real interval, $|z(r)| \geq e^{-C_1 r^2}$ and its first four derivatives have modulus at most 1. The finite chain-rule formula for $z(r)^{-p}$ therefore gives

$$\left| \frac{d^j}{dr^j} z(r)^{-p} \right| \leq C_j (p+1)^j e^{C_2 p r^2} \quad (0 \leq j \leq 4).$$

The finitely many additional inverse powers of z are harmless since z stays away from zero. The derivatives through degree four of $r_p \overline{r_q}$ are consequently bounded by $C(1+p+q)^4 e^{C_3(p+q)r^2}$. Its real part is even in r . Taylor's theorem through the cubic term, whose odd coefficients vanish, proves the pointwise estimate

$$\left| \Re(r_p \overline{r_q}) - 1 + \frac{p(p-1) + q(q-1)}{24} Q \right| \leq C(1+p+q)^4 Q^2 e^{C_3(p+q)Q}. \quad (16.4)$$

Equation (15.7) integrates this to $O((1+p+q)^4/m^2)$ for sufficiently small ε , while (15.6) integrates the quadratic term. Its expectation equals $-(a_p + a_q)/m + O((1+p+q)^2/m^2)$. Conjugation invariance permits replacing the full integrand by its real part.

There is no singular-denominator assumption outside the local chart: use the original polynomial integrand in (15.1). Since $|h_p| \leq c_p$, its modulus is at most

$$9^m d_p d_q \left(\frac{|P|}{3} \right)^{2m-p-q}.$$

Off the center neighborhood this is bounded by $9^m d_p d_q \rho^{(2-\varepsilon)m}$. After division by A_m its contribution is $O(d_p d_q m^4 e^{-cm})$, absorbed in the displayed remainder. The constant and quadratic approximants have exponentially small complement corrections as well. This proves (16.1). $\square$

As exact normalization checks, $h_0 = 1$ and $h_1 = P$, so $F_m(p, q) = A_m$ for $p, q \in \{0, 1\}$. These agree with $d_0 = d_1 = 1$ and $a_0 = a_1 = 0$. For example the next case is $F_m(2, 0)/A_m = (2/3)(1 - 1/m + O(m^{-2}))$.

16.1 The uniform half expansion

Let $\alpha = -11/2$. Combining (15.5) and (16.1) with a fixed shift $h \in \{1, 2\}$ gives

$$F_{s+h}(p, q) = C_0 9^s s^{-4} 9^h d_p d_q \left(1 + \frac{\alpha - 4h - a_p - a_q}{s} + O\left(\frac{(1+p+q)^4}{s^2}\right) \right). \quad (16.5)$$

The constants remain uniform in the range required by Proposition 16.1. Set $w_i = (i+1)^6 3^{-i}$. Subtraction as in (12.1) yields, uniformly when $i+j+2 \leq \varepsilon' s$,

$$H_s(i, j) = C_0 9^s s^{-4} \left(C(i, j) + \frac{\alpha C(i, j) + B(i, j)}{s} + O\left(\frac{w_i w_j}{s^2}\right) \right). \quad (16.6)$$

The sixth powers dominate the quadratic growth of d_i times the fourth-degree uniform error. With an absolute constant, $w_i w_j$ also majorizes $|C(i, j)|$ and $|B(i, j)|$.

17 Passing the correction through the convolution

We prove Theorem 11.2 with a remainder summable over the whole parameter range. First, for any fixed nonnegative integers r, s ,

$$\sum_{i, k \geq 0} (i+1)^r (k+1)^s 3^{-i-k} \binom{i+k}{i} < \infty. \quad (17.1)$$

Indeed, grouping by $v = i+k$ bounds the contribution by $(v+1)^{r+s} (2/3)^v$. Polynomial enlargements of the old boundary majorant therefore remain summable.

17.1 Removing large boundaries uniformly

Let $L_n = \lceil 20 \log n \rceil$, with natural logarithm. In the exact convolution omit terms with any of i, j, k, l exceeding L_n . By (12.7) and (17.1), the omitted total, normalized by A_n , is

$$O(\text{poly}(L_n) (2/3)^{L_n}) = O(n^{-3}). \quad (17.2)$$

Here poly denotes a fixed polynomial bound, not a quantity with degree depending on n . To check uniformity in half sizes, (12.6) gives

$$\sum_{s+t=n-4} \frac{A_s A_t}{A_n} = O(1). \quad (17.3)$$

Split at $(n-4)/2$, bound the larger half by a constant times n^{-4} , and sum $(s+1)^{-4}$ for the smaller half. If an omitted index is, say, $i > L_n$, then $i+k > L_n$; grouping that pair gives the geometric tail in (17.2), and the other pair is fully summable. Since $20 \log(3/2) > 8$, the logarithmic polynomial is harmless for the stated $O(n^{-3})$ estimate.

17.2 A uniform endpoint expansion

Put $N = n - 4$ and write the larger half as $s = N - t$. By symmetry we may double $0 \leq t \leq \lfloor N/2 \rfloor$. If N is even this double-counts its central term. Its normalized contribution is $O(n^{-4})$, since its aggregate is at most a constant times $A_{N/2}^2/A_n$ by (12.7) and (17.1).

Throughout the retained range $s \geq N/2$, and all boundaries are at most L_n , so (16.6) applies for large n . Combining it with the denominator expansion (15.5) gives

$$\begin{aligned} \frac{H_{n-4-t}(i, j)}{A_n} &= 9^{-t-4} \left(C(i, j) + \frac{4(t+4)C(i, j) + B(i, j)}{n} \right. \\ &\quad \left. + O\left(\frac{w_i w_j (t+4)^2}{n^2}\right) \right). \end{aligned} \quad (17.4)$$

For a uniform justification, let $x = (t+4)/n$. For all sufficiently large n , $0 \leq x \leq 3/4$, so Taylor's theorem on that fixed interval gives

$$(n/s)^4 = (1-x)^{-4} = 1 + \frac{4(t+4)}{n} + O\left(\frac{(t+4)^2}{n^2}\right).$$

Also $|1/s - 1/n| = O((t+4)/n^2)$ there. The terms $\alpha C(i, j)$ from numerator and denominator cancel at order $1/n$. All remaining coefficients are dominated by a constant times $w_i w_j$, proving (17.4) on the entire smaller-half range, not only for fixed t .

Multiply (17.4) by $H_t(k, l)$ and the two binomial kernels. The total absolute remainder is bounded by

$$\frac{C}{n^2} \sum_{t \leq N/2} 9^{-t} A_t (t+4)^2 \left(\sum_{i, k \geq 0} w_i b_k \binom{i+k}{i} \right)^2 = O(n^{-2}). \quad (17.5)$$

The boundary factor converges by (17.1). The half-size sum converges uniformly because $9^{-t} A_t (t+4)^2 = O((t+1)^{-2})$. This is the required weighted uniform error estimate.

17.3 Restoring the infinite coefficient series

The absolute t -th terms of the leading endpoint series are $O((t+1)^{-4})$. Their tail beyond $N/2$ is $O(n^{-3})$. The coefficient of $1/n$ in (17.4) has absolute t -th terms $O((t+1)^{-3})$, so its tail is $O(n^{-2})$ before multiplication by $1/n$ and $O(n^{-3})$ afterward. This proves absolute convergence of (11.5) independently of its transform evaluation. Boundary tails in both coefficient series have the same exponential-polynomial estimate as (17.2): (17.1) permits the extra polynomial factors, and the needed zeroth and first half-size moments are finite.

Restoring all cutoffs, the leading term is exactly (12.8), the first coefficient is (11.5), and the remainder is $O(n^{-2})$ by (17.5). This proves (11.6). Multiplication by (15.5) proves (11.7) with $\beta = S/R - 11/2$, completing the proof of Theorem 11.2.

18 Exact transforms and a finite-boundary series

Define four absolutely convergent binomial transforms:

$$\begin{aligned} D_k &= \sum_{i \geq 0} \binom{i+k}{i} d_i, & E_k &= \sum_{i \geq 0} \binom{i+k}{i} d_{i+1}, \\ J_k &= \sum_{i \geq 0} \binom{i+k}{i} a_i d_i, & K_k &= \sum_{i \geq 0} \binom{i+k}{i} a_{i+1} d_{i+1}. \end{aligned}$$

(The scalar J_k is distinct from the Vandermonde factor $J(X)$.) Their exact evaluations are

$$\begin{aligned} D_k &= \frac{3}{16}(k^2 + 11k + 18) \left(\frac{3}{2}\right)^k, \\ E_k &= \frac{1}{16}(k^2 + 15k + 38) \left(\frac{3}{2}\right)^k, \\ J_k &= \frac{3}{128}(k+1)(k+2)(k^2 + 23k + 108) \left(\frac{3}{2}\right)^k, \\ K_k &= \frac{1}{128}(k+1)(k^3 + 33k^2 + 290k + 648) \left(\frac{3}{2}\right)^k. \end{aligned} \tag{18.1}$$

Here is a direct symbolic derivation. For $G_k(x) = (1-x)^{-k-1} = \sum_{i \geq 0} \binom{i+k}{i} x^i$ and $\vartheta = x \, d/dx$,

$$\begin{aligned} D_k &= \frac{1}{2}(\vartheta^2 + 3\vartheta + 2)G_k|_{x=1/3}, \\ E_k &= \frac{1}{6}(\vartheta^2 + 5\vartheta + 6)G_k|_{x=1/3}, \\ J_k &= \frac{1}{4}(\vartheta^4 + 2\vartheta^3 - \vartheta^2 - 2\vartheta)G_k|_{x=1/3}, \\ K_k &= \frac{1}{12}(\vartheta^4 + 6\vartheta^3 + 11\vartheta^2 + 6\vartheta)G_k|_{x=1/3}. \end{aligned}$$

The identity $\vartheta^r = \sum_{j=0}^r \{r \atop j\} x^j d^j/dx^j$, with Stirling numbers of the second kind, reduces each evaluation to a polynomial in k times $(3/2)^k$ and gives (18.1). It is also an exact finite algebraic verification route; no truncation of an infinite sum is needed. At $k=0$ the checks are $D_0 = 27/8$, $E_0 = 19/8$ and $J_0 = K_0 = 81/16$.

Substitute (11.3)–(11.4) in (11.5) and use these transforms. The result is the finite-boundary expression

$$\begin{aligned} S &= \frac{2}{81} \sum_{t \geq 0} 9^{-t} \sum_{k+l \leq t} H_t(k, l) \left\{ (4t+8)E_k E_l - K_k E_l - E_k K_l \right. \\ &\quad \left. - \frac{1}{9}[(4t+12)D_k D_l - J_k D_l - D_k J_l] \right\}. \end{aligned} \tag{18.2}$$

Every inner sum is finite and every $H_t(k, l)$ is exactly computable from (12.3)–(12.5). For comparison the established positive amplitude has the form

$$R = \frac{2}{81} \sum_{t \geq 0} 9^{-t} \sum_{k+l \leq t} H_t(k, l) \left(E_k E_l - \frac{1}{9} D_k D_l \right). \tag{18.3}$$

The absolute tail bound proved above for S is $O(T^{-2})$ after truncation at half size T , with an unspecified constant. Thus the finite rational partial sums in the reproducibility archive are exact finite data, but they are not certified upper or lower bounds for S , nor are they certified approximations with known decimal error.

19 The second correction and the first logarithm

The finite second half-size moment permits one further coefficient. The next moment diverges, but its leading harmonic part can be evaluated rather than discarded. This produces a nonzero logarithmic term. To keep it distinct from the global boundary weight b_i in (12.7), write

$$\zeta_p = \frac{p(p-1)(2p^2-6p+5)}{16}, \quad \zeta(p, q) = \zeta_p + \zeta_q + \frac{5}{4}a_p a_q. \quad (19.1)$$

Use $\alpha = -11/2$ as before and define the new boundary array

$$\begin{aligned} \mathcal{D}_2(i, j) = & 81d_{i+1}d_{j+1}[51 + 10(a_{i+1} + a_{j+1}) + \zeta(i+1, j+1)] \\ & - 9d_i d_j [31/2 + 5(a_i + a_j) + \zeta(i, j)]. \end{aligned} \quad (19.2)$$

For $z = t + 4$ put

$$P_2(t; i, j) = (10z^2 + \alpha z)C(i, j) + 5zB(i, j) + \mathcal{D}_2(i, j). \quad (19.3)$$

Here z is a temporary scalar half-size shift, not the generating variable used in Sections 13–14.

Theorem 19.1. *The signed series*

$$S_2 = 2 \sum_{t \geq 0} 9^{-t-4} \sum_{k+l \leq t} H_t(k, l) \sum_{i, j \geq 0} \binom{i+k}{i} \binom{j+l}{j} P_2(t; i, j) \quad (19.4)$$

converges absolutely, and

$$\frac{u_n}{A_n} = R + \frac{S}{n} + \frac{S_2}{n^2} + \kappa \frac{\log n}{n^3} + O(n^{-3}), \quad \kappa = \frac{43020\sqrt{3}}{\pi} > 0. \quad (19.5)$$

Equivalently, with $\delta = 20 + \alpha S/R + S_2/R$ and $\eta = \kappa/R$,

$$u_n = C_0 R 9^n n^{-4} \left(1 + \frac{\beta}{n} + \frac{\delta}{n^2} + \eta \frac{\log n}{n^3} + O(n^{-3}) \right). \quad (19.6)$$

Thus a pure relative expansion in integer powers of $1/n$ fails at its third correction. The finite coefficient at n^{-3} has not been computed or proved to exist here: the remainder in (19.5) is only bounded after multiplication by n^3 . The value and sign of S_2 or δ are not numerically certified. The explicit positive logarithmic coefficient is different in this respect from those series-defined constants. **[write] Superseded in part, 5 October 2026.** Part III proves that the limit r_3 exists, gives an absolutely convergent regularized-series recipe for it, and shows that every further coefficient has the form $r_j + \kappa_j \log n$ (Theorem 23.1, (26.5)). The values and signs of S and S_2 remain uncertified.

19.1 Local expansion through second order

In the centered chart of Section 15, put $T_3 = \sum \theta_j^3$. In addition to the earlier symmetric identities,

$$\sum_{a < b} (\theta_a - \theta_b)^6 = \frac{33}{4} Q^3 - 9T_3^2.$$

Expanding the trace logarithm and the factored Weyl density gives

$$\log \frac{|P|^2}{9} = -\frac{Q}{3} - \frac{Q^2}{72} - \frac{13Q^3}{12960} + \frac{T_3^2}{360} + O(Q^4), \quad (19.7)$$

$$\prod_{a < b} |e^{i\theta_a} - e^{i\theta_b}|^2 = \Delta^2 \left(1 - \frac{Q}{4} + \frac{9Q^2}{320} + O(Q^3) \right). \quad (19.8)$$

The sinc factorization still gives uniformity through the lines $\Delta = 0$. The leading scaled measure has radial moments

$$\mathbb{E}_0 Q = 12, \quad \mathbb{E}_0 Q^2 = 180, \quad \mathbb{E}_0 Q^3 = 3240, \quad \mathbb{E}_0 Q^4 = 68040.$$

Choose the polar angle ϕ on the centered plane so that $T_3^2 = Q^3 \cos^2(3\phi)/6$ and $\Delta^2 = Q^3 \sin^2(3\phi)/2$. With angular weight $\sin^2(3\phi)$, the average of $\cos^2(3\phi)$ is $1/4$, and hence $\mathbb{E}_0 T_3^2 = \mathbb{E}_0 Q^3/24 = 135$.

After scaling, the first two relative integrand corrections are

$$K_1 = -Q/4 - Q^2/72, \quad K_2 = 9Q^2/320 + Q^3/405 + T_3^2/360 + Q^4/10368.$$

Their expectations are $-11/2$ and $81/16 + 8 + 3/8 + 105/16 = 20$. Therefore

$$A_m = C_0 9^m m^{-4} \left(1 - \frac{11}{2m} + \frac{20}{m^2} + O(m^{-3}) \right). \quad (19.9)$$

To justify the error, omitted logarithm terms begin at mQ^4 and omitted density terms at Q^3 . The remaining exponential products have scaled order at least m^{-3} . On a sufficiently small fixed chart their absolute values are bounded by a polynomial of these orders times a weaker Gaussian. Radial integration gives relative $O(m^{-3})$; the complementary region remains exponentially small. This repeats the uniform argument of Lemma 15.1 one order farther. The same calculation with extra Q factors gives

$$\mathbb{E}_m[Q; \text{local}] = \frac{12}{m} - \frac{24}{m^2} + O(m^{-3}), \quad \mathbb{E}_m[Q^2; \text{local}] = \frac{180}{m^2} + O(m^{-3}). \quad (19.10)$$

Indeed, the new coefficient of the first moment is

$$\text{Cov}_0(Q, K_1) = -\frac{180 - 144}{4} - \frac{3240 - 12 \cdot 180}{72} = -24.$$

The exponentially weighted moment estimate (15.7) is unchanged.

19.2 The uniform second boundary coefficient

The refined boundary estimate is, for sufficiently small absolute $\varepsilon > 0$ and $p + q \leq \varepsilon m$,

$$\frac{F_m(p, q)}{A_m} = d_p d_q \left(1 - \frac{a_p + a_q}{m} + \frac{\zeta(p, q)}{m^2} + O\left(\frac{(1 + p + q)^6}{m^3}\right) \right). \quad (19.11)$$

We supply both its coefficient and its uniform remainder. For uniform weak compositions of p ,

$$\mathbb{E}(\delta_1 - \delta_2)^4 = \frac{p(p+3)(2p^2 + 6p - 3)}{30}.$$

This follows by differentiating the same composition generating function used for (16.2). Every symmetric homogeneous quartic on the centered three-coordinate plane is a multiple of Q^2 ; evaluation at $(x, -x, 0)$ therefore gives

$$\begin{aligned} h_p/c_p &= 1 - \frac{p(p+3)}{24}Q + \text{an imaginary cubic} \\ &\quad + \frac{p(p+3)(2p^2 + 6p - 3)}{2880}Q^2 + O_p(|\theta|^5), \\ (P/3)^{-p} &= 1 + \frac{p}{6}Q + \text{an imaginary cubic} + \frac{p(2p+1)}{144}Q^2 + O_p(|\theta|^5). \end{aligned}$$

Consequently, for $v_p = p(p-1)(2p^2 - 6p - 11)/2880$,

$$r_p = 1 - a_p Q/12 + \text{an imaginary cubic} + v_p Q^2 + O_p(|\theta|^5).$$

There are no linear terms. The real product has expansion

$$\Re(r_p \bar{r}_q) = 1 - \frac{a_p + a_q}{12} Q + \left(v_p + v_q + \frac{a_p a_q}{144} \right) Q^2 + \text{remainder}. \quad (19.12)$$

The directional derivative argument in Proposition 16.1 extends through degree six: for $j \leq 6$, $|g_p^{(j)}| \leq p^j$, while derivatives of z^{-p} are bounded by $C_j(p+1)^j e^{C_2 p r^2}$ on a fixed interval. The sixth derivative of the real product is at most $C(1+p+q)^6 e^{C_3(p+q)r^2}$. Evenness in r and Taylor's theorem through degree five therefore bound the omitted part of (19.12) by

$$C(1+p+q)^6 Q^3 e^{C_3(p+q)Q}.$$

Its integral is $O((1+p+q)^6/m^3)$ by (15.7). The two moments in (19.10) give second coefficient

$$2(a_p + a_q) + 180(v_p + v_q) + \frac{5}{4}a_p a_q = \zeta(p, q).$$

Off the chart the original polynomial integrand is bounded by $9^m d_p d_q \rho^{2m-p-q}$, as before. This is uniformly exponentially small after normalization and introduces no division by a vanishing trace. Thus (19.11) is proved. For example $(p, q) = (2, 0)$ gives $(2/3)(1 - 1/m + 1/(8m^2) + O(m^{-3}))$; the four cases $p, q \in \{0, 1\}$ remain exact.

19.3 Half expansion and extraction of the divergent moment

For fixed $h \in \{1, 2\}$, equations (19.9) and (19.11) imply

$$\begin{aligned} \frac{F_{s+h}(p, q)}{A_s} &= 9^h d_p d_q \left[1 + \frac{-4h - a_p - a_q}{s} \right. \\ &\quad \left. + \frac{10h^2 - \alpha h + 5h(a_p + a_q) + \zeta(p, q)}{s^2} + O\left(\frac{(1+p+q)^6}{s^3}\right) \right]. \end{aligned} \quad (19.13)$$

The avoidance coefficient 20 cancels in the shifted ratio. Taking the half difference gives exactly the array (19.2):

$$\frac{H_s(i, j)}{A_s} = C(i, j) + \frac{B(i, j)}{s} + \frac{\mathcal{D}_2(i, j)}{s^2} + O\left(\frac{\tilde{w}_i \tilde{w}_j}{s^3}\right), \quad \tilde{w}_i = (i+1)^8 3^{-i}, \quad (19.14)$$

uniformly when $i + j + 2 \leq \varepsilon' s$. The same polynomial weight majorizes $C, B, \mathcal{D}_2$.

For concise notation define the bilinear convolution pairing

$$\langle F, G \rangle_{\mathcal{K}} = \sum_{i, j, k, l \geq 0} F(i, j) G(k, l) \binom{i+k}{i} \binom{j+l}{j}.$$

All instances below converge absolutely by (17.1). Put $N = n - 4$ and again double $0 \leq t \leq \lfloor N/2 \rfloor$. The possible central double count is $O(n^{-4})$ after normalization. Use the boundary cutoff $\lceil 40 \log n \rceil$; by the global majorant, (17.1) and (17.3), the omitted normalized convolution mass is $O(n^{-5})$. The same cutoff controls all extra polynomial boundary weights needed here.

With $z = t + 4$ and $s = n - z$, expansion on $0 \leq z/n \leq 3/4$ gives

$$\begin{aligned} \frac{H_{n-z}(i, j)}{A_n} = 9^{-z} \left[C(i, j) + \frac{4zC(i, j) + B(i, j)}{n} + \frac{P_2(t; i, j)}{n^2} \right. \\ \left. + \frac{20z^3C(i, j)}{n^3} + E_{n,t}(i, j) \right], \end{aligned} \quad (19.15)$$

where

$$|E_{n,t}(i, j)| \leq C\tilde{w}_i\tilde{w}_j \left(\frac{z^2}{n^3} + \frac{z^4}{n^4} \right). \quad (19.16)$$

To check the second coefficient, use $A_{n-z}/A_n = 9^{-z}[1 + 4z/n + (10z^2 + \alpha z)/n^2 + \dots]$ and $B/(n-z) = B/n + zB/n^2 + \dots$. The only cubic-in- z term of order n^{-3} is $20z^3C$ from $(1 - z/n)^{-4}$. All other terms at that order have z -degree at most two. The next remainder of the leading power is $O(z^4/n^4)$, and discarded half and avoidance errors are $O(\tilde{w}_i\tilde{w}_j/n^3)$. Since $z \geq 4$, these facts prove (19.16) without a third boundary coefficient.

After multiplication by the smaller half and summation, the two error terms in (19.16) contribute at most

$$O(n^{-3}) \sum_{t \leq N/2} (t+4)^2 A_t 9^{-t} + O(n^{-4}) \sum_{t \leq N/2} (t+4)^4 A_t 9^{-t} = O(n^{-3}).$$

The first sum is bounded and the second is $O(n)$. The leading, first and second coefficient series have absolute tails $O(n^{-3}), O(n^{-2}), O(n^{-1})$ after truncation at $N/2$. Restoring them, with factors $1, n^{-1}, n^{-2}$, costs $O(n^{-3})$ in each case. Their full sums are R, S, S_2 , and this proves the absolute convergence asserted in (19.4).

It remains to identify the harmonic term. Set

$$e_t = 9^{-t-4} \langle C, H_t \rangle_{\mathcal{K}}.$$

The first uniform half expansion gives

$$e_t = at^{-4} + O(t^{-5}), \quad a = C_0 9^{-4} \langle C, C \rangle_{\mathcal{K}}. \quad (19.17)$$

Indeed, truncate all four boundaries at $\lceil 40 \log(t+2) \rceil$. The global majorant bounds the omitted part by less than order t^{-5} . On the retained range the first half expansion has a polynomially weighted uniform error; summing it via (17.1) gives $O(t^{-5})$. Restoring the leading C boundaries costs $o(t^{-5})$ in e_t . Thus (19.17) does not interchange uncontrolled fixed-index limits with infinite sums. It follows that

$$\sum_{t \leq N/2} (t+4)^3 e_t = a \log n + O(1).$$

The doubled cubic term of (19.15) is therefore $40a \log n/n^3 + O(n^{-3})$.

19.4 Exact evaluation of the logarithmic coefficient

Put $e_i = d_{i+1}$ for this paragraph and define $M_{fg} = \sum_{i,k \geq 0} f_i g_k \binom{i+k}{i}$ for $f, g \in \{d, e\}$. Apply the quadratic operators $\frac{1}{2}(\vartheta^2 + 3\vartheta + 2)$ and $\frac{1}{6}(\vartheta^2 + 5\vartheta + 6)$ in the two variables of

$$\sum_{i,k \geq 0} \binom{i+k}{i} x^i y^k = \frac{1}{1-x-y},$$

then put $x = y = 1/3$. Finite differentiation gives

$$M_{dd} = 99, \quad M_{de} = M_{ed} = 49, \quad M_{ee} = 25.$$

The double series and their polynomially weighted derivatives converge absolutely there. Since $C = 81(e \otimes e) - 9(d \otimes d)$,

$$\begin{aligned} \langle C, C \rangle_{\mathcal{K}} &= 81^2 \cdot 25^2 - 2 \cdot 81 \cdot 9 \cdot 49^2 + 9^2 \cdot 99^2 \\ &= 1\,393\,848. \end{aligned}$$

Consequently

$$40a = 40 \frac{81\sqrt{3}}{16\pi} 9^{-4} \cdot 1\,393\,848 = \frac{43020\sqrt{3}}{\pi} > 0.$$

This proves (19.5). Multiplication by (19.9) gives (19.6), completing Theorem 19.1. There is no cancellation of this first logarithm.

19.5 Two more exact transforms

The infinite boundary sums in S_2 may also be removed. Alongside (18.1), define

$$\begin{aligned} L_k &= \sum_{i \geq 0} \binom{i+k}{i} \zeta_i d_i \\ &= \frac{3}{2048} (k+1)(k+2)(k^4 + 46k^3 + 657k^2 + 3424k + 5616) \left(\frac{3}{2}\right)^k, \\ M_k &= \sum_{i \geq 0} \binom{i+k}{i} \zeta_{i+1} d_{i+1} \\ &= \frac{1}{2048} (k+1)(k^5 + 60k^4 + 1197k^3 + 9726k^2 + 31784k + 33696) \left(\frac{3}{2}\right)^k. \end{aligned} \tag{19.18}$$

Expand the polynomial in i in falling factorials and use

$$\sum_{i \geq 0} \binom{i+k}{i} i_{\bar{r}} x^i = (k+1) \bar{r} x^r (1-x)^{-k-1-r}$$

at $x = 1/3$. This derives (19.18) by finite algebra for every k . The resulting finite-boundary expression is

$$\begin{aligned} S_2 &= \frac{2}{81} \sum_{t \geq 0} 9^{-t} \sum_{k+l \leq t} H_t(k, l) \left\{ \left(10t^2 + \frac{69}{2}t + 29 \right) E_k E_l \right. \\ &\quad - (5t+10)(K_k E_l + E_k K_l) + M_k E_l + E_k M_l + \frac{5}{4} K_k K_l \\ &\quad - \frac{1}{9} \left[\left(10t^2 + \frac{109}{2}t + \frac{147}{2} \right) D_k D_l \right. \\ &\quad \left. \left. - (5t+15)(J_k D_l + D_k J_l) + L_k D_l + D_k L_l + \frac{5}{4} J_k J_l \right] \right\}. \end{aligned} \tag{19.19}$$

The scalar transform D_k and array $\mathcal{D}_2(i, j)$ are distinct. Every inner sum is finite, but this is a signed series. Its partial sums do not automatically bound S_2 in either direction. The absolute half-size tail is $O(T^{-1})$ with an unspecified constant.

20 Sharper inverse thresholds

For real $T > 0$, define $\tau(T) = \min\{n \in \mathbb{N}_0 : u_n \geq T\}$ and put

$$\lambda = \log 9, \quad L = \log T, \quad M_0 = C_0 R, \quad \mu = \delta - \frac{\beta^2}{2}.$$

Prepending a new largest value cannot create an increasing subsequence of length at least two containing that value. It preserves old occurrences and is injective, so u_n is nondecreasing. It is unbounded by (19.6), and $\tau(T)$ is well defined. Taking logarithms of (19.6) gives

$$\log u_n = \lambda n - 4 \log n + \log M_0 + \frac{\beta}{n} + \frac{\mu}{n^2} + \eta \frac{\log n}{n^3} + O(n^{-3}). \quad (20.1)$$

The finite cubic terms from the logarithm are absorbed by the remainder; the displayed logarithmic coefficient is unaffected.

Let $y(T)$ be the unique sufficiently large solution of

$$h_*(y) := \lambda y - 4 \log y + \log M_0 + \frac{\beta}{y} + \frac{\mu}{y^2} + \eta \frac{\log y}{y^3} = L. \quad (20.2)$$

The phrase “sufficiently large” selects its increasing branch: $h'_*(y) \rightarrow \lambda > 0$, and $y(T) \sim L/\lambda$.

Corollary 20.1. *There is a finite constant $M > 0$ such that, for all sufficiently large T ,*

$$\left\lceil y(T) - \frac{M}{L^3} \right\rceil \leq \tau(T) \leq \left\lceil y(T) + \frac{M}{L^3} \right\rceil. \quad (20.3)$$

The root $y(T)$ may be replaced by the explicit center $x_3(T)$ in (20.5), increasing M if necessary. Neither M nor the onset of this statement is made effective.

Proof. On $n \geq y(T)/2$, the error in (20.1) is at most a constant times L^{-3} , while h'_* is eventually bounded below by $\lambda/2$. Choose M so large that a displacement M/L^3 changes h_* by strictly more than this error. The mean value theorem then gives $u_n < T$ for integers $y(T)/2 \leq n \leq y(T) - M/L^3$, and $u_n > T$ for integers $n \geq y(T) + M/L^3$. Earlier integers have $u_n < T$ as well: the equivalent gives $\log u_{\lceil y(T)/2 \rceil} - L \rightarrow -\infty$, and monotonicity applies. Taking ceilings gives (20.3), with its uncertainty inside the two ceilings. The explicit-center assertion follows from the next calculation. $\square$

For an explicit expansion define

$$\begin{aligned} s &= 4 \log L - 4 \log \lambda - \log M_0, \\ t &= 4s - \beta \lambda, \\ v &= 4t - 2s^2 + \beta \lambda s - \mu \lambda^2, \\ w &= 4v - 4st + \frac{4}{3}s^3 - \beta \lambda(s^2 - t) + 2\mu \lambda^2 s - \eta \lambda^3 (\log L - \log \lambda). \end{aligned} \quad (20.4)$$

Then set

$$x_3(T) = \frac{1}{\lambda} \left(L + s + \frac{t}{L} + \frac{v}{L^2} + \frac{w}{L^3} \right). \quad (20.5)$$

With $Z = \lambda x_3$, direct expansion gives

$$\begin{aligned} \log x_3 &= \log(L/\lambda) + s/L + (t - s^2/2)/L^2 + (v - st + s^3/3)/L^3 + O((\log L)^4/L^4), \\ \beta/x_3 &= \beta \lambda/L - \beta \lambda s/L^2 + \beta \lambda(s^2 - t)/L^3 + O((\log L)^3/L^4), \\ \mu/x_3^2 &= \mu \lambda^2/L^2 - 2\mu \lambda^2 s/L^3 + O((\log L)^2/L^4), \\ \eta \log x_3/x_3^3 &= \eta \lambda^3 (\log L - \log \lambda)/L^3 + O((\log L)^2/L^4). \end{aligned}$$

The definitions in (20.4) cancel the constant and the first three inverse powers of L in $h_*(x_3) - L$. Its derivative is bounded away from zero, so

$$y(T) - x_3(T) = O((\log L)^4/L^4) = o(L^{-3}). \quad (20.6)$$

This proves the same width in (20.3) for x_3 .

There is a useful hierarchy of what the successive count results justify. Using only the corrected model $\lambda y - 4 \log y + \log M_0 + \beta/y = L$ and the first correction gives an inside-ceiling width $O(L^{-2})$. Including μ/y^2 but omitting the known logarithmic term gives width $O((\log L)/L^3)$. The full model (20.2) gives $O(L^{-3})$. Its displayed w/L^3 is an expansion of that continuous model. The unspecified finite n^{-3} count term can still shift the actual inverse by order L^{-3} ; no exact ceiling formula is asserted.

These are asymptotic statements, not certified numerical threshold algorithms. Even a very small centering error can change a ceiling near an integer. A finite threshold requires exact counts or an effective error bound. Substitution of uncertified partial sums for R, S, S_2 does not supply such a bound.

[write] 5 October 2026. Corollary 20.1 is an instance of parts (1)–(2) of `p0:thm:staircase` (the front-matter section Relation to neighbouring reports), with $x_3(T)$ as approximate inverse and $\eta = M/L^3$. Part III's Corollary 27.1 gives the sandwich to every fixed order; it uses the identified finite coefficient r_3 that question 4 below asks for, but still no effective remainder.

21 Position relative to prior work

The general prescribed-occurrence conjecture of Noonan and Zeilberger [12, Introduction] includes P-recursiveness for exactly one classical 1234. Nakamura and Zeilberger [11] give catalytic functional equations and polynomial-time enumeration for fixed small occurrence counts, including the present sequence. Their accompanying computations and later extensions supply many terms; polynomial-time enumeration by itself does not establish a finite-order univariate polynomial-coefficient recurrence.

Bóna and Burstein [9, Theorem 6.2] already prove nonalgebraicity for exactly one monotone pattern of even length greater than two. Reversal or complement identifies their length-four decreasing case with the present one. Therefore nonalgebraicity is not a new conclusion of this report. Together with Theorem 11.1, the OGF is D-finite and nonalgebraic.

Conway and Guttmann [8, Section 5.1] study this exactly-one sequence through available coefficients and differential approximants. The approximating differential equations are not certified annihilators of the exact OGF; their introduction's known D-finiteness statement for 1234 concerns avoidance. Waite's recent work [10] extends asymptotic comparison and nonrationality or nonalgebraicity results for fixed monotone occurrence counts. It does not supply the specific P-recursiveness proof used here.

The prior-art check covered those primary works and the OEIS entry, with attention to the difference between classical and consecutive patterns, zero and one occurrences, exact equations and approximants. No earlier proof of the particular P-recursiveness assertion was located in those sources. This is a scoped assessment, not a worldwide priority claim or a comprehensive bibliography. No author correspondence or exhaustive citation-graph search was performed. The broad finite-pattern-set P-recursiveness conjecture is false by Garrabrant and Pak [18]; the present theorem depends on the explicit fixed-dimensional formula and does not revive that general conjecture.

[write] Checked, 5 October 2026. Against the arXiv texts: [9, Theorem 6.2] is stated for even $k > 2$ and proved by its Lemma 6.3; [10, Corollary 4.4] extends the order comparison to r occurrences and gives

nonrationality for odd k and nonalgebraicity for even k . Neither proves P-recursiveness. “No earlier proof located” is the manuscript’s scoped assessment; the quick search at intake found none either.

22 Reproducibility and remaining research

The accompanying archive contains this source and PDF, the unchanged Report 134 companion, reusable finite exact data, and a standard-library Python verification suite. The README specifies normal and optimized Python verification, external-output builds, inventory and SHA-256 checks, and deterministic PDF and ZIP regeneration. All numerical acceptance decisions for finite algebra and counts use exact integers or rational numbers. Verification does not write into the release bundle; PDF build products and check outputs go to explicitly separate locations.

The new checks compare the six-term binomial dimension with both hook and branching constructions; verify the indicator and CT exponent encoding; replay all six polynomial transform identities and the exact logarithmic contraction; and check finite exact count, angular-moment and boundary data. Symbolic and numerical inverse checks address the displayed continuous-model expansion. They do not compute an exact threshold using unknown infinite constants. Any reported partial sum of S or S_2 is labeled as a finite rational sum with uncertified tail. Finite checks support transcription and implementation accuracy; the infinite claims rest on the proofs in this article and its companion.

Several specific questions remain.

1. *Effective evaluation of the constants.* The positive series for R and absolutely convergent signed series for S, S_2 are exact specifications. Useful explicit tail constants could provide certified enclosures, determine the signs of the series-defined corrections, and turn the inverse bounds into an effective algorithm. No such constants are supplied here.
2. *A usable annihilating equation.* Rational diagonality proves existence. Eliminating variables from the compressed kernel or finding a smaller integral may yield an explicit differential equation or recurrence, but the 42-variable representation gives no practical complexity guarantee. Coefficient guessing alone would still require a proof and adequate initial data.
3. *The finite third coefficient and further logarithms.* The first logarithm has been proved and evaluated. The bounded remainder in (19.5), after multiplication by n^3 , is not proved to converge. Identifying its finite coefficient requires a matched endpoint and middle analysis. Higher character expansions can be constructed with stronger polynomial weights, but divergent half-size moments must be subtracted and matched rather than treated as convergent moments. No full expansion in powers and logarithms is inferred from local analyticity alone.
4. *Sharper discrete inversion.* An identified finite third count coefficient and an effective remainder could improve the centering and the rounding guarantee. The w/L^3 term of the continuous model cannot stand in for that unknown finite coefficient.

The results here concern exactly one classical 1234. They make no assertion for arbitrary patterns, arbitrary occurrence counts, or all-orders asymptotic expansions.

22.1 Further questions and research

[write] **Status of the questions above, 5 October 2026.** Question 1 (effective evaluation of R, S, S_2): open (Section 66.1, items 1 and 4). Question 2 (a usable annihilating equation): open (item 5); the 42-variable constant term of Theorem 14.1 is an input, and no elimination has been performed. Question 3 (the finite

third coefficient and further logarithms): *answered* by Part III, Theorem 23.1 (r_3 exists; $\kappa_4 = -5041845\sqrt{3}/\pi$; every $\kappa_j \in (\sqrt{3}/\pi)\mathbb{Q}$). Question 4 (sharper discrete inversion): answered in part by Part III's Corollary 27.1 (every fixed order, with r_3 and later coefficients identified); an effective remainder is open. The proof of Theorem 11.1 rests on the cited closure theorem; Remark 13.2 records that its hypotheses hold. This Part states no other claim without proof, and nothing in it was found to be wrong.

Part III

All orders logarithmic asymptotics for A217057

Bundle Report 137, the base of the merge, archive `A217057_All_Orders_Logarithmic_Asymptotics_and_Inversion_Source.zip`, delivered as `Report137.tex` (13 pages), dated 2 October 2026; title block “All orders logarithmic asymptotics for A217057”, author line “Report 137”. Printed as delivered, with **[write]** notes; a section number is Report 137’s plus 22, and equations are renumbered within sections. Section 29.1 is added in the write.

Reading conventions of Part III	Elsewhere in the report
$u_n, A_n, H_s(i, j), C, B$ as in Parts I–II	
U_j arrays in E_r ; $a_n = 9^{-n} H_n$	Parts V–VI: U_n a count; Part VI: $u_j, a_s = 16^{-s} H_s$
r the Banach radius; r_j coefficients; $r_n^{(K)}$	Part VI: r_j its coefficients
H_m harmonic numbers in (25.9), (26.4)	Parts I–II: half counts $H_s(i, j)$
$b_k(n) = [z^n](1 - z)^k \log(1 - z)$; $V_k; T_j; v_n; \Phi_m$	Part VI: $q_j(n), v_j, T_j, \mu_\ell$
κ_j logarithmic coefficients	Part II: $\kappa = \kappa_3$
$N(T), y = \log T, d, Q_j, L_j$	Parts I–II: $\tau(T), L = \log T$; Part VI: $N(X), P_j$
$\langle \cdot, \cdot \rangle_K, \mathcal{B}$	Parts V–VI: $\langle \cdot, \cdot \rangle_K$ (four binomials)

Abstract of this Part (as delivered). For the number u_n of permutations with exactly one classical occurrence of 1234, and its avoidance counterpart A_n , we prove a complete Poincaré expansion whose coefficients are affine functions of $\log n$. The first logarithm occurs at relative order three. Every logarithmic coefficient is a computable rational multiple of $\sqrt{3}/\pi$; the next one after Report 135 is $-5041845\sqrt{3}/\pi$ at relative order four. Every nonlogarithmic coefficient has an absolutely convergent regularized series prescription. In particular, the previously undetermined finite third coefficient exists. The proof uses uniform all-order character estimates in a weighted Banach space and a coefficient-level convolution lemma. No analytic continuation or singularity-analysis hypothesis for a remainder generating function is assumed. Inverse thresholds have expansions to every fixed order as well.

[write] 5 October 2026. “Report 134” and “Report 135” [1, 2] are Parts I and II of this report; the companions this Part calls unchanged are those Parts and the `134-amplitude-` and `135-rational-` files.

23 Exact inputs and the conclusion

An occurrence of the classical pattern 1234 is a choice of four positions with increasing values, without any adjacency requirement. Let u_n count permutations of $[n]$ with exactly one such choice; set $u_0 = u_1 = u_2 = u_3 = 0$. This is OEIS A217057. The corresponding avoidance count is A_n , with $A_0 = 1$.

Report 134 [1] established an exact two-half decomposition and a positive leading ratio. Report 135 [2] obtained its first two power corrections and its first logarithmic correction, with an $O(n^{-3})$ remainder in the ratio. The present report proves the finite third constant and extends the expansion to every fixed order. All new uniform estimates and convolution arguments are supplied below. The unchanged final Report 135 bundle, including its Report 134 companion, is included with the reproducibility archive for the exact combinatorial inputs. The advance claimed here is relative to those reports; an exhaustive literature-priority claim is not made.

Let $F_m(p, q)$ count 1234-avoiders with the final p entries decreasing and with the q smallest values occurring in decreasing order. Write $A_m = F_m(0, 0)$, and put

$$H_s(i, j) = F_{s+2}(i+1, j+1) - F_{s+1}(i, j) \quad (i+j \leq s), \quad H_s(i, j) = 0 \quad (i+j > s).$$

For $0 \leq p, q \leq m$, we use the following proved inputs from Report 135, whose combinatorial foundations are supplied in its Report 134 companion:

$$u_n = \sum_{s+t=n-4} \langle H_s, H_t \rangle_{\mathcal{K}}, \quad (23.1)$$

$$\langle X, Y \rangle_{\mathcal{K}} = \sum_{i,j,k,l \geq 0} X(i, j) Y(k, l) \binom{i+k}{i} \binom{j+l}{j}, \quad (23.2)$$

$$0 \leq H_s(i, j) \leq K A_s (i+1)^4 (j+1)^4 3^{-i-j}, \quad (23.3)$$

$$c \frac{9^m}{(m+1)^4} \leq A_m \leq C \frac{9^m}{(m+1)^4}, \quad (23.4)$$

$$F_m(p, q) = \int_{U(3)} P^{m-p} h_p \overline{P^{m-q} h_q} dU. \quad (23.5)$$

Here $P = \text{tr } U$, h_p is the complete homogeneous character, and Haar measure has mass one. Set

$$C_0 = \frac{81\sqrt{3}}{16\pi}, \quad d_p = \binom{p+2}{2} 3^{-p}, \quad a_p = \frac{p(p-1)}{2},$$

$$C(i, j) = 81d_{i+1}d_{j+1} - 9d_i d_j, \quad (23.6)$$

$$B(i, j) = -81d_{i+1}d_{j+1}(8 + a_{i+1} + a_{j+1}) + 9d_i d_j(4 + a_i + a_j). \quad (23.7)$$

The avoidance and half expansions of Report 135 start as

$$A_m = C_0 9^m m^{-4} \left(1 - \frac{11}{2m} + \frac{20}{m^2} + O(m^{-3}) \right), \quad \frac{H_m(i, j)}{A_m} = C(i, j) + \frac{B(i, j)}{m} + \dots$$

The positivity $R > 0$ of its ratio limit is another inherited input.

[write] **The inputs, proved in this report, 5 October 2026.** (23.1)–(23.2) are Lemma 2.1; (23.3) is Lemma 7.1; (23.4) is (7.5); (23.5) is (15.1); the displayed expansions are (19.9) and (16.6); $R > 0$ is Theorem 1.1. The condition under which Report 137’s 137-allorders-SOURCES.md calls this Part’s theorem “conditional on the explicitly inherited . . . inputs” is thus discharged inside this report, apart from the external results those Parts cite (Regev’s constant C_0 , Weyl integration and character orthogonality).

Theorem 23.1. *There are uniquely determined real numbers r_j, κ_j such that, for every fixed integer $J \geq 0$,*

$$\frac{u_n}{A_n} = \sum_{j=0}^J \frac{r_j + \kappa_j \log n}{n^j} + O_J \left(\frac{1 + \log n}{n^{J+1}} \right). \quad (23.8)$$

They satisfy

$$\begin{aligned} r_0 &= R, & r_1 &= S, & r_2 &= S_2, & \kappa_0 &= \kappa_1 = \kappa_2 = 0, \\ \kappa_3 &= \frac{43020\sqrt{3}}{\pi}, & \kappa_4 &= -\frac{5041845\sqrt{3}}{\pi}. \end{aligned} \quad (23.9)$$

Every κ_j belongs to $(\sqrt{3}/\pi)\mathbb{Q}$ and is computable by finite rational operations. Each r_j is specified by finitely many absolutely convergent Banach-valued regularized series followed by continuous pairings and finite algebra with universal harmonic constants. In particular,

$$\lim_{n \rightarrow \infty} \left[n^3 \left(\frac{u_n}{A_n} - R - \frac{S}{n} - \frac{S_2}{n^2} \right) - \kappa_3 \log n \right] = r_3$$

exists; an explicit convergent-series recipe is given in Section 26.

This is an all-orders asymptotic statement at each fixed order. It makes no assertion of convergence or summability of the infinite asymptotic series, or of an expansion into exponentially small sectors. It supplies neither certified decimal values for the regularized constants nor an effective numerical onset.

[write] **5 October 2026.** r_0, r_1, r_2 are Part I's R and Part II's S, S_2 , by uniqueness (Section 26), and κ_3 is Part II's κ . The theorem answers Part II's question 3 and, in part, its question 4.

24 All orders estimates in a weighted Banach space

Fix $1/3 < r < 1/2$, and let E_r be the space of real arrays with norm

$$\|X\|_r = \sup_{i,j \geq 0} |X(i,j)| r^{-i-j}.$$

It is a Banach space, by its isometric identification with bounded real arrays after multiplying coordinate (i,j) by r^{-i-j} . The pairing in (23.2) is symmetric and continuous, since

$$|\langle X, Y \rangle_{\mathcal{K}}| \leq (1 - 2r)^{-2} \|X\|_r \|Y\|_r. \quad (24.1)$$

Indeed $\sum_{i,k \geq 0} r^{i+k} \binom{i+k}{i} = (1 - 2r)^{-1}$. Put $a_n = 9^{-n} H_n \in E_r$. The global estimates imply $\|a_n\|_r = O((n+1)^{-4})$, since a fixed polynomial times $(3r)^{-i-j}$ is bounded. We reserve A_m for avoidance counts and use U_j for the Banach coefficient arrays below.

Proposition 24.1. *There are arrays $U_j \in E_r$, independent of r , such that for every fixed $M \geq 0$,*

$$\left\| a_n - n^{-4} \sum_{j=0}^M U_j n^{-j} \right\|_r = O_{M,r}(n^{-M-5}). \quad (24.2)$$

They have the form $U_j(i,k) = C_0 3^{-i-k} P_j(i,k)$ for a rational polynomial P_j . The first two are

$$U_0 = C_0 C, \quad U_1 = C_0 \left(-\frac{11}{2} C + B \right). \quad (24.3)$$

The avoidance sequence has a complete pure-power expansion

$$A_m = C_0 9^m m^{-4} \left(\sum_{j=0}^M \alpha_j m^{-j} + O_M(m^{-M-1}) \right), \quad \alpha_j \in \mathbb{Q}, \quad (24.4)$$

where $\alpha_0 = 1$, $\alpha_1 = -11/2$, $\alpha_2 = 20$, and $\alpha_3 = -965/16$.

24.1 The fixed order Laplace estimates

Common central phase cancels in (23.5). On the quotient of the eigenangle torus by common phase, $|P| = 3$ has a single maximum. Use a small centered chart $\theta_1 + \theta_2 + \theta_3 = 0$, and put $Q = \sum \theta_i^2$ and $\Delta = \prod_{i < j} (\theta_i - \theta_j)$. The phase and factored Weyl density are real analytic there:

$$\phi(\theta) = \log(|P|^2/9) = -Q/3 + O(Q^2), \quad \prod_{i < j} |e^{i\theta_i} - e^{i\theta_j}|^2 = \Delta(\theta)^2 w(\theta), \quad w(0) = 1.$$

Both ϕ and w are even under $\theta \mapsto -\theta$. In a sufficiently small chart $\phi \leq -cQ$; outside it the trace ratio is at most some $\rho < 1$. The quotient has a constant local Jacobian. Rescaling $\theta = y/\sqrt{m}$ gives leading density $e^{-Qy/3} \Delta(y)^2 dy$ and a leading factor m^{-4} .

Here is a fixed-order remainder argument. Write $\phi = -Q/3 + v$, where $v = O(Q^2)$, and fix an integer $M \geq 0$. Choose the chart small enough that $|v| \leq Q/12$. Truncate v through degree $2M + 2$, leaving a remainder $O(Q^{M+2})$, and truncate w through degree $2M$, leaving $O(Q^{M+1})$. For $M = 0$ the first truncation is the empty polynomial. Changing the phase to that polynomial creates an error bounded by $CmQ^{M+2}e^{-c'mQ}$, by the mean value formula for the exponential; its integral is of relative order m^{-M-1} .

For the polynomial phase perturbation $v_{\leq 2M+2}$, Taylor's integral formula bounds the remainder after its exponential is expanded through power M by

$$C_M m^{M+1} Q^{2M+2} e^{m|v_{\leq 2M+2}|}.$$

A smaller chart absorbs this exponential into $e^{-mQ/3}$, and its integral has relative order m^{-M-1} . Expand the retained finite polynomial products and retain exactly the monomials of relative weight at most M , where m^a times a homogeneous polynomial of degree $2b$ has weight $b - a$. Each discarded term has weight at least $M + 1$. In fact, direct radial scaling on the two-dimensional plane gives

$$\int Q^b \Delta^2 e^{-c'mQ} d\theta = O(m^{-4-b});$$

thus $m^a Q^b$ has the claimed relative order. The amplitude remainder is handled by the same bound. Extending the resulting Gaussian polynomial integrals to the whole plane changes them by an exponentially small quantity. Odd terms integrate to zero on an inversion-symmetric chart. This proves the fixed-order expansion with an actual remainder, rather than only a formal series.

The argument also applies with an arbitrary fixed polynomial insertion. For a homogeneous insertion of degree $2h$, scaling first extracts m^{-h} , and the proof gives any requested number of further integer powers. Linearity and finite-dimensional norm equivalence make the remainder constant uniform on a coefficient-norm unit ball of any fixed polynomial space. The complementary region is exponentially small, using the original integrand there. The same argument gives, for fixed c_1, h and sufficiently small $\varepsilon > 0$,

$$\mathbb{E}_m[Q^h e^{c_1 L Q}; \text{local}] \leq C_h m^{-h} \quad (0 \leq L \leq \varepsilon m), \quad (24.5)$$

where $\mathbb{E}_m$ is normalized trace-moment expectation. These arguments prove (24.4) and all polynomial-insertion moment expansions, uniformly on the unit ball of each fixed finite-dimensional polynomial space.

The normalized Gaussian moments are rational. For example, write $\theta = (x, y, -x - y)$: the Gaussian $e^{-Q/3}$ has covariance $\begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}$, and the additional weight is the rational polynomial Δ^2 . The Gaussian moment recurrence of Isserlis [24], often called Wick's rule, then gives rational normalized moments. All Taylor coefficients above are rational, proving $\alpha_j \in \mathbb{Q}$. The constant C_0 is the already established leading constant.

24.2 Uniformity in both boundary parameters

Put $c_p = \binom{p+2}{2}$ and

$$r_p(\theta) = \frac{h_p(e^{i\theta})}{c_p(P(\theta)/3)^p}, \quad h_p/P^p = d_p r_p.$$

For a centered unit vector v , set $g_p(t) = h_p(e^{itv})/c_p$ and $z(t) = P(tv)/3$. For every fixed derivative order D ,

$$|g_p^{(h)}(t)| \leq (p+1)^h \quad (0 \leq h \leq D), \quad \left| \frac{d^h}{dt^h} z(t)^{-p} \right| \leq C_D (p+1)^h e^{C_D p t^2}$$

on a fixed sufficiently small interval, uniformly in v . The first bound follows by averaging the c_p weak-composition terms; the second is the finite chain rule, using $|z(t)| \geq e^{-C t^2}$. Consequently, for $L = p + q$ and any fixed M , the real even function $\Re(r_p \bar{r}_q)$ has a homogeneous Taylor expansion

$$\Re(r_p \bar{r}_q) = \sum_{h=0}^M D_{2h}^{p,q}(\theta) + \mathcal{R}_{M,p,q}(\theta), \quad (24.6)$$

with

$$|D_{2h}^{p,q}(\theta)| \leq C_M (1+L)^{2h} Q^h, \quad |\mathcal{R}_{M,p,q}(\theta)| \leq C_M (1+L)^{2M+2} Q^{M+1} e^{C_M L Q}.$$

The real product is even because the characters conjugate when the angles are negated. Taylor's theorem through degree $2M+1$ proves the remainder bound; its last odd homogeneous coefficient vanishes.

Integrate the remainder using (24.5). Expand the expectation of $D_{2h}^{p,q}$ through relative order $M-h$ by the polynomial-insertion argument above. Uniformity on that finite-dimensional space, and the displayed polynomial bound for its coefficients, show that for some $\varepsilon_M > 0$ and uniformly when $p+q \leq \varepsilon_M m$,

$$\frac{F_m(p,q)}{A_m} = d_p d_q \left[\sum_{j=0}^M f_j(p,q) m^{-j} + O_M \left((1+p+q)^{2M+2} m^{-M-1} \right) \right]. \quad (24.7)$$

The f_j are rational polynomials of degree at most $2j$, with $f_0 = 1$. To see polynomiality directly, weak compositions satisfy

$$\frac{1}{c_p} \sum_{\delta_1 + \delta_2 + \delta_3 = p} (\delta_1)_a (\delta_2)_b (\delta_3)_c = \frac{2a!b!c!}{(a+b+c+2)!} p^{\frac{a+b+c}{2}}.$$

All Taylor coefficients of the averaged character, and of z^{-p} , are therefore rational polynomials in the parameters.

There is no division by P outside the chart. The original polynomial integrand in (23.5) has modulus bounded by

$$9^m d_p d_q \left(\frac{|P|}{3} \right)^{2m-p-q}.$$

After division by A_m its off-chart part is $O(d_p d_q m^4 \rho^{(2-\varepsilon_M)m})$, absorbed in the remainder. Thus (24.7) is uniform on its entire stated range.

Combining (24.7), (24.4) and the fixed shifts $m = n+1, n+2$ yields the claimed half coefficients and, whenever $i+j+2 \leq \varepsilon'_M n$, a bound

$$\left| a_n(i,j) - n^{-4} \sum_{h=0}^M U_h(i,j) n^{-h} \right| \leq C_M n^{-M-5} (1+i+j)^{D_M} 3^{-i-j}$$

for a fixed finite D_M . Dividing by r^{i+j} makes the boundary polynomial uniformly bounded. In the complementary range, (23.3) and the polynomial coefficient bounds give an exponentially small norm tail, since $(3r)^{-i-j}$ decays geometrically and $i+j \geq \varepsilon'_M n - 2$. This includes the triangular zero region $i+j > n$. It proves (24.2) and completes the uniform estimates.

24.3 A finite algorithm for every local coefficient

The construction above can be carried out with rational polynomials at each fixed order. Let ϕ_{2h} and w_{2h} be the homogeneous parts of the local phase and amplitude, where $\phi_2 = -Q/3$ and $w_0 = 1$. With t a formal variable, define

$$K_j(y) = [t^j] \left\{ \left(\sum_{h=0}^j t^h w_{2h}(y) \right) \exp \left(\sum_{h=2}^{j+1} t^{h-1} \phi_{2h}(y) \right) \right\}. \quad (24.8)$$

Only finitely many terms of the exponential can contribute. If $\mathbb{E}_0$ is expectation for the normalized density $e^{-Q/3} \Delta^2$, then $\alpha_j = \mathbb{E}_0 K_j$. The normalization agrees with $\alpha_0 = 1$; the leading factor C_0 is the inherited one.

For the boundary insertion in (24.6), put

$$N_j(p, q) = \sum_{h=0}^j \mathbb{E}_0 [D_{2h}^{p,q} K_{j-h}].$$

The finite division rule

$$f_0 = 1, \quad f_j = N_j - \sum_{h=1}^j \alpha_h f_{j-h} \quad (24.9)$$

gives the coefficients in (24.7). The averaged character Taylor coefficients are obtained from the displayed weak-composition factorial moments, converting ordinary powers to falling powers if necessary. The reciprocal trace is expanded by its finite binomial series. Thus this is an explicit rational algorithm for the polynomials f_j , including f_2 and f_3 , without needing a separate closed-form third-boundary formula.

For clarity, it also gives the half arrays directly. Write $g_j(p, q) = \sum_{h=0}^j \alpha_h f_{j-h}(p, q)$, and define, for a fixed integer shift s ,

$$E_j(s; p, q) = \sum_{h=0}^j g_h(p, q) \binom{-4-h}{j-h} s^{j-h}.$$

Expanding $(n+s)^{-4-h}$ proves

$$U_j(i, k) = C_0 [81d_{i+1}d_{k+1}E_j(2; i+1, k+1) - 9d_i d_k E_j(1; i, k)]. \quad (24.10)$$

These formulas use only finitely many exact Gaussian moments at any fixed j . One elementary implementation, in coordinates $(x, y, -x-y)$, sets $G(a, b) = \mathbb{E}_G(x^a y^b)$ and uses

$$G(a, b) = (a-1)G(a-2, b) - \frac{b}{2}G(a-1, b-1) \quad (a \geq 1),$$

with $G(0, 2b) = (2b-1)!!$, $G(0, 2b+1) = 0$, $G(0, 0) = 1$, and negative indices interpreted as zero. This recurrence follows directly by Gaussian integration by parts. Multiplying by Δ^2 and dividing by $\mathbb{E}_G \Delta^2$ then evaluates $\mathbb{E}_0$.

24.4 A reproducible check of the third avoidance coefficient

Let $D_1 = x - y$, $D_2 = x + 2y$, $D_3 = 2x + y$, and let L_{2h} be the coefficient of t^h in

$$\log \left(1 + \frac{2}{9} \sum_{h \geq 1} \frac{(-1)^h t^h}{(2h)!} \sum_{i=1}^3 D_i^{2h} \right).$$

Let W_{2h} be the coefficient of t^h in

$$\prod_{i=1}^3 \left(\frac{\sin(\sqrt{t} D_i / 2)}{\sqrt{t} D_i / 2} \right)^2.$$

Expectation below is with Gaussian covariance just specified and extra weight $\Delta^2 = D_1^2 D_2^2 D_3^2$. Its normalizing Gaussian moment is $\mathbb{E}_G \Delta^2 = 81/2$. Then

$$\alpha_3 = \mathbb{E}_0[W_6 + W_4 L_4 + W_2(L_6 + L_4^2/2) + L_8 + L_4 L_6 + L_4^3/6] = -965/16.$$

These are finite polynomial calculations. The companion replay evaluates the Gaussian moments exactly over $\mathbb{Q}$, reproducing $1, -11/2, 20, -965/16$.

25 A coefficient level convolution theorem

The following elementary argument works for any real Banach space E with a continuous symmetric bilinear form $\mathcal{B} : E \times E \rightarrow \mathbb{R}$. Assume a sequence $a_n \in E$ has (24.2), with the same leading power n^{-4} . Let

$$v_n = \sum_{t=0}^n \mathcal{B}(a_t, a_{n-t}).$$

For $k \geq 3$ define the scalar model sequence

$$b_k(n) = [z^n](1-z)^k \log(1-z). \quad (25.1)$$

For $n > k$ the coefficient is exactly

$$b_k(n) = \frac{(-1)^{k+1} k!}{n^{k+1}}. \quad (25.2)$$

For all $n \geq 0$ the coefficient is also given by the finite sum

$$b_k(n) = - \sum_{h=0}^{\min(k, n-1)} \frac{(-1)^h \binom{k}{h}}{n-h}, \quad b_k(0) = 0,$$

with the sum used when $n \geq 1$. For example, $(b_3(0), b_3(1), b_3(2), b_3(3)) = (0, -1, 5/2, -11/6)$. The finitely many coefficients at $n \leq k$ retain their definition in (25.1); replacing them by another extension would change the regularized constants and is not permitted.

Lemma 25.1. *There are unique vectors $V_k \in E$, $k \geq 3$, such that for every $K \geq 3$,*

$$r_n^{(K)} = a_n - \sum_{k=3}^K V_k b_k(n) = O_E((n+1)^{-K-2}). \quad (25.3)$$

If $h_d(1, \dots, k)$ denotes the complete homogeneous polynomial of degree d in $1, \dots, k$, they are computed by

$$V_k = \frac{U_{k-3} - \sum_{\ell=3}^{k-1} (-1)^{\ell+1} \ell! h_{k-\ell}(1, \dots, \ell) V_\ell}{(-1)^{k+1} k!}. \quad (25.4)$$

Proof. Expand the rational function (25.2) as $(-1)^{k+1} k! n^{-k-1} \prod_{j=1}^k (1 - j/n)^{-1}$ to any fixed order. Its leading coefficient is nonzero. Triangular matching against the assumed Banach expansion gives (25.4) and (25.3). $\square$

For $j \geq 0$ define a regularized Taylor coefficient by

$$T_j = \frac{(-1)^j}{j!} \sum_{n \geq 0} n^j \left(a_n - \sum_{k=3}^j V_k b_k(n) \right). \quad (25.5)$$

An empty inner sum means zero. These series converge absolutely in E : for $j \leq 2$, use $a_n = O_E(n^{-4})$; for $j \geq 3$, apply (25.3) with $K = j$. In the latter case the weighted summand is $O_E(n^{-2})$.

Lemma 25.2. *For a fixed $K \geq 3$ set*

$$S_K(z) = \sum_{k=3}^K V_k (1-z)^k \log(1-z), \quad P_K(z) = \sum_{j=0}^{K-1} T_j (1-z)^j.$$

Then, as $n \rightarrow \infty$,

$$v_n = [z^n] (2\mathcal{B}(P_K, S_K) + \mathcal{B}(S_K, S_K)) + O(n^{-K-2}). \quad (25.6)$$

Proof. Let $R_K(z) = \sum r_n^{(K)} z^n$. Its derivatives through order K at $z = 1$ are absolutely convergent coefficient sums. For $k > j$, the j th derivative of $(1-z)^k \log(1-z)$ tends to zero at $z = 1$ and its coefficient derivative series is absolutely convergent. Therefore the Taylor polynomial of R_K through degree $K-1$, expressed in powers of $1-z$, is exactly P_K from (25.5). Let $q_n = [z^n](R_K - P_K)$. Then

$$\|q_n\| = O((n+1)^{-K-2}), \quad \sum_{n \geq 0} n^j q_n = 0 \quad (0 \leq j \leq K-1), \quad \sum_{n \geq 0} n^K \|q_n\| < \infty. \quad (25.7)$$

The zero falling-factorial moments also imply zero ordinary moments through $K-1$.

All generating functions converge absolutely for $|z| < 1$, so their products equal coefficient convolutions. Terms involving P_K twice are polynomial and vanish for sufficiently large n . Convolution of q with P_K is $O(n^{-K-2})$. Convolution of q with itself has the same bound: split at $n/2$, bound the larger index by $O(n^{-K-2})$, and sum the other sequence absolutely.

It remains to bound $q * b_k$, for fixed $3 \leq k \leq K$. In the part $t > n/2$, $\|q_t\| = O(n^{-K-2})$ and $\sum_s |b_k(s)| < \infty$. In the part $t \leq n/2$, use the rational extension in (25.2): its h th derivative on $[n/2, n]$ is $O(n^{-k-1-h})$ for all large n . Taylor expansion of $b_k(n-t)$ through degree $K-1$ has remainder $O(t^K n^{-k-1-K})$. Its convolution with q is bounded using the last condition in (25.7). Each retained moment, after extending the sum to all t , vanishes. The truncation tail for its degree h is

$$O(n^{-k-1-h}) \sum_{t > n/2} t^h \|q_t\| = O(n^{-k-K-2}).$$

Thus the entire convolution is $O(n^{-K-2})$. Continuity of $\mathcal{B}$ handles its vector coefficients. Expansion of $\mathcal{B}(P_K + S_K + (R_K - P_K), P_K + S_K + (R_K - P_K))$ proves the lemma. No continuation of R_K across $|z| = 1$ has been used. $\square$

Theorem 25.3. *The convolution has a complete expansion*

$$v_n = \sum_{j=0}^J (c_j + \ell_j \log n) n^{-4-j} + O_J(n^{-5-J}(1 + \log n)) \quad (25.8)$$

for every fixed J , with $\ell_0 = \ell_1 = \ell_2 = 0$. Every coefficient is obtained by expanding the finite expression in (25.6), with any $K \geq J + 4$. In particular, each ℓ_j depends only on the local tail vectors U_h , never on the regularized moments T_h .

Proof. Products $P_K S_K$ are finite sums of the single-logarithm models in (25.1). Products S_K^2 are finite sums of $(1 - z)^m \log^2(1 - z)$ with $m \geq 6$. For $n > m$, differentiation twice in α of the finite polynomial $[z^n](1 - z)^\alpha = (-1)^n \alpha^n / n!$ gives the exact formula

$$[z^n](1 - z)^m \log^2(1 - z) = 2b_m(n)(H_m - H_{n-m-1}). \quad (25.9)$$

Here $H_m = \sum_{h=1}^m 1/h$ and $H_0 = 0$; these scalar harmonic numbers are distinct from the earlier half arrays $H_s(i, j)$. For a fixed integer $p \geq 0$, Euler–Maclaurin gives

$$H_N = \log N + \gamma + \frac{1}{2N} - \sum_{h=1}^p \frac{B_{2h}}{2hN^{2h}} + O_p(N^{-2p-2}),$$

where B_{2h} are Bernoulli numbers. The finite remainder follows from the integral remainder in Euler–Maclaurin applied to $1/x$; the periodic Bernoulli function is bounded and the appropriate derivatives are integrable on $[N, \infty)$. Equivalently, see the exact harmonic–digamma identity and its fixed-order expansion in DLMF [25, 26]. Expanding $N = n - m - 1$ with fixed m and using (25.2) gives full expansions involving only a constant and a single $\log n$. The earliest log-squared model has $m = 6$, so its coefficient logarithm first occurs at n^{-7} . No higher logarithmic powers occur. Taking $K \geq J + 4$ makes the error of (25.6) smaller than the error in (25.8). $\square$

[write] **5 October 2026.** Part VI’s Theorem 61.1 is the counterpart of Theorem 25.3 for the half-integer exponent $15/2$, with fractional-binomial models whose products are polynomials, so that no logarithm arises there. Remark 66.1 compares the two.

26 The constants and the first two logarithms

Apply Theorem 25.3 to E_r , $\mathcal{B} = \langle \cdot, \cdot \rangle_K$ and $a_n = 9^{-n} H_n$. By (23.1),

$$u_n = 9^{n-4} v_{n-4}. \quad (26.1)$$

The fixed shift, the pure-power avoidance expansion, and ordinary formal division yield (23.8), with its claimed error. The coefficients are unique: at the first differing order, multiplying the difference by that power of n would give a nonzero affine function of $\log n$ tending to zero. Thus the first three constants agree with R, S, S_2 from Report 135.

Here is an explicit recipe for the finite third coefficient. The first four singular vectors are

$$\begin{aligned} V_3 &= U_0/6, & V_4 &= (6U_0 - U_1)/24, \\ V_5 &= (35U_0 - 10U_1 + U_2)/120, & V_6 &= (225U_0 - 85U_1 + 15U_2 - U_3)/720. \end{aligned} \quad (26.2)$$

Compute U_2, U_3 by the finite character/Gaussian algorithm in Section 24.3. Form the absolutely convergent series

$$\begin{aligned} T_0 &= \sum_{n \geq 0} a_n, & T_1 &= - \sum_{n \geq 0} n a_n, \\ T_2 &= \frac{1}{2} \sum_{n \geq 0} n(n-1) a_n, & T_3 &= -\frac{1}{6} \sum_{n \geq 0} n(n-1)(n-2)(a_n - V_3 b_3(n)). \end{aligned} \quad (26.3)$$

For $m = 3, 4, 5, 6$, put

$$\Phi_m = 2 \sum_{k=3}^m \langle T_{m-k}, V_k \rangle_{\mathcal{K}}.$$

The finite coefficients of v_n through the first log are

$$\begin{aligned} c_0 &= 6\Phi_3, \\ c_1 &= 36\Phi_3 - 24\Phi_4, \\ c_2 &= 150\Phi_3 - 240\Phi_4 + 120\Phi_5, \\ \ell_3 &= 1440\langle V_3, V_3 \rangle_{\mathcal{K}} = 40\langle U_0, U_0 \rangle_{\mathcal{K}}, \\ c_3 &= 540\Phi_3 - 1560\Phi_4 + 1800\Phi_5 - 720\Phi_6 + \ell_3(\gamma - H_6), \end{aligned} \quad (26.4)$$

where γ is Euler's constant. These formulas use the finite coefficients of b_k at small n exactly as in (25.1). The inverse avoidance series starts

$$(1 - 11/(2n) + 20/n^2 - 965/(16n^3) + \cdots)^{-1} = 1 + \frac{11}{2n} + \frac{41}{4n^2} + \frac{107}{16n^3} + \cdots.$$

Expanding the shift $n - 4$ in (26.1) now gives

$$r_3 = \frac{9^{-4}}{C_0} \left(c_3 + \frac{59}{2}c_2 + \frac{1441}{4}c_1 + \frac{37291}{16}c_0 \right). \quad (26.5)$$

Equations (26.3)–(26.5), together with finite local polynomial calculations, are the promised absolutely convergent regularized-series prescription. Existence and convergence, not a numerically certified evaluation, are claimed here.

The next logarithmic coefficient of the unshifted convolution is

$$\ell_4 = 30240\langle V_3, V_3 \rangle_{\mathcal{K}} - 20160\langle V_3, V_4 \rangle_{\mathcal{K}} = 140\langle U_0, U_1 \rangle_{\mathcal{K}}.$$

The shift and division therefore give

$$\begin{aligned} \kappa_3 &= \frac{9^{-4}}{C_0} \ell_3, \\ \kappa_4 &= \frac{9^{-4}}{C_0} \left(\ell_4 + \frac{67}{2} \ell_3 \right) = C_0 9^{-4} (570\langle C, C \rangle_{\mathcal{K}} + 140\langle C, B \rangle_{\mathcal{K}}). \end{aligned} \quad (26.6)$$

For transparency, both needed pairings are exact rational calculations:

$$\langle C, C \rangle_{\mathcal{K}} = 1393848, \quad \langle C, B \rangle_{\mathcal{K}} = -52348032. \quad (26.7)$$

A convenient general evaluation rule is this: if $f_i = 3^{-i}f(i)$ and $g_i = 3^{-i}g(i)$ for rational polynomials, write $f(i) = \sum_p f_p i^p$ and $g(i) = \sum_q g_q i^q$. Then

$$\sum_{i,k \geq 0} f_i g_k \binom{i+k}{i} = 3 \sum_{p,q} f_p g_q (p+q)!. \quad (26.8)$$

It follows by differentiating $(1 - x - y)^{-1}$ and putting $x = y = 1/3$. Absolute convergence licenses every differentiation. Rank-one factorizations of C, B reduce (26.7) to this rule; the companion replay carries out the rational arithmetic. Substitution proves (23.9).

More generally, every V_k is C_0 times an array of the form 3^{-i-j} times a rational polynomial. Thus all pairings of singular vectors are C_0^2 times rationals by (26.8). The log coefficient of (25.9) is the rational function $-2b_m(n)$. Shift and division use rational coefficients. Consequently every κ_j is a rational multiple of C_0 , proving the final algebraic assertion of Theorem 23.1.

27 All orders inverse thresholds

Let $M_0 = C_0 R > 0$, $\lambda = \log 9$, and

$$N(T) = \min\{n \geq 0 : u_n \geq T\}.$$

The ratio $u_{n+1}/u_n \rightarrow 9$ makes the sequence eventually strictly increasing. For sufficiently large thresholds, this controls the inverse irrespective of finitely many early values.

The direct expansion implies, to every fixed order,

$$\log u_n = \lambda n - 4 \log n + \log M_0 + \sum_{j=1}^J \frac{L_j(\log n)}{n^j} + O_J \left(\frac{(1 + \log n)^{J+1}}{n^{J+1}} \right), \quad (27.1)$$

where L_j is a polynomial with $\deg L_j \leq \lfloor j/3 \rfloor$. Indeed every logarithm in the normalized direct expansion costs at least three powers of $1/n$; expand the ordinary scalar logarithm by its finite Taylor formula. This degree bound concerns $\log u_n$, where higher powers of logarithms may occur; it does not contradict the affine-log structure of u_n/A_n .

Corollary 27.1. *For every fixed $J \geq 0$ there are explicitly recursive polynomials Q_j , $1 \leq j \leq J$, of degree at most j , such that, with $y = \log T$ and $d = \log M_0 + 4 \log \lambda$,*

$$x_J(T) = \frac{1}{\lambda} \left[y + 4 \log y - d + \sum_{j=1}^J \frac{Q_j(\log y)}{y^j} \right]$$

satisfies the eventual sandwich

$$\left[x_J(T) - C_J \frac{(1 + \log y)^{J+1}}{y^{J+1}} \right] \leq N(T) \leq \left[x_J(T) + C_J \frac{(1 + \log y)^{J+1}}{y^{J+1}} \right]. \quad (27.2)$$

The constants and onset are not asserted effective. No convention of rounding a truncated asymptotic expansion to a single exact integer is implied when its error interval crosses an integer.

Proof. Set $w = \lambda x$ and formally substitute $w = y + 4 \log y - d + \sum_{j=1}^J Q_j(\log y) y^{-j}$ in (27.1). At each new order y^{-j} the new polynomial Q_j appears with coefficient one; all other terms have already been computed. This determines Q_j successively. Expansion of $\log(1 + (w - y)/y)$ and $(1 + (w - y)/y)^{-h}$ shows inductively that its degree is at most j . Finite Taylor remainders on $(w - y)/y = o(1)$ give residual $O_J((1 + \log y)^{J+1} y^{-J-1})$.

For a rigorous threshold conclusion, regard the right side of the finite expansion (27.1) as a smooth function of a positive real variable x . Its derivative is $\lambda + o(1)$, bounded below by $\lambda/2$ for all large x . The discrete remainder and the substitution residual have the stated bound uniformly for integers within a bounded distance of x_J . Increasing the constant C_J makes the lower and upper shifts dominate both errors. For full precision about rounding, put $\epsilon = C_J(1 + \log y)^{J+1} y^{-J-1}$. The lower comparison is at $k_- = \lceil x_J - \epsilon \rceil - 1 < x_J - \epsilon$, and the upper one is at $k_+ = \lceil x_J + \epsilon \rceil \geq x_J + \epsilon$. The derivative bound and the chosen error margin give $\log u_{k_-} < \log T$ and $\log u_{k_+} \geq \log T$. All these integers lie within a bounded distance of x_J . Eventual monotonicity, with the finite early maximum below T , gives $k_- < N(T) \leq k_+$, exactly (27.2). $\square$

For example, write $u_n = M_0 9^n n^{-4} (1 + \beta/n + \delta/n^2 + \dots)$, where

$$\beta = -\frac{11}{2} + \frac{S}{R}, \quad \delta = 20 - \frac{11S}{2R} + \frac{S_2}{R}.$$

If $a = 4z - d$, the first two inverse polynomials are

$$Q_1(z) = 4a - \lambda\beta, \quad Q_2(z) = 4Q_1(z) - 2a^2 + \lambda\beta a - \lambda^2(\delta - \beta^2/2).$$

At higher orders the same recursion includes the logarithmic coefficients and the regularized constants from the direct expansion.

[write] **5 October 2026.** Corollary 27.1 is an instance of parts (1)–(2) of `p0:thm:staircase` (the front-matter section Relation to neighbouring reports), with $x_J(T)$ as approximate inverse; the recursion for the Q_j is a finite reversion. It contains Corollaries 9.1 and 20.1.

28 Reproducibility and the scope of the computations

The companion archive contains this article in editable L^AT_EX and PDF, the exact arithmetic replay, typed reference fixtures, integrity checks, and documented clean-build and deterministic-pack commands. It also includes the unchanged final Report 135 companion and its Report 134 materials. The latter retain their own verifiers and seals; the new replay checks their byte integrity before relying on them. No third-party paper PDFs are redistributed.

The new finite checks cover the single-log coefficient identities and squared-log coefficient products, triangular singular-vector matching, the Gaussian coefficients through α_3 , the two exact kernel pairings and κ_4 , the finite r_3 shift weights, and the first inverse-polynomial cancellations. Additional regularization checks use explicit synthetic scalar sequences with exactly known model tails. They check the implementation of subtraction and convergent weighted sums. They are not numerical evaluations or error certificates for the actual A217057 constants.

The verifier rejects undeclared or ill-typed fixture data and checks the sealed source inventory. Normal and optimized Python runs are required to produce the same result; the tests do not rely on removable `assert` statements. Separate output locations and explicit path guards protect the frozen package. Fresh clean builds and repacking are checked for deterministic output. These finite computations support the displayed algebra and reproducibility. The uniform estimates, infinite-series convergence, and all-orders claims are established by the proofs in this article, not by testing finitely many coefficients.

29 Further questions

The all-orders statement, existence of the finite third constant, convergent regularization algorithm, next exact logarithmic coefficient, and arbitrary fixed-order inverse sandwich are proved here from the exact inputs of Report 135. The following questions remain separate.

1. Can the regularized series be evaluated with explicit tail bounds to certify useful decimal enclosures for R, S, S_2, r_3 and later nonlogarithmic constants? Absolute convergence alone supplies no numerical certificate at a stated truncation.
2. Can the local and off-chart estimates be made effective enough to give practical constants and onset thresholds in the direct and inverse remainders? A shrinking asymptotic error interval still does not resolve exact integer rounding if it crosses an integer.
3. What is the growth of the coefficients as their order increases? Convergence, Gevrey bounds, or summability of the infinite formal series would require additional analysis. No exponentially small sector expansion is established here.

4. Can the exact rational-diagonal description from Report 135 be turned into a usable differential equation or recurrence that supports certified large-index computation of the regularized constants?
5. Which other exact-pattern decompositions admit a comparable weighted Banach expansion and a bilinear convolution reduction? The coefficient theorem applies whenever its explicit hypotheses hold; verifying those hypotheses for another pattern is a new problem.

29.1 Further questions and research

[write] **Status of the questions above, 5 October 2026.** Questions 1–4: open (Section 66.1, items 1, 4, 6 and 5). Question 5: advanced. Part VI verifies the hypotheses of the analogous convolution theorem for one 12345 (Theorem 58.1), with a half-integer exponent and the opposite outcome (Remark 66.1); Part IV’s spine convolution for one 1243 has not been brought into this form (item 7). Report 137’s `137-allorders-SOURCES.md` says that the underlying proof source was “frozen and independently reviewed” (SHA-256 0070de0a...); that source and its review were not delivered, so the repository cannot check them, and nothing here rests on them. This Part states no claim without proof, and nothing in it was found to be wrong.

Part IV

Permutations with exactly one classical 1243 occurrence

Bundle Report 136, archive A224179_Leading_Amplitude_Diffusive_Cancellation_and_Inversion_Source.zip, delivered as Report136.tex (24 pages), dated 2 October 2026; title block “Permutations with exactly one classical 1243 occurrence”, author line “Report 136”. Printed as delivered, with [write] notes, except that its Proposition 6 and Lemma 7, which are Report 134’s Proposition 4 and Lemma 5 word for word, are printed once, in Part I (Sections 33 and 34 keep their titles and point there), and its Appendix A is printed as Section 43. Report 136 numbers its statements consecutively; here they are numbered within sections, with section numbers Report 136’s plus 29. Section 42.1 is added in the write.

Reading conventions of Part IV	Elsewhere in the report
b_n : exactly one 1243 (A224179); $B(z) = \sum b_n z^n$ $R = \lim b_n/A_n$, written R_{1243} by the write $H_h(i, j)$, d_p , $C(i, j)$ as in Part I; $B_p = \binom{p+2}{2}$ $J_k(s)$ spine half sums; γ_k ; e_t ; $w(t, k)$ d_i allowed counts and a_i choices (Section 31) $M(\sigma)$ eligible rows; $M_r(\mu)$; $M_r(t)$ moments D : the constant (41.1), $D(x)$, and $ p - q $ $\kappa = \log 9$, $C = C_0 R$, $N(x)$, $L = \log x$, $g(x)$ A, B, C integration variables in Section 43	Parts I–III: u_n ; Parts V–VI: U_n Parts I–III: $R = R_{1234}$ Part I: c_p ; Parts II–III: $B(i, j)$ Part II: transforms J_k ; Part I: its own e_t d_p from Section 36 on Part II: $\kappa = \kappa_3$

Abstract of this Part (as delivered). Let b_n count permutations with exactly one classical 1243 occurrence, and let A_n count 1234-avoiders. An exact marked-avoider bijection and a shared decreasing spine give finite three-row tableau formulas for b_n . A uniform Gaussian estimate for mismatched horizontal strips makes the spine sum summable and proves $b_n \sim RA_n \sim (81\sqrt{3}/(16\pi))R9^n n^{-4}$, where R is an explicit convergent positive series. An everywhere-defined 23-index binomial encoding proves that the generating function is a rational diagonal and hence D-finite; its coefficient scale implies nonalgebraicity. A diffusive spine profile gives an exact endpoint-tail equivalent and proves cancellation of the square-root correction, improving the relative error to $o(n^{-1/2})$ and the inside-ceiling inverse uncertainty to $o((\log x)^{-1/2})$. The value $R = 149/160$ suggested in the literature remains unresolved. No certified decimal enclosure, first nonzero correction, or claim of worldwide historical priority is made.

[write] **5 October 2026.** “Report 134” and “Report 135” [1, 2] are Parts I and II of this report. Conway and Guttman’s value 149/160 is still open; see the note after (37.2) and Section 66.1, item 2.

30 The sequence and the conclusions

An occurrence of 1243 in $\pi \in S_n$ is a quadruple of positions $i < j < k < l$ with $\pi_i < \pi_j < \pi_l < \pi_k$. The positions need not be consecutive. We count permutations with exactly one such quadruple; additional 1234 occurrences are allowed. Set $b_n = 0$ for $0 \leq n < 4$. This is OEIS A224179 [5]; for $n = 4, \dots, 15$ its values are

1, 11, 88, 638, 4478, 31199, 218033, 1535207,
10910759, 78310579, 567588264, 4152765025.

Write $A_n = |\text{Av}_n(1234)| = |\text{Av}_n(1243)|$ and $A_0 = 1$. West's classical bijection gives the equality of these avoidance counts; we recall it, with attribution, in Section 31. Regev's three-row, exponent-two strip theorem gives

$$A_n \sim C_0 \frac{9^n}{n^4}, \quad C_0 = \frac{81\sqrt{3}}{16\pi}. \quad (30.1)$$

This is the $d = 1$, three-row, $\alpha = 2$ specialization explicitly recorded in [15, Section 3.1.4], confirming the ordinary strip case of [14]. In particular there are positive constants a_0, a_1 with

$$a_0 \frac{9^n}{(n+1)^4} \leq A_n \leq a_1 \frac{9^n}{(n+1)^4} \quad (n \geq 0). \quad (30.2)$$

Here are the exact quantities in the answer. Let $F_m(p, q)$ count 1234-avoiders of size m whose final p entries decrease and whose q smallest values occur decreasingly. Empty and one-element conditions are vacuous. Put

$$H_h(i, j) = F_{h+2}(i+1, j+1) - F_{h+1}(i, j) \quad (i+j \leq h), \quad (30.3)$$

$$J_k(s) = \sum_{j=0}^s H_{s+k}(k, j), \quad \gamma_k = \frac{27}{8}(5k+24)(k+2)3^{-k}. \quad (30.4)$$

Set $H_h(i, j) = 0$ when $i+j > h$. All indices are nonnegative unless stated otherwise. Section 33 gives finite tableau evaluations of F and proves the nonnegativity of the relevant H values.

Theorem 30.1. *The exact enumeration is*

$$b_n = \sum_{\substack{s, t, k \geq 0 \\ s+t+k=n-4}} J_k(s) J_k(t) \quad (n \geq 4). \quad (30.5)$$

The amplitude ratio exists and has the positive convergent representation

$$\begin{aligned} R &:= \lim_{n \rightarrow \infty} \frac{b_n}{A_n} = \frac{2}{9^4} \sum_{t, k \geq 0} 9^{-t} \gamma_k J_k(t) \\ &= \frac{1}{972} \sum_{t, k \geq 0} 9^{-t} 3^{-k} (5k+24)(k+2) J_k(t), \quad 0 < R < \infty. \end{aligned} \quad (30.6)$$

Consequently

$$b_n \sim C_0 R \frac{9^n}{n^4}. \quad (30.7)$$

The ordinary generating function $B(z) = \sum_{n \geq 0} b_n z^n$ is the diagonal of a rational power series over $\mathbb{Q}$. It is D -finite but not algebraic over $\mathbb{Q}(z)$.

No scalar differential equation, recurrence, order bound, or efficient elimination algorithm is computed here. The rational-diagonal assertion is a structural consequence of the exact finite formula, not a guess from initial coefficients. The equivalent alone gives the inverse statement in Section 39. The additional uniform argument of Section 40 gives

$$b_n = C_0 R 9^n n^{-4} (1 + O(n^{-1/2}))$$

and an inside-ceiling width $O((\log x)^{-1/2})$. The diffusive analysis of Section 41 improves both remainders to little- o at these scales, proving a zero square-root coefficient. It does not identify the first nonzero correction or an all-orders expansion.

What is inherited and what is established here

The rank-at-most-two skeleton bijection is West’s prior work; the exposition in Bóna [20, Section 2, Lemma 2.2] supplies an accessible source. Regev’s avoidance equivalent is also prior work. The tableau formula for F and its fixed-boundary limit were used in the companion Report 134 on one 1234 occurrence [1]; their proofs are reproduced here so that this report does not require a reader to obtain that companion. The marked extension, shared-spine extraction, 23-index encoding, and uniform growing-spine analysis are the steps proved for the present target. Standard RSK, Pieri, and hook-length identities are used explicitly. The nonalgebraicity deduction uses the general analytic lemma of Bóna–Burstein [9, Lemma 6.3]; their monotone-pattern combinatorial injection is not being applied to 1243.

[write] 5 October 2026. In this report Report 134’s tableau formula and fixed-boundary lemma are printed once, as Proposition 4.1 and Lemma 5.1; Sections 33 and 34 below point there (Report 136 writes B_p for Part I’s c_p). Report 135 is Part II; Lemma 40.1 below is a weaker, global second route to Part II’s Proposition 16.1. [9, Lemma 6.3] was checked against arXiv:2101.00332v3: it is as quoted in Section 39.

31 From one occurrence to a marked avoider

For an entry of a permutation, its *rank* is the length of a longest increasing subsequence ending there. Call ranks at most two *low* and the remaining entries *high*. Fix the positions and values of the low entries; this is the low skeleton. List the remaining positions as $h_1 < \dots < h_m$ and remaining values as $v_1 < \dots < v_m$. A cell (h_i, y) is allowed if some two low entries at positions $a < b < h_i$ have values $\pi_a < \pi_b < y$.

The allowed board is northeast closed: increasing position or value preserves allowance. Every high entry has a supporting low pair. Indeed, a chain ending there can be started at a rank-one point and continued to a rank-two point before its high part. Conversely, any full matching of high positions with high values using allowed cells preserves the low skeleton. A filled high point has rank at least three. It cannot precede and lie below a low point, since then that low point would also have a low supporting pair and rank at least three. Thus the ranks of low points remain unchanged.

Let d_i be the number of high values globally allowed at h_i , with $d_0 = 0$. These numbers are nondecreasing. Every value placed in an earlier high position is allowed at h_i , so the number of remaining choices is

$$a_i = d_i - i + 1 \geq 1. \quad (31.1)$$

The inequality follows from the existence of the original matching and is independent of earlier choices. Choosing the $(e_i + 1)$ st smallest remaining allowed value, where $0 \leq e_i < a_i$, is therefore an exact code for all matchings of the skeleton.

An *active inversion* consists of high positions $h_i < h_j$ with values $x > y$, where y is already allowed at h_i . Exactly e_i active inversions start at h_i : the e_i smaller allowed values skipped there all appear later. Thus their number is $\sum_i e_i$. An active inversion with smaller value y has exactly

$$s_\pi(h_i, y) = \#\{a < b < h_i : \pi_a < \pi_b < y\}$$

supporting pairs, and this positive integer is its number of 1243 occurrences. Every 1243 occurrence arises in this way: its last two entries are high, and its supporting pair either is low or has an earlier low supporting pair below it. Thus the smaller top value is already allowed at the earlier high row. Consequently

$$N_{1243}(\pi) = \sum_{\text{active inversions } (h_i, h_j)} s_\pi(h_i, \pi_{h_j}). \quad (31.2)$$

The all-zero code is the unique 1243-avoiding matching: fill each high position from left to right with the smallest remaining allowed value. Call it the *greedy avoider* σ . There is also a unique 1234-avoiding matching, obtained by filling the high values in decreasing order. It is allowed because the largest remaining value is allowed whenever any remaining value is. It avoids 1234 because the last two entries of an increasing four-chain would be increasing high points; conversely every increasing pair of high points extends the low support of its first point to such a chain. This is West's classical skeleton-preserving bijection [20].

At step i of the greedy filling, write y for the smallest remaining allowed value. If a second choice exists, write z for it. Call row i *eligible* when $a_i \geq 2$ and $s_\sigma(h_i, y) = 1$.

Theorem 31.1 (The marked extension). *Swapping y and z in a greedy avoider at an eligible row is a bijection from eligible marked 1243-avoiders to permutations containing exactly one 1243. In particular, if $M(\sigma)$ is the number of eligible rows,*

$$b_n = \sum_{\sigma \in \text{Av}_n(1243)} M(\sigma). \quad (31.3)$$

Proof. By (31.2), a matching with one occurrence has one active inversion of support one. Its code has one digit equal to one and every other digit zero. Compare it with the greedy matching. At its unique exceptional row it selects z rather than y . Until the greedy matching selects z , both matchings then choose the same entries: newly admitted values are below y , and no value already admitted at the exceptional row lies strictly between y and z . At the row where the greedy matching selects z , the exceptional one selects y . Thereafter their remaining sets agree. They differ exactly by that transposition. Their prefixes before the exceptional row agree, so the unique active inversion has the same support count $s_\sigma(h_i, y)$. Hence it gives exactly one occurrence precisely at an eligible row. Conversely this transposition is valid, since y remains allowed later, and its code has the required single exceptional digit. The unique occurrence identifies the active inversion; swapping it back recovers the greedy avoider and the mark. This proves both directions and uniqueness. $\square$

Lemma 31.2 (A local corner criterion). *Let y_i be the smallest high value globally allowed at row h_i . A row is eligible if and only if*

$$d_i \geq i + 1, \quad d_i > d_{i-1}, \quad \text{exactly one low increasing pair lies southwest of } (h_i, y_i). \quad (31.4)$$

Proof. After the preceding greedy step, the minimum high value used so far is the smallest globally allowed value at that preceding row. This follows inductively: a newly admitted minimum is immediately chosen, and without a new minimum the old minimum stays unchanged. If $d_i > d_{i-1}$, the new global minimum y_i is unused and below every earlier high value. It is the greedy choice. Thus no earlier high point lies southwest of (h_i, y_i) and its supporting pairs are all low. The second-choice condition $a_i \geq 2$ is exactly $d_i \geq i + 1$.

If $d_i = d_{i-1}$, the global minimum was used earlier. The present greedy candidate y lies above that earlier high value x . A low supporting pair $a < b < x$ for x precedes the present row and lies below y ; the three increasing pairs $(a, b), (a, x), (b, x)$ then support y . Such a row cannot be eligible. This proves the criterion, including the initial row with $d_0 = 0$. $\square$

For clarity, the southwest condition is genuinely local in the low skeleton. Rank-one points decrease, and rank-two points decrease. Let b be the last rank-two point before h_i , and b' the preceding such point if it exists. Let a be the last rank-one point preceding b , and a' its predecessor. The unique-low-pair condition says $b < y_i < b'$ and $a < b < a'$, where an absent predecessor has value $+\infty$. Every increasing low pair ends at a rank-two point and starts at a rank-one point, which

proves these equivalences. At an eligible row, b lies after the preceding high position; otherwise the board threshold could not fall.

This mark is not uniformly bounded on individual permutations. For example, the avoider $(1, 2q, 2q+1, 2q-2, 2q-1, \dots, 2, 3, 2q+2)$ has q eligible rows. A limiting average therefore requires a uniform estimate; fixed-boundary limits by themselves do not suffice.

32 The shared decreasing spine

Transport an eligible mark to the 1234-avoider with the same skeleton. In that avoider, all high values decrease. At row h_i write z for its value and $y = y_i$ for the newly allowed minimum. The smallest among the d_i globally allowed high values occurs at high position h_{d_i} . Since $d_i \geq i+1$, it occurs after z . Write a, b for the unique low support in increasing position and value order. The marked core is a, b, z, y in position order and $a < b < y < z$ in value order.

Equivalently: the rectangle before z and below y contains exactly one increasing pair, namely a, b ; a has rank one and b rank two; z is the first high point after b ; and y is the smallest high value above b . The corner criterion proves these assertions. In particular the last rank-two point before z is b , which appears after the previous high row; the allowed values at z are precisely the high values above b . Conversely these four assertions and a later high point $y < z$ recover (31.4).

Use cell coordinates relative to the four marked positions a, b, z, y and the four marked values a, b, y, z : coordinate (u, v) means u marked positions lie to the left and v marked values lie below. The only cells for extra points are

$$\begin{aligned}\mathcal{N} &= \{(0, 2), (0, 3), (0, 4), (1, 3), (1, 4)\}, \\ \mathcal{T} &= \{(2, 0), (3, 0), (3, 1), (4, 0), (4, 1)\}, \\ &\text{and the central cell } (3, 3).\end{aligned}\tag{32.1}$$

Here is the necessity check. Before a , values below b contradict rank one of a or uniqueness of the southwest support. Between a, b , values below y create another support. Between b, z , values in (a, y) do likewise, while values above y would be earlier high points after b . Between z, y , values in (b, y) contradict minimality of the high value y , and values above z complete an increasing four-chain. After y , values in (b, y) again contradict minimality, and values above y complete the four-chain a, b, y , followed by that point. These exclusions leave exactly (32.1).

The points of cell $(0, 2)$ decrease, as do those of $(2, 0)$; otherwise each increasing pair there would be another southwest support. The central points, together with z, y , form a decreasing spine W of length $r \geq 2$. They all have rank three, since they are above and after a, b and no increasing four-chain is allowed.

Every $\mathcal{N}$ point is before b and above b ; every $\mathcal{T}$ point is after b and below b . Thus these two sets are mutually inverted, and no increasing chain uses both types. The whole permutation avoids 1234 if and only if both restrictions

$$\mathcal{N} \cup \{a, b\} \cup W, \quad \mathcal{T} \cup \{a, b\} \cup W$$

avoid it. Call these the northern and southern halves. They share the entire core $12 \oplus \delta_r$, where δ_r is a decreasing word of length r . Inversion exchanges the two types and fixes the standardized core.

Lemma 32.1 (Unique amalgamation). *Two valid halves with the same spine length r amalgamate uniquely into an eligible marked 1234-avoider. No shuffle factor is needed.*

Proof. The north half supplies all comparisons involving northern positions and spine values; the south half supplies all comparisons involving southern values and spine positions. Northern versus southern comparisons are forced by their mutual inversion. The interleavings of southern points among spine positions and northern values among spine values already belong to the two half words. Consequently there is one amalgamation.

No four-chain can use both types, so the two avoidance conditions ensure global avoidance. The other points of the southwest support rectangle form a decreasing group before a with values between b, y , and a decreasing group after b but before z with values below a . Neither group supplies an increasing pair internally, with the other group, or with a, b . Thus a, b is its sole increasing pair. Points between b, z are decreasing new left-to-right minima; thus z is the first high point after b . Points whose values are between b, y lie decreasingly before a and have rank one; hence y is the smallest high value above b . The reconstructed mark satisfies the equivalent corner assertions above. This proves sufficiency as well as uniqueness. Necessity follows by taking restrictions of the original marked avoider. $\square$

Let $N_r(s)$ count northern halves with s extra points outside the spine and a, b . The preceding bijections yield

$$b_n = \sum_{r \geq 2} \sum_{s+t=n-r-2} N_r(s) N_r(t). \quad (32.2)$$

Lemma 32.2 (Identifying the half count). *For $r \geq 2$ and $s \geq 0$,*

$$N_r(s) = \sum_{q=1}^{s+1} (F_{s+r}(r-1, q) - F_{s+r-1}(r-2, q-1)) = J_{r-2}(s). \quad (32.3)$$

Proof. A standardized northern half has $a = 1, b = 2$ and word $U1V2W$, where W is a decreasing suffix of length r , ending at y . All V values exceed y , while the j values $3, \dots, j+2$ below $y = j+3$ occur decreasingly in U . Put $q = j+1$, so $1 \leq q \leq s+1$. Delete $b = 2$ and the final point y , then standardize. The result has size $s+r$, avoids 1234, has a decreasing suffix of length $r-1$, and has its q smallest values decreasing. That suffix is disjoint from those q values.

Conversely, take any avoider with these properties. Append a new final value of rank $q+1$, shifting larger values up. It cannot complete a four-chain because the q smaller values decrease. Insert a new value 2 immediately before the designated suffix (now of length r), keeping the old minimum 1 and shifting the other values. This insertion cannot create a four-chain: there is only one smaller value and the later spine decreases. The decreasing q smallest entries end at the old minimum and occur before the suffix, so the result has precisely the required word $U1V2W$, geometry, and boundary decrease. The operations are inverse.

For a nonempty decreasing suffix and a decreasing set of q smallest values, the two sets intersect if and only if the final value is 1. Indeed an intersection puts the minimum later in the decreasing suffix, where it must be final. Deleting that final 1 is a bijection to $F_{s+r-1}(r-2, q-1)$. Subtracting the intersecting cases gives the summand in (32.3). With $k = r-2$ and $j = q-1$, it is $H_{s+k}(k, j)$. Summing proves the assertion. $\square$

Substituting (32.3) into (32.2) proves the exact formula (30.5). In particular, as an identity of formal series,

$$B(z) = z^4 \sum_{k \geq 0} z^k J_k(z)^2, \quad J_k(z) = \sum_{s \geq 0} J_k(s) z^s. \quad (32.4)$$

Every coefficient here uses finitely many terms. No analytic convergence is needed for this exact identity.

33 The exact tableau formula

[write] **Printed in Part I.** Report 136's Section 4 states and proves its Proposition 6, which is Report 134's Proposition 4 word for word, with B_p in place of c_p . It is printed as Proposition 4.1, with its displays (4.1), (4.2) and (4.3); references to them below point there. Report 136 closes the section with the following paragraph, printed as delivered.

These formulas are finite and exact. They also show $A_m = F_m(0, 0) = \sum_{\ell(\rho) \leq 3} (f^\rho)^2$. These inherited finite and fixed-boundary formulas are reproduced from Report 134 [1]. For $i + j \leq h$, the difference in (30.3) counts exactly the size- $h + 2$ avoiders with decreasing suffix of length $i + 1$ and decreasing $j + 1$ smallest values for which those two sets are disjoint. The final-minimum deletion argument in Lemma 32.2 proves this assertion, hence $H_h(i, j) \geq 0$. The remaining argument requires a bound uniform in every boundary parameter.

34 Fixed boundary asymptotics

[write] **Printed in Part I.** Report 136's Section 5, with its definition $B_p = \binom{p+2}{2}$, $d_p = B_p 3^{-p}$ and its Lemma 7 and proof, is Report 134's Section 5 word for word, with B_p in place of c_p . It is printed as (5.1) and Lemma 5.1; the limit $C(i, j)$ used below is (5.3).

35 Uniform Gaussian estimates

For a three-row partition μ of r , put

$$M_r(\mu) = \max_i |\mu_i - r/3|, \quad B_p = \binom{p+2}{2}.$$

All constants $K, c > 0$ below are absolute; their values may change from line to line. Their existence, rather than numerical optimization, is what the limit argument needs.

Lemma 35.1 (Global three-row Gaussian tail). *There are constants $K, c > 0$ such that, for all integers $r \geq 0$ and all real $x \geq 0$,*

$$\frac{1}{A_r} \sum_{M_r(\mu) \geq x} (f^\mu)^2 \leq K \exp \left(-\frac{cx^2}{r+1} \right). \quad (35.1)$$

In particular, this statement is uniform at empty and small shapes.

Proof. We first record the uniform local multinomial bound

$$\binom{N}{x_1, x_2, x_3} \leq \frac{K 3^N}{N+1} \exp \left(-c \frac{\sum_i (x_i - N/3)^2}{N} \right) \quad (N \geq 1). \quad (35.2)$$

Here x_i are nonnegative integers summing to N . For completeness, put $q_i = x_i/N$ and $D(q) = \sum_i q_i \log(3q_i)$. Entropy convexity gives $D(q) \geq \frac{1}{2} \sum_i (q_i - 1/3)^2$: on the interior of the simplex the Hessian of $\sum_i q_i \log q_i$ is diagonal with entries $1/q_i \geq 1$, and the boundary follows by continuity. Also $3^{-N} \binom{N}{x_1, x_2, x_3} \leq e^{-ND(q)}$, since the multinomial probability at parameter q of that count vector is at most one. If all $x_i \geq N/6$, uniform Stirling bounds supply the additional factor K/N , proving (35.2). Otherwise $S := \sum_i (x_i - N/3)^2 \geq N^2/36$, and

$$e^{-S/(2N)} \leq \frac{K}{N} e^{-S/(4N)},$$

because $Ne^{-N/144}$ is bounded. This proves (35.2) throughout the simplex.

For the hook formula set

$$(x_1, x_2, x_3) = (a + 2, b + 1, c), \quad N = r + 3.$$

Then

$$f^{(a,b,c)} = \binom{N}{x_1, x_2, x_3} \frac{(x_1 - x_2)(x_1 - x_3)(x_2 - x_3)}{(r + 1)(r + 2)(r + 3)}.$$

The shifted deviations satisfy $x_i - N/3 = \mu_i - r/3 + (2 - i)$. Thus, with $M = M_r(\mu)$,

$$\max_i |x_i - N/3| \geq (M - 1)_+, \quad \prod_{i < j} |x_i - x_j| \leq 8(M + 1)^3.$$

Since $N \leq 3(r + 1)$ and $(M - 1)_+^2 \geq M^2/2 - 1$, equation (35.2) gives

$$f^\mu \leq K3^r(r + 1)^{-4}(M + 1)^3 e^{-cM^2/(r+1)}.$$

Squaring, writing $T = M/\sqrt{r + 1}$, and absorbing the factor $(1 + T)^6$ into a weaker Gaussian yields the pointwise bound

$$(f^\mu)^2 \leq K9^r(r + 1)^{-5} e^{-cM_r(\mu)^2/(r+1)}. \quad (35.3)$$

All these inequalities apply also to $r = 0$, where $N = 3$.

For $M \geq x$, split the exponential in (35.3) into two equal factors and extract $e^{-cx^2/(2(r+1))}$. The sum of the remaining factor over partitions of r is $O(r + 1)$. Indeed, extend the first two row coordinates to all integers and use

$$M^2 \geq \frac{1}{2}((a - r/3)^2 + (b - r/3)^2), \quad \sup_{u \in \mathbb{R}} \sum_{h \in \mathbb{Z}} e^{-c(h-u)^2/(r+1)} \leq K\sqrt{r + 1}.$$

The tail sum is therefore at most $K9^r(r + 1)^{-4} e^{-cx^2/(r+1)}$. Dividing by the lower bound in (30.2) proves the result. $\square$

Lemma 35.2 (Gaussian mismatch for strip sums). *For all $m \geq 0$ and $0 \leq p, q \leq m$, put $r = m - p$, $u = m - q$, and $D = |p - q| = |r - u|$. Then*

$$F_m(p, q) \leq KB_p B_q \sqrt{A_r A_u} \exp\left(-\frac{cD^2}{\max(r, u) + 1}\right). \quad (35.4)$$

Replacing the denominator by $m + 1$ is also valid. For $D = 0$, one may instead use the sharp elementary prefactor bound $F_m(p, p) \leq B_p^2 A_{m-p}$.

Proof. Let T_p be the zero-one horizontal-strip incidence matrix, with rows indexed by outer shapes of size m and columns by inner shapes of size $r = m - p$. Every row and every column has at most B_p ones: the numbers of boxes removed from or added to the three rows form a weak composition of p . Its Euclidean operator norm is at most B_p , either by the row/column Schur bound or by two applications of Cauchy–Schwarz. If v_r is the vector with entries f^μ , then $W_p = T_p v_r$ and $\|v_r\|_2^2 = A_r$.

Suppose that μ, ν are inner shapes of sizes r, u admitting the same outer shape λ . By symmetry assume $u \geq r$. Interlacing implies $\mu_1 \geq \lambda_2 \geq \nu_2$, and consequently

$$\frac{u - r}{3} \leq (u/3 - \nu_2) + (\mu_1 - r/3) \leq M_u(\nu) + M_r(\mu).$$

Hence $M_r(\mu) \geq D/6$ or $M_u(\nu) \geq D/6$. Let $v_r^{\text{tail}}, v_u^{\text{tail}}$ be the vectors restricted to these respective tail sets. Positivity and the preceding support implication give

$$\begin{aligned} F_m(p, q) &\leq \langle T_p v_r^{\text{tail}}, T_q v_u \rangle + \langle T_p v_r, T_q v_u^{\text{tail}} \rangle \\ &\leq B_p B_q \sqrt{A_r A_u} \left(\frac{\|v_r^{\text{tail}}\|_2}{\sqrt{A_r}} + \frac{\|v_u^{\text{tail}}\|_2}{\sqrt{A_u}} \right). \end{aligned}$$

Lemma 35.1, followed by square roots, bounds the parentheses by

$$K \left(e^{-cD^2/(r+1)} + e^{-cD^2/(u+1)} \right) \leq K e^{-cD^2/(\max(r,u)+1)}.$$

This also covers $D = 0$ and empty inner shapes. $\square$

Lemma 35.3 (Triangular normalized bound). *There are constants $K, c > 0$ such that, whenever $p, q \geq 0$ and $p + q \leq m$,*

$$\frac{F_m(p, q)}{A_m} \leq K B_p B_q 3^{-p-q} \exp \left(-\frac{c(p-q)^2}{m+1} \right). \quad (35.5)$$

Proof. By (30.2) and Lemma 35.2, the right side before simplification has the additional rational factor

$$R_{m,p,q} = \frac{(m+1)^4}{(m-p+1)^2(m-q+1)^2}.$$

Write $r = m - p$, $u = m - q$; the support assumption gives $r + u \geq m$. If $\min(r, u) \geq m/4$, then $R_{m,p,q} \leq 4^4$. Otherwise $D = |u - r| \geq m/2$ and $R_{m,p,q} \leq (m+1)^4$. For $m \geq 1$, a half of the available Gaussian exponent absorbs this polynomial, since

$$\sup_{m \geq 1} (m+1)^4 \exp \left(-\frac{cm^2}{8(m+1)} \right) < \infty.$$

The case $m = 0$ is immediate. Weakening c proves the assertion. $\square$

36 Summability along the shared spine

Recall

$$H_h(i, j) = F_{h+2}(i+1, j+1) - F_{h+1}(i, j) \quad \text{for } i+j \leq h,$$

and zero otherwise. By Lemma 32.2 and the discussion after the tableau formula, H is nonnegative, so $0 \leq H_h(i, j) \leq F_{h+2}(i+1, j+1)$. Define

$$J_k(s) = \sum_{j=0}^s H_{s+k}(k, j), \quad d_p = B_p 3^{-p}, \quad C(k, j) = 81d_{k+1}d_{j+1} - 9d_k d_j,$$

and

$$\gamma_k = \sum_{j \geq 0} C(k, j) = \frac{27}{8} (5k+24)(k+2)3^{-k}. \quad (36.1)$$

The formula follows from $\sum_j d_j = 27/8$ and $\sum_j d_{j+1} = 19/8$. In particular $\gamma_k \asymp (k+1)^2 3^{-k}$, uniformly for $k \geq 0$.

Proposition 36.1 (Uniform J bounds and endpoint summability). *For every $s, k \geq 0$,*

$$J_k(s) \leq K A_{s+k} \gamma_k. \quad (36.2)$$

For every fixed k , the known fixed-boundary limits of H imply

$$\frac{J_k(s)}{A_{s+k}} \longrightarrow \gamma_k \quad (s \longrightarrow \infty). \quad (36.3)$$

Finally, for

$$e_t = 9^{-t} \sum_{k \geq 0} \gamma_k J_k(t), \quad (36.4)$$

one has

$$0 \leq e_t \leq K(t+1)^{-3/2}. \quad (36.5)$$

In particular the positive endpoint series is finite, and its tail above T is $O((T+1)^{-1/2})$.

Proof. In the j th summand of $J_k(s)$, take $m = s + k + 2$, $p = k + 1$, $q = j + 1$. Since $j \leq s$, one has $p + q \leq m$. Lemma 35.3, the bounded ratio A_{h+2}/A_h , and the summability of $B_{j+1}3^{-j}$ show

$$\frac{H_{s+k}(k, j)}{A_{s+k}} \leq K B_{k+1} B_{j+1} 3^{-k-j}, \quad \frac{J_k(s)}{A_{s+k}} \leq K(k+1)^2 3^{-k} \leq K \gamma_k.$$

The same summable majorant in j permits dominated summation of the fixed-boundary limits $H_{s+k}(k, j)/A_{s+k} \longrightarrow C(k, j)$, proving (36.3).

For endpoint summability it is important to retain the original, un-normalized denominator factors rather than only (36.2). Lemma 35.2 and (30.2) give, for $j \leq t$,

$$9^{-t} \gamma_k H_{t+k}(k, j) \leq K \frac{(k+1)^4 (j+1)^2 3^{-j}}{(t+2)^2 (t+k-j+2)^2} \exp\left(-\frac{c(k-j)^2}{t+k+3}\right). \quad (36.6)$$

Indeed the two inner sizes are $t+1$ and $t+k+1-j$, and their exponential factor $3^{2t+k+2-j}$ cancels $9^t 3^k$, up to an absolute constant.

First sum terms with $j \geq k/2$. Discard the exponential and the second inverse-square denominator. Then $k \leq 2j$, and

$$\sum_{j \geq 0} \sum_{0 \leq k \leq 2j} (k+1)^4 (j+1)^2 3^{-j} < \infty.$$

This region contributes $O((t+2)^{-2})$.

For $j < k/2$, one has $k-j > k/2$ and $t+k-j+2 \geq (t+k+3)/2$. Sum the geometrically weighted j factor to bound this region by

$$\frac{K}{(t+2)^2} \sum_{k \geq 0} \frac{(k+1)^4}{(t+k+3)^2} \exp\left(-\frac{ck^2}{t+k+3}\right). \quad (36.7)$$

For $k \leq t+1$, the two inverse-square denominators together are $O((t+1)^{-4})$, and the exponential is at most $e^{-c'k^2/(t+1)}$. The elementary Gaussian-moment estimate

$$\sum_{k \geq 0} (k+1)^4 e^{-c'k^2/(t+1)} \leq K(t+1)^{5/2}$$

therefore gives $O((t+1)^{-3/2})$. For $k > t+1$, one has $t+k+3 \leq 3k$, so this portion of the sum in (36.7) is at most

$$\frac{K}{(t+2)^2} \sum_{k \geq 1} (k+1)^2 e^{-c'k} = O((t+1)^{-2}).$$

Combining the three bounds proves (36.5). □

37 Proof of the leading equivalent

The exact identity (30.5) has now been proved, and all uniform estimates needed to pass to its limit are available. Orient the sum by requiring $s \geq t$. Equivalently, sum over $(t, k) \in \mathbb{N}_0^2$, set $s = n - k - t - 4$, keep the term only when $s \geq t$, and give it the factor $2 - \mathbf{1}_{s=t}$. This counts all triples exactly, including the diagonal. On its support,

$$\frac{n+1}{s+k+1} \leq 5.$$

The global bound (30.2) and (36.2) therefore give

$$(2 - \mathbf{1}_{s=t}) \frac{J_k(s) J_k(t)}{A_n} \leq K 9^{-t} \gamma_k J_k(t). \quad (37.1)$$

The right side is summable over all t, k by (36.5). For each fixed t, k , eventually $s > t$ and $s \rightarrow \infty$. The fixed-spine limit (36.3) and Regev's ratio $A_{n-t-4}/A_n \rightarrow 9^{-t-4}$ give

$$(2 - \mathbf{1}_{s=t}) \frac{J_k(n-k-t-4) J_k(t)}{A_n} \longrightarrow 2 \cdot 9^{-t-4} \gamma_k J_k(t).$$

Extend summands by zero off the oriented support. Dominated convergence proves (30.6), including every diagonal and middle term. The limit is finite by endpoint summability. It is positive because $J_0(0) = H_0(0, 0) = 1$ and $\gamma_0 = 162$. Multiplying by Regev's equivalent proves (30.7).

What the series says about the amplitude

Conway–Guttmann [8, Section 5.2, printed pages 20–23] numerically estimate a ratio near 0.93125 and tentatively suggest 149/160. The positive series here establishes a ratio's existence and finiteness, but does not determine whether it equals that rational number. For every integer $T \geq 0$ the rational sum

$$L_T = \frac{1}{972} \sum_{k+t \leq T} 9^{-t} 3^{-k} (5k+24)(k+2) J_k(t) \quad (37.2)$$

is a rigorous lower bound for R . The included exact certificate computes

$$L_{35} = \frac{55999844944982350869187286355724963}{90113598179756697647360591499090564},$$

whose numerical rendering is about 0.621436121475. This is a rendering of a lower bound, not a certified approximation to R . Equation (36.5) gives $\sum_{t>T} e_t = O((T+1)^{-1/2})$ with an unspecified absolute constant; it gives no convenient certified numerical error bar for (37.2). It also does not by itself yield an asymptotic correction to b_n . Section 41 subsequently proves a tail equivalent and, by a separate matched-interior argument, cancellation at the square-root scale.

[write] On 149/160, 5 October 2026. Conway and Guttmann (arXiv:2306.12682v2, Section 5.2) report that the ratio tends to an apparent limit 0.93125 ± 0.00015 and write that $149/160 = 0.93125$ would be a likely candidate if the limit were a simple fraction; they prove nothing about it. The lower bound $L_{35} \approx 0.6214$ is far below, and the tail equivalent of Theorem 41.1, of order $T^{-1/2}$, explains the slow convergence of finite sums (it concerns the tail in t only, not the triangle $k+t > T$ omitted by L_T). The intake dossier for this batch extrapolated the exact ratios $b_n/A_n = 0.7548, 0.8106, 0.8299$ at $n = 40, 60, 72$ naively in powers of $1/n$ and obtained $0.93157\text{--}0.93177$, consistent with $149/160$. That is not a certificate: the correction structure beyond the vanishing $n^{-1/2}$ term is unknown. The question is open (Section 66.1, item 2).

38 An everywhere defined binomial encoding

We now derive the rational-diagonal claim directly from the exact formula. The result used is Bostan–Lairez–Salvy [16], Definition 1.1, Theorem 3.5 and Corollary 3.6 in arXiv version 2: their algebra of binomial sums is closed under products, integral affine substitutions, and directed partial summation. A univariate binomial sum supported on $\mathbb{N}_0$ has a rational-diagonal generating function and a nonzero polynomial-coefficient recurrence. It is important to exhibit this closure construction, rather than invoke an unrestricted claim about finite hypergeometric sums.

For all integers a, b , define

$$\mathcal{B}(a, b) = [z^b](1 + z)^a.$$

The formal expansion at zero is intended. The value is zero for $b < 0$; for $b \geq 0$ it is the falling-factorial polynomial in a , divided by $b!$. Negative upper arguments are allowed. Define

$$h(r) = \mathcal{B}(r, r), \quad \delta(r) = \mathcal{B}(0, r).$$

These are the indicators of $r \geq 0$ and $r = 0$, respectively. All expressions below are therefore defined at every integer tuple.

For an arbitrary integer triple $x = (x_1, x_2, x_3)$ write $M = |x|$ and $e_i = x_i - i + \sigma(i)$ for $\sigma \in S_3$. Put

$$D(x) = \sum_{\sigma \in S_3} \text{sgn}(\sigma) \mathcal{B}(M, e_1) \mathcal{B}(M - e_1, e_2) \mathcal{B}(e_3, e_3). \quad (38.1)$$

For a nonnegative partition x this equals f^x . To check this, $e_1 + e_2 + e_3 = M$; a negative e_i makes the corresponding product zero, and otherwise the product is $M!/(e_1!e_2!e_3!)$. Thus the signed sum is $M! \det(1/(x_i - i + j)!)$, with negative reciprocal factorials interpreted as zero only in this verification. Factoring with $X_i = x_i + 3 - i$ leaves falling-factorial columns of degrees 2, 1, 0, hence a Vandermonde and exactly the hook formula (4.3). Formula (38.1) itself uses no potentially undefined factorials.

For integer triples λ, μ define

$$I(\lambda, \mu) = h(\lambda_1 - \mu_1)h(\mu_1 - \lambda_2)h(\lambda_2 - \mu_2) \\ \cdot h(\mu_2 - \lambda_3)h(\lambda_3 - \mu_3)h(\mu_3).$$

It is the indicator of $\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \lambda_3 \geq \mu_3 \geq 0$, which forces both triples to be partitions and is precisely horizontal-strip interlacing. Set

$$E(m, p, q; \lambda, \mu, \nu) = \delta(|\lambda| - m)\delta(|\mu| - m + p)\delta(|\nu| - m + q) \\ \cdot I(\lambda, \mu)I(\lambda, \nu)D(\mu)D(\nu). \quad (38.2)$$

For $0 \leq p, q \leq m$ and every integer $N \geq m$,

$$F_m(p, q) = \sum_{\lambda, \mu, \nu \in \{0, \dots, N\}^3} E(m, p, q; \lambda, \mu, \nu). \quad (38.3)$$

Indeed the size indicators and interlacings reproduce the tableau formula and every surviving coordinate is at most m .

Use 23 summation variables, grouped as

$$z = (k, s, t, j, l, \lambda, \mu, \nu, \Lambda, \eta, \theta),$$

where the final six symbols denote triples. For $\alpha, \beta \in \{0, 1\}$, define

$$\begin{aligned} Q_{\alpha, \beta}(n, z) = & \delta(s + t + k - n + 4)h(s - j)h(t - l) \\ & \cdot E(s + k + 2 - \alpha, k + 1 - \alpha, j + 1 - \alpha; \lambda, \mu, \nu) \\ & \cdot E(t + k + 2 - \beta, k + 1 - \beta, l + 1 - \beta; \Lambda, \eta, \theta). \end{aligned} \quad (38.4)$$

The complete formula is

$$b_n = \sum_{\alpha, \beta \in \{0, 1\}} (-1)^{\alpha + \beta} \sum_{z \in \{0, \dots, n+2\}^{23}} Q_{\alpha, \beta}(n, z) \quad (n \geq 0). \quad (38.5)$$

The first indicator enforces $s + t + k = n - 4$, and the next two enforce $j \leq s, l \leq t$. The binary choices expand the two differences defining H , giving exactly (30.5). For $n < 4$ the size condition is impossible. For $n \geq 4$, all five outer indices are at most $n - 4$, the outer tableau sizes are at most $n - 2$, and the bound $n + 2$ contains every surviving coordinate. All surviving F calls have $0 \leq p, q \leq m$.

Expanding the four D factors leaves at most $4 \cdot 6^4 = 5184$ signed products of $\mathcal{B}$ at integral affine forms. This number and the 23 summation indices are fixed independently of n . Implement the 23 box sums by directed partial sums with fresh upper-bound variables, then set every upper bound to $n + 2$. This defines a binomial sum for all integer n ; multiply by $h(n)$ to give support on $\mathbb{N}_0$. For $n \geq 0$ the directed sums are the ordinary finite sums in (38.5). The cited Bostan–Lairez–Salvy theorem now gives a rational diagonal and P-recursiveness. Equivalently $B(z)$ is D-finite. No statement about arbitrary patterns or higher prescribed occurrence counts follows.

[write] **5 October 2026.** The check of Remark 13.2 (Definition 1.1, Theorem 3.5 and Corollary 3.6 of the cited arXiv version) applies to this encoding without change.

39 Nonalgebraicity and inverse thresholds

The positive equivalent gives $b_n \asymp 9^n/n^4$ for large n . Bóna–Burstein [9, Lemma 6.3] prove the following general analytic obstruction: a nonnegative power series whose coefficients lie between positive multiples of K^n/n^m for every positive index, where integer $m > 1$, is not algebraic. The initial zeros of b_n require care. Apply their lemma to

$$\tilde{B}(z) = B(z) + z + z^2 + z^3.$$

For $n \geq 4$, $b_n > 0$ because prepending a new maximum to 1243 repeatedly preserves its unique occurrence. After adjusting constants for finitely many indices, the coefficients of $\tilde{B}$ satisfy the displayed bounds for every $n \geq 1$, with $m = 4, K = 9$. Thus $\tilde{B}$ is nonalgebraic. Adding a polynomial preserves algebraicity, so B is nonalgebraic as well. This argument uses the general analytic lemma, not that paper’s injection for one decreasing pattern.

Prepending a new maximum to any permutation cannot create a 1243 occurrence containing that new point: its first position cannot have any role except the smallest value in such a pattern. This is an injection preserving the occurrence count, so b_n is nondecreasing. For $x > 0$, define the inverse threshold

$$N(x) = \min\{n \geq 0 : b_n \geq x\}, \quad C = C_0 R, \quad \kappa = \log 9.$$

The minimum is finite, by (30.7).

Corollary 39.1 (Rounding-safe inversion). *Let $L = \log x$ and, for sufficiently large x , put*

$$g(x) = \frac{L + 4 \log L - 4 \log \kappa - \log C}{\kappa}. \quad (39.1)$$

There exists $\varepsilon(x) \rightarrow 0$ such that

$$\lceil g(x) - \varepsilon(x) \rceil \leq N(x) \leq \lceil g(x) + \varepsilon(x) \rceil. \quad (39.2)$$

In particular $N(x) = g(x) + O(1)$.

Proof. Taking logarithms of the equivalent gives $\log b_n = f(n) + o(1)$, where $f(u) = \kappa u - 4 \log u + \log C$. For sufficiently large u , $f'(u) = \kappa - 4/u > \kappa/2$, so f is increasing and its inverse is locally Lipschitz with constant $2/\kappa$. Direct substitution gives $f(g(x)) = L + O(\log L/L) = L + o(1)$; consequently the real solution $r(x)$ of $f(r) = L$ satisfies $r(x) = g(x) + o(1)$.

Choose a decreasing envelope $\eta(m) \rightarrow 0$ such that $|\log b_n - f(n)| \leq \eta(m)$ for every $n \geq m$. Let $m(x) = \lfloor r(x)/2 \rfloor$; then $m(x) \rightarrow \infty$. Since $f(m(x)) + \eta(m(x)) < L$ eventually, monotonicity implies $N(x) > m(x)$. For all indices in this relevant range, the two inverse solutions of $f(u) = L \pm \eta(m(x))$ bracket the threshold after taking ceilings: below the lower solution one has $b_n < x$, and at or above the upper solution one has $b_n \geq x$. Both real solutions differ from $r(x)$ by $o(1)$, giving (39.2) after enlarging a positive $\varepsilon(x)$ tending to zero. $\square$

The equivalent supplies no explicit rate for ε . In particular, removing the ceilings and replacing $O(1)$ by $o(1)$ would lose the integer staircase and is not justified. A rate or higher asymptotic correction requires an additional theorem.

[write] 5 October 2026. Corollary 39.1 and the inverse parts of Theorems 40.3 and 41.1 are instances of parts (1)–(2) of `p0:thm:staircase` (the front-matter section Relation to neighbouring reports), with $g(x)$ as approximate inverse.

40 A quantitative error bound

The leading-equivalent argument above uses fixed-boundary limits together with global bounds. A modest uniform rate can also be proved. We include the required unitary-integral argument, adapted from the boundary method in the companion Report 135 [2], rather than use a fixed-parameter expansion uniformly without justification.

Lemma 40.1 (A global boundary approximation). *For $p, q \geq 0$ with $p + q \leq m$,*

$$\left| \frac{F_m(p, q)}{A_m} - d_p d_q \right| \leq K d_p d_q \min \left\{ 1, \frac{(1 + p + q)^2}{m + 1} \right\}. \quad (40.1)$$

Also

$$A_m = C_0 9^m m^{-4} (1 + O(m^{-1})), \quad \frac{A_{h+a}}{A_h} = 9^a (1 + O((h+1)^{-1})) \quad (a = 1, 2). \quad (40.2)$$

Proof. Let $P(U) = \text{tr} U$ on $U(3)$ and let $h_p = s_{(p)}$ be the complete homogeneous symmetric polynomial of degree p in the three eigenvalues. Schur–Weyl decomposition and horizontal Pieri give $P^{m-p} h_p = \sum_{\lambda} W_p(\lambda) s_{\lambda}$. Orthogonality of unitary characters, with Haar probability measure, gives the exact identity

$$F_m(p, q) = \int_{U(3)} P^{m-p} h_p \overline{P^{m-q} h_q} dU. \quad (40.3)$$

For $p = q = 0$, this is $A_m = \int |P|^{2m} dU$; the trace-moment identity is also recorded by Rains [17].

The common eigenvalue phase cancels because each factor has total degree m . After quotienting by this phase, $|P| = 3$ has a unique maximum. In a fixed small chart write the centered angles as $\theta_1 + \theta_2 + \theta_3 = 0$ and $Q = \sum \theta_i^2$. Weyl's density, up to a constant Jacobian, is

$$\prod_{a < b} |e^{i\theta_a} - e^{i\theta_b}|^2 = \Delta(\theta)^2 (1 + O(Q)), \quad \log(|P|^2/9) = -Q/3 + O(Q^2).$$

The remainders are uniform also on $\Delta = 0$, by factoring out the analytic sinc factors. Off this neighborhood, compactness gives $|P|/3 \leq \rho < 1$. In the local chart $|P|^{2m}/9^m \leq e^{-cmQ}$. Scaling $\theta = y/\sqrt{m}$, the two-dimensional volume contributes m^{-1} and the degree-six Vandermonde square contributes m^{-3} . The errors Q and mQ^2 integrate to relative $O(m^{-1})$: shrinking the chart absorbs their exponential remainder into a weaker Gaussian. Off-chart contributions are exponentially small. Thus $A_m = D9^m m^{-4} (1 + O(m^{-1}))$; comparison with (30.1) identifies $D = C_0$. The fixed-shift assertion follows, with finitely small indices absorbed into the constant. The same Gaussian scaling gives

$$\mathbb{E}_m[Qe^{c_1 L Q}; \text{local}] = O(m^{-1}) \quad (0 \leq L \leq \epsilon m), \quad (40.4)$$

for each fixed c_1 , some $\epsilon > 0$, and density $|P|^{2m} dU/A_m$. This uses (30.2) for normalization.

In the chart define

$$r_p(\theta) = \frac{h_p(e^{i\theta})}{B_p(P(\theta)/3)^p}, \quad h_p/P^p = d_p r_p.$$

Fix a centered unit vector v , put $\theta = rv$, and let $z(r) = P(rv)/3$, $g_p(r) = h_p(e^{irv})/B_p$. Averaging over weak compositions of p shows $|g_p^{(j)}(r)| \leq p^j$ for $j = 1, 2$, and $|g_p| \leq 1$. In a fixed small interval $|z(r)| \geq e^{-Cr^2}$; its first two derivatives are bounded, and the finite chain rule gives

$$|(z^{-p})^{(j)}(r)| \leq K_j (p+1)^j e^{C' p r^2} \quad (0 \leq j \leq 2).$$

Extra inverse powers of z are uniformly bounded. The real part of $r_p(rv)\overline{r_q(rv)}$ is even in r , has value one at zero, and has second derivative bounded by $K(1+p+q)^2 e^{C''(p+q)r^2}$. Taylor's theorem gives

$$|\Re(r_p \overline{r_q}) - 1| \leq K(1+p+q)^2 Q e^{C''(p+q)Q}.$$

Conjugation invariance allows taking the real part under the integral. If $1+p+q \leq \sqrt{m+1}$, equation (40.4) integrates this bound to $O((1+p+q)^2/m)$ for sufficiently large m .

No division by P is used off the chart. The original integrand in (40.3), since $|h_p| \leq B_p$, is bounded by $9^m d_p d_q (|P|/3)^{2m-p-q}$. Its normalized off-chart contribution is $O(d_p d_q m^4 \rho^{2m-p-q})$, exponentially small in the range in question; the constant approximant has the same negligible complement. This proves (40.1) in the small-boundary range. Otherwise the minimum on its right is one, and the triangular bound (35.5) supplies $F_m(p, q)/A_m \leq K d_p d_q$. Enlarging K handles the finitely many small m . $\square$

Lemma 40.2 (Uniform spine approximation). *For every $s, k \geq 0$,*

$$\left| \frac{J_k(s)}{A_{s+k}} - \gamma_k \right| \leq K \gamma_k \min \left\{ 1, \frac{(k+1)^2}{s+k+1} \right\}. \quad (40.5)$$

For each fixed nonnegative integer r , the endpoint moments satisfy

$$M_r(t) := 9^{-t} \sum_{k \geq 0} (k+1)^{2r} \gamma_k J_k(t) \leq K_r (t+1)^{r-3/2}. \quad (40.6)$$

Proof. Put $h = s + k$. Apply (40.1) to both terms of $H_h(k, j)$, using the fixed-shift ratio (40.2). Since $d_{k+1} \asymp d_k$ uniformly, the result is

$$\left| \frac{H_h(k, j)}{A_h} - C(k, j) \right| \leq \frac{K d_k d_j (k + j + 1)^2}{h + 1} \quad (k + j \leq h).$$

Summing $0 \leq j \leq s$ and using geometric moments gives $K d_k (k + 1)^2 / (h + 1)$. The missing tail of γ_k is at most $K d_k (s + 1)^2 3^{-s}$, which is bounded by the same expression, since $(s + k + 1) / (k + 1)^2 \leq s + 1$ and $(s + 1)^3 3^{-s}$ is bounded. Now $\gamma_k \asymp d_k$, so this gives the second bound in the minimum. The first follows from (36.2) and positivity.

For the moment claim, multiply (36.6) by $(k + 1)^{2r}$. The part $j \geq k/2$ is still $O_r((t + 1)^{-2})$, because its geometric-polynomial double sum converges. The part $j < k/2$ is at most

$$\frac{K_r}{(t + 2)^2} \sum_{k \geq 0} \frac{(k + 1)^{4+2r}}{(t + k + 3)^2} e^{-ck^2/(t+k+3)}.$$

For $k \leq t + 1$, the denominators supply $O((t + 1)^{-4})$ and the Gaussian moment is $O_r((t + 1)^{(5+2r)/2})$. For $k > t + 1$, exponential decay in k gives $O_r((t + 1)^{-2})$. Both are bounded as in (40.6). In particular $M_0(t) = e_t = O((t + 1)^{-3/2})$ and $M_1(t) = O((t + 1)^{-1/2})$. $\square$

Theorem 40.3 (Quantitative leading estimate). *As $n \rightarrow \infty$,*

$$\frac{b_n}{A_n} = R + O(n^{-1/2}), \quad b_n = C_0 R 9^n n^{-4} (1 + O(n^{-1/2})). \quad (40.7)$$

With g as in (39.1), there is $K < \infty$ such that, for all sufficiently large x ,

$$\left| g(x) - K(\log x)^{-1/2} \right| \leq N(x) \leq \left| g(x) + K(\log x)^{-1/2} \right|. \quad (40.8)$$

The constants are not numerically certified. No nonzero coefficient at the relative $n^{-1/2}$ scale is asserted.

Proof. For $n \geq 32$ put

$$D_n = \{(t, k) \in \mathbb{N}_0^2 : 2t + k \leq n - 4\}, \quad s = n - 4 - t - k, \quad h = s + k = n - 4 - t, \quad w(t, k) = 9^{-t} \gamma_k J_k(t).$$

On D_n , $h \geq (n - 4)/2$. From (40.2),

$$\frac{A_h}{A_n} = 9^{-t-4} \left(1 + O\left(\frac{t+1}{n}\right) \right).$$

Indeed $(n/h)^4$ stays bounded and differs from one by $O((t + 1)/n)$; the relative avoidance errors contribute $O(1/n)$. Combining this with (40.5) gives

$$\left| \frac{J_k(s) J_k(t)}{A_n} - 9^{-4} w(t, k) \right| \leq \frac{K}{n} w(t, k) (t + 1 + (k + 1)^2).$$

Summing over D_n , equation (40.6) bounds the total by

$$\frac{K}{n} \sum_{0 \leq t \leq n/2} ((t + 1) e_t + M_1(t)) = O(n^{-1/2}).$$

Both the complement of D_n and its diagonal $2t + k = n - 4$ are contained in $\{t > n/8\} \cup \{t \leq n/8, k > n/2\}$ for $n \geq 32$. The w mass of this union is at most

$$\sum_{t > n/8} e_t + \frac{4}{n^2} \sum_{t \leq n/8} M_1(t) = O(n^{-1/2}) + O(n^{-3/2}).$$

The normalized convolution diagonal is bounded by K times its w mass by (37.1). Thus replacing the oriented factor $2 - \mathbf{1}_{s=t}$ by two and extending to all t, k changes the answer by $O(n^{-1/2})$. The complete endpoint sum is R , proving the first assertion. Multiplication by (40.2) gives the second.

Finally, taking logarithms gives $\log b_n = \kappa n - 4 \log n + \log(C_0 R) + O(n^{-1/2})$. Its smooth main term has derivative tending to $\kappa > 0$, so its real inverse has an uncertainty $O((\log x)^{-1/2})$. The monotonicity and ceiling argument of Corollary 39.1 applies with this explicit asymptotic error envelope. Substitution in (39.1) shows that the smooth inverse differs from $g(x)$ by $O(\log \log x / \log x) = o((\log x)^{-1/2})$, proving (40.8). $\square$

The bound strengthens the existence theorem and its rounding width. It is not an expansion with a known next term: cancellation could make the true error smaller. It supplies no certified decimal digits of R .

41 The diffusive spine profile and cancellation

The $O(n^{-1/2})$ estimate supplies uniform control, but its leading half-power coefficient actually vanishes. We now prove this from a profile at spine lengths of order $\sqrt{n}$. Write

$$a = \frac{135}{8}, \quad D = C_0 a^2 \int_0^\infty u^4 e^{-u^2/2} du = \frac{4428675}{2048} \sqrt{\frac{6}{\pi}}. \quad (41.1)$$

Theorem 41.1 (Profile and improved remainder). *If $h = s + k \rightarrow \infty$ and $k/\sqrt{h} \rightarrow u \in [0, \infty)$, then*

$$\frac{J_k(s)}{A_h \gamma_k} \longrightarrow e^{-u^2/2}. \quad (41.2)$$

The endpoint density and its amplitude tail satisfy

$$e_t \sim Dt^{-3/2}, \quad \frac{2}{9^4} \sum_{t > T} e_t \sim \frac{675}{512} \sqrt{\frac{6}{\pi}} T^{-1/2}. \quad (41.3)$$

Moreover,

$$\frac{b_n}{A_n} = R + o(n^{-1/2}), \quad b_n = C_0 R 9^n n^{-4} (1 + o(n^{-1/2})). \quad (41.4)$$

Consequently the width in (40.8) can be replaced by some $\varepsilon(x) = o((\log x)^{-1/2})$ inside both ceilings.

The last assertion identifies a zero coefficient at the relative $n^{-1/2}$ scale. It neither proves an $O(n^{-1})$ error nor identifies the first nonzero correction. The tail equivalent is asymptotic and has no certified finite- T error bound. Each e_t includes the full sum over k ; this t -only tail is distinct from the omitted triangular region $k + t > T$ in L_T .

41.1 A character limit at a growing boundary

We use the unitary formula (40.3) and the local coordinates in Lemma 40.1. Under $y = \sqrt{m}\theta$, the normalized trace-weighted probability measure converges on the centered plane $y_1 + y_2 + y_3 = 0$ to density

$$Z^{-1} e^{-\sum_i y_i^2/3} \Delta(y)^2 dy. \quad (41.5)$$

The same Gaussian domination used in that lemma justifies convergence against bounded functions with a small Gaussian tilt. Suppose $p/\sqrt{m} \rightarrow u \geq 0$ and q is fixed. Locally the normalized integrand is $r_p(\theta)\overline{r_q(\theta)}$. On compact y sets, $(P(y/\sqrt{m})/3)^{-p} \rightarrow 1$, since $\log(P/3) = O(|y|^2/m)$; also $r_q \rightarrow 1$. The factor $h_p(e^{iy/\sqrt{m}})/B_p$ is the characteristic function of a uniform weak composition $\delta_1 + \delta_2 + \delta_3 = p$, evaluated at $y/\sqrt{m}$. If $u > 0$, triangular-grid Riemann sums show that δ/p tends to the uniform probability law on the simplex, $X \sim \text{Dirichlet}(1, 1, 1)$. Hence

$$\frac{h_p(e^{iy/\sqrt{m}})}{B_p} \longrightarrow \mathbb{E}_X e^{iuX \cdot y}.$$

For $u = 0$, the same limit follows from $|\delta \cdot y|/\sqrt{m} \leq p|y|/\sqrt{m} \rightarrow 0$, including bounded p .

This passage is global. On the small chart, $|h_p|/B_p \leq 1$ and $|(P/3)^{-p-q}| \leq e^{C(p+q)Q}$; as $p+q = O(\sqrt{m})$, a fixed part of the saddle Gaussian absorbs this tilt. Off the chart the original polynomial integrand has normalized bound $Km^4 \rho^{2m-p-q}$ and is exponentially negligible. Thus dominated convergence gives

$$\frac{F_m(p, q)}{A_m d_p d_q} \longrightarrow \mathbb{E}_Y \mathbb{E}_X e^{iuX \cdot Y}, \quad (41.6)$$

where Y has density (41.5).

To evaluate the transform, let G be a traceless Hermitian 3 by 3 matrix with density proportional to $e^{-\text{tr}(G^2)/3}$ on its eight-dimensional real vector space, equipped with inner product $\text{tr}(AB)$. The Weyl change of variables gives precisely (41.5) as its eigenvalue density. For an independent Haar-uniform complex unit vector v , the squared coordinates in an orthonormal basis have the Dirichlet law above. This follows by normalizing three independent complex Gaussian coordinates: their squared moduli are independent exponentials, and division by their sum gives the uniform simplex law. Conditional on the eigenvalues Y , $v^* G v$ therefore has the law of $X \cdot Y$.

Unitary invariance makes its averaged law the law of G_{11} . The orthonormal real coordinates of G are centered Gaussians of variance $3/2$, so

$$\text{Var}(\text{tr}(AG)) = \frac{3}{2} \text{tr}(A_0^2), \quad A_0 = A - \frac{\text{tr} A}{3} I.$$

For $A = \text{diag}(1, 0, 0)$, the final trace is $2/3$. Thus G_{11} is a real centered Gaussian of variance one, and

$$\frac{F_m(p, q)}{A_m d_p d_q} \longrightarrow e^{-u^2/2} \quad (p/\sqrt{m} \rightarrow u, \ q \text{ fixed}). \quad (41.7)$$

This convergence is uniform when $p/\sqrt{m}$ ranges over a fixed compact interval: a contrary sequence would have a subsequence with convergent ratios, contradicting the sequential limit and continuity of the target.

Apply (41.7) to the two terms defining $H_h(k, j)$ for fixed j . The fixed-shift avoidance ratios give

$$\frac{H_h(k, j)}{A_h} = C(k, j) e^{-u^2/2} + o(\gamma_k).$$

Here d_{k+1}, d_k, γ_k are uniformly comparable. The triangular bound gives the uniform summable majorants

$$\frac{H_h(k, j)}{A_h \gamma_k} \leq K(j+1)^2 3^{-j}, \quad \frac{|C(k, j)|}{\gamma_k} \leq K(j+1)^2 3^{-j}.$$

Since $k = O(\sqrt{h})$, $s = h - k \rightarrow \infty$. Truncate at a fixed j , pass to the limit, then remove the truncation by these bounds. This proves (41.2), again compact-uniformly in $k/\sqrt{h}$.

41.2 The endpoint density

Recall $w(t, k) = 9^{-t} \gamma_k J_k(t)$. If $k/\sqrt{t} \rightarrow u > 0$, the profile, Regev's equivalent, and $\gamma_k \sim ak^2 3^{-k}$ give

$$t^2 w(t, k) \longrightarrow C_0 a^2 u^4 e^{-u^2/2}. \quad (41.8)$$

For $u = 0$, the global bound $w(t, k) \leq K(k+1)^4/(t+k+1)^4$ gives the same zero limit. The convergence is compact-uniform by sequential compactness.

Splitting the j sum of (36.6) at $j = k/2$ yields

$$w(t, k) \leq \frac{K(k+1)^4}{(t+2)^2(t+k+3)^2} e^{-ck^2/(t+k+3)} + \frac{K(k+1)^6 3^{-k/2}}{(t+2)^2}. \quad (41.9)$$

Indeed the first region uses the geometrically summable j factor; in the other, the tail of $(j+1)^2 3^{-j}$ supplies the displayed geometric factor in k . The second term sums to $O(t^{-2})$. For $k \leq t+1$, the first is at most

$$Kt^{-2} \left(\frac{k+1}{\sqrt{t}} \right)^4 e^{-c'k^2/t},$$

with bounded small t treated separately. For $k > t+1$, its $t^{3/2}$ -scaled total tends to zero exponentially. These estimates justify the full Riemann sum, giving $t^{3/2} e_t \rightarrow C_0 a^2 \int_0^\infty u^4 e^{-u^2/2} du = D$. The integral is $3\sqrt{\pi/2}$. Summing the positive power tail gives $\sum_{t>T} e_t \sim 2DT^{-1/2}$ and the coefficient in (41.3).

41.3 Matching the interior to the endpoints

Fix $\delta \in (0, 1/4)$ and orient the convolution by $t \leq s$ as before. For its interior part take $t \geq \delta n$. If $t/n \rightarrow x \in [\delta, 1/2]$ and $k/\sqrt{n} \rightarrow u$, both half indices $h_1 = s + k = n - t - 4$ and $h_2 = t + k$ tend to infinity linearly, and (41.2) gives

$$n^2 \frac{J_k(s) J_k(t)}{A_n} \longrightarrow C_0 a^2 9^{-4} u^4 [x(1-x)]^{-4} e^{-u^2/(2x(1-x))}. \quad (41.10)$$

For $u = 0$, the global bound gives $n^2 J_k(s) J_k(t)/A_n \leq K_\delta (k+1)^4/n^2 \rightarrow 0$, so this boundary case has the same limit. The convergence is uniform on compact x, u ranges. The exponential size factor leaves $9^{-4} 9^k \gamma_k^2$, explaining both the constant and the scale in this display.

By (37.1), the summand is bounded by $Kw(t, k)$. For $\delta n \leq t \leq n/2$, the first term of (41.9) with $k \leq t+1$ is at most $K_\delta n^{-2} (1 + k/\sqrt{n})^4 e^{-c_\delta k^2/n}$. The sum over $k > M\sqrt{n}$ and this t range, multiplied by $\sqrt{n}$, tends uniformly to zero as $M \rightarrow \infty$. The geometric term of (41.9) has total $O_\delta(n^{-1})$, and $k > t+1$ is exponentially negligible. Thus the two-dimensional Riemann sum is justified with its entire k range. Its diagonal $s = t$ has only $O(M\sqrt{n})$ terms for $k \leq M\sqrt{n}$, each $O_{\delta, M}(n^{-2})$; the same tail argument removes the rest. The upper bound $t \leq (n - 4 - k)/2$ shifts the scaled endpoint only by $o(1)$ for bounded $k/\sqrt{n}$.

Let $B_{\text{mid}}(n, \delta)$ denote the normalized oriented sum over $t \geq \delta n$, with multiplicity $2 - \mathbf{1}_{s=t}$. Integration in u in (41.10) gives

$$\sqrt{n} B_{\text{mid}}(n, \delta) \longrightarrow 2D9^{-4} \int_{\delta}^{1/2} [x(1-x)]^{-3/2} dx. \quad (41.11)$$

For the endpoint part, the quantitative proof of Theorem 40.3, now summed only to $t < \delta n$, gives

$$\left| B_{\text{end}}(n, \delta) - 2 \cdot 9^{-4} \sum_{t < \delta n} e_t \right| \leq K \frac{\sqrt{\delta}}{\sqrt{n}} + O(n^{-1}), \quad (41.12)$$

with K independent of $\delta \in (0, 1/4)$. To see every omitted part explicitly, the local approximation error is at most $(K/n) \sum_{t < \delta n} ((t+1)e_t + M_1(t))$, which has the displayed bound. Missing k values satisfy $k \geq n/3$ for large n and have w mass at most $Kn^{-2} \sum_{t < \delta n} M_1(t) = O(\sqrt{\delta} n^{-3/2}) + O(n^{-2})$. The endpoint diagonal satisfies the same estimate.

The endpoint equivalent gives $\sqrt{n} \sum_{t \geq \delta n} e_t \rightarrow 2D\delta^{-1/2}$. Subtract $R = 2 \cdot 9^{-4} \sum_t e_t$ and combine this with (41.11)–(41.12). The possible scaled discrepancy is bounded by the absolute value of

$$2D9^{-4} \left(\int_{\delta}^{1/2} [x(1-x)]^{-3/2} dx - 2\delta^{-1/2} \right)$$

plus $O(\sqrt{\delta})$. An antiderivative is $2(2x-1)/\sqrt{x(1-x)}$, so the parenthesis equals

$$\frac{2(1-2\delta)}{\sqrt{\delta(1-\delta)}} - 2\delta^{-1/2} = O(\sqrt{\delta}).$$

Letting $\delta \downarrow 0$ proves $\limsup_{n \rightarrow \infty} \sqrt{n} |b_n/A_n - R| = 0$. The avoidance rate (40.2) gives the coefficient statement in (41.4). Finally the logarithmic error is $o(n^{-1/2})$, and the inverse argument gives an inside-ceiling width $o((\log x)^{-1/2})$; the smooth-center discrepancy $O(\log \log x / \log x)$ is smaller still. This proves Theorem 41.1 without assigning a value to a divergent beta integral or assuming an unproved correction expansion.

42 Source status and reproducibility

Nakamura [13, Section 2.2, Theorem 2] gives exact catalytic functional equations and truncation algorithms for fixed 1243 occurrence counts. The first 15 values agree with the finite formula here. Conway–Guttmann [8, Section 5.2] conjecture the leading scale and estimate the amplitude numerically. Their differential approximants and extended series are numerical methods, not exact annihilating equations. No finite-dimensional scalar representation, present spine formula, or leading-amplitude proof was located in those reviewed sources. This is a scoped source assessment, not an exhaustive historical-priority claim.

A nearby target requires particular care. A224292 additionally avoids 1234 while having one 1243. Hou–Mansour [21, Example 6 and Theorem 7] already proved an algebraic generating function and an orientation swap for that class in 2008. It is not the unrestricted sequence treated here, and its generating function is not claimed as a new result. Likewise the general Noonan–Zeilberger P-recursiveness conjecture cannot be substituted for the concrete binomial proof above.

The source review covered the exact OEIS identifier, exactly-one-1243 queries, D-finiteness and amplitude queries, the cited primary papers, and the nearby joint-pattern result. It did not include

author correspondence, a complete MathSciNet review, or all unpublished theses. An absent OEIS formula alone does not establish novelty.

The reader bundle contains this PDF and its editable LaTeX source, a standard-library Python certificate, typed exact fixtures, and a manifest. The certificate independently enumerates small permutation objects and compares their marked images, reconstructed halves, and product counts; it evaluates the tableau sums, the support-reduced 23-index algebra, and the exact positive partial (37.2). It also checks finite strip interlacing and Gaussian-support diagnostics. The bundled README states the exact finite ranges and replay commands. These checks detect implementation and indexing errors. They do not prove an infinite limit, universal Gaussian constants, or historical priority; those roles belong to the proofs and primary-source assessment. Normal and optimized Python replay must give the same exact results. No third-party PDF or private research note is required to reproduce the bundle's checks.

Further questions

Four concrete tasks remain open. First, resolve whether $R = 149/160$, or obtain a rigorous numerical enclosure that distinguishes it. Second, make the finite- T endpoint tails effective with explicit constants and onset. Third, determine the first nonzero correction after the proved square-root cancellation; neither an $O(n^{-1})$ rate nor its coefficient has been established here. Fourth, compute an annihilating equation of usable order from the rational structure. The present formulas provide inputs to these questions, not their solutions.

42.1 Further questions and research

[write] **Status, 5 October 2026.** The four questions above are open: Section 66.1, items 2, 4, 7 and 5. Two further points of this Part are stated without proof or were not checked. The constant term of Section 43 is not asserted to be the rational diagonal whose existence Section 38 proves; whether it is one is open. The attribution of the algebraic generating function of A224292 to Hou and Mansour [21, Example 6 and Theorem 7] is the manuscript's own and was not checked in the write; the OEIS entry for A224292 cites Nakamura only (intake). Nothing in this Part was found to be wrong.

43 An explicit rational constant term

[write] **5 October 2026.** Delivered as Report 136's Appendix A; printed here as the last section of Part IV.

The BLS argument proves existence of an origin-regular rational diagonal. For a concrete rational integrand, we give a separate 40-variable constant term with explicit contour semantics. This Laurent-period integrand is not asserted to be that particular rational diagonal.

For a triple X put

$$\Sigma(X) = X_1 + X_2 + X_3, \quad \Delta(X) = (X_1 - X_2)(X_1 - X_3)(X_2 - X_3), \quad V(X) = \frac{\Delta(X)}{X_1^2 X_2}, \quad c(v) = \frac{1 + v}{v}.$$

One block uses the 20 variables

$$A, B, C; \ b; \ p_1, \dots, p_5; \ q_1, \dots, q_5; \ X_1, X_2, X_3; \ Y_1, Y_2, Y_3.$$

Take also an independent primed copy of these variables. In this appendix A, B, C are integration variables, unrelated to the earlier sequence or amplitude notation. Set $H = c(b)$, $T_r = c(p_r)$ and

$U_r = c(q_r)$. Define 11 rational weights by

$$\begin{aligned} R_s &= \frac{zH}{ABC}, & R_j &= \frac{C}{H}, \\ R_{\lambda_1} &= AT_1U_1, & R_{\lambda_2} &= \frac{AT_3U_3}{T_2U_2}, & R_{\lambda_3} &= \frac{AT_5U_5}{T_4U_4}, \\ R_{\mu_1} &= \frac{B\Sigma(X)T_2}{X_1T_1}, & R_{\mu_2} &= \frac{B\Sigma(X)T_4}{X_2T_3}, & R_{\mu_3} &= \frac{B\Sigma(X)}{X_3T_5}, \\ R_{\nu_1} &= \frac{C\Sigma(Y)U_2}{Y_1U_1}, & R_{\nu_2} &= \frac{C\Sigma(Y)U_4}{Y_2U_3}, & R_{\nu_3} &= \frac{C\Sigma(Y)}{Y_3U_5}. \end{aligned}$$

Let $\mathcal{R}$ be this list and put

$$K(z) = \frac{(1-A)V(X)V(Y)}{A^2BC} \prod_{R \in \mathcal{R}} \frac{1}{1-R}.$$

Define $K'(z)$ with the primed variables, and put $R_k = z/(ACA'C')$. Then

$$B(z) = z^4 \text{CT} \frac{K(z)K'(z)}{1-R_k}. \quad (43.1)$$

The constant term is over all 40 internal variables; z is not integrated. There are 23 geometric factors, eleven per block and one for the shared spine.

Extraction

Expand all geometric factors in nonnegative powers. The first block indices are s, j and its nine tableau coordinates; the shared factor has index k . Expand $1-A$ using $\alpha \in \{0, 1\}$ with sign $(-1)^\alpha$. The powers of the equality variables are

$$A : |\lambda| - s - k - 2 + \alpha, \quad B : |\mu| - s - 1, \quad C : |\nu| - s - k + j - 1.$$

Their constant terms enforce the three sizes in $F_{s+k+2}(k+1, j+1) - F_{s+k+1}(k, j)$. For every integer d , on any circle $|v| < 1$,

$$\text{CT}_v c(v)^d = [v^d](1+v)^d = \mathbf{1}_{d \geq 0}.$$

Thus H^{s-j} imposes $s \geq j$. The five T powers have exponents

$$\lambda_1 - \mu_1, \quad \mu_1 - \lambda_2, \quad \lambda_2 - \mu_2, \quad \mu_2 - \lambda_3, \quad \lambda_3 - \mu_3,$$

and the U powers give the corresponding exponents with ν . They impose both interlacings; $\mu_3, \nu_3 \geq 0$ already hold for geometric-series indices. Finally,

$$\begin{aligned} \text{CT}_X V(X) \prod_{r=1}^3 \left(\frac{\Sigma(X)}{X_r} \right)^{\mu_r} &= [X_1^{\mu_1+2} X_2^{\mu_2+1} X_3^{\mu_3}] \Delta(X) \Sigma(X)^{|\mu|} \\ &= f^\mu \end{aligned}$$

on partition inputs, by the six-term determinant calculation in Section 38. The strip indicators ensure such inputs. The Y extraction gives f^ν . The primed block provides the second H factor with indices t, l and the same k . The power of z is $4 + s + t + k$, proving (43.1) from (30.5).

Convergent contours

Give all dimension variables X_r, Y_r, X'_r, Y'_r radius one; every boundary or strip variable radius $1/2$; and each equality variable A, B, C, A', B', C' radius $\epsilon = 1/100$. Take $|z| < \epsilon^4/6$. On these circles $1 \leq |c(v)| \leq 3$ and $|\Sigma(X)/X_r| \leq 3$. Hence

$$|R_s| \leq 3|z|\epsilon^{-3} < \epsilon/2, \quad |R_j| \leq \epsilon, \quad |R_{\lambda_r}|, |R_{\mu_r}|, |R_{\nu_r}| \leq 9\epsilon < 1, \quad |R_k| < 1/6.$$

The primed estimates agree. All 23 geometric series converge absolutely and uniformly on this product torus. The remaining numerator is bounded there. The constant term means the normalized integral with measure $\prod_v dv/(2\pi i v)$; sum–integral interchange is justified. The integral is analytic near $z = 0$. At any fixed power z^n , the size conditions and $s + t + k = n - 4$ leave finitely many surviving indices, so analytic and formal interpretations agree. No arbitrary expansion at the simultaneous origin of the 40 variables is intended. This gives an explicit elimination input, without a recurrence-order or complexity claim and without evaluating R .

Part V

Exact formulas and leading asymptotics for unique 12345

Bundle Report 138, archive A224248_Exact_Formulas_Leading_Asymptotics_and_Inversion_Source.zip, delivered as Report138.tex (17 pages), dated 2 October 2026; title block “Exact formulas and leading asymptotics for unique 12345”, author line “Report 138”. Printed as delivered, with [write] notes; a section number is Report 138’s plus 43, and equations are renumbered within sections. Its table of contents is replaced by the report’s. Section 54.1 is added in the write.

Reading conventions of Part V	Elsewhere in the report
U_n : exactly one 12345 (A224248);	Parts I–IV: $A_n = \text{Av}_n(1234) $
$A_n = \text{Av}_n(12345) $ (A047889)	
R_5 the amplitude; $C_5 = C_A R_5$;	Parts I–III: R ; Part IV: R_{1243}
$C_A = 1536/\pi^{3/2}$	
$v = (i_2, i_3; j_2, j_3)$; P, R, Q, T (44.2)	Part I: P_k, Q_k ; R the amplitude elsewhere
$H_s(v)$ four-index half counts	Parts I–IV: $H_s(i, j)$
$K(v, w), \langle \cdot, \cdot \rangle_K; \langle \cdot, \cdot \rangle_4$	Parts II–III: $\langle \cdot, \cdot \rangle_K$ (two binomials)
$h(v)$; $w(v)$ a weight, w a vector	Part VI: $B(v), \omega(v)$
$p = 15/2$ from Section 49; $p = i_2 + i_3 + 2$ in Section 46	Parts I–IV: a boundary parameter
alphabets A, B, C, D (Section 47);	
$E = 2M - P - Q$	
$n_*(x), F(x), L = \log x$	other inverses

Abstract of this Part (as delivered). Let U_n count permutations with exactly one classical increasing subsequence of length five, OEIS A224248, and let A_n count permutations avoiding that pattern. We give an exact decomposition into two halves with a four-binomial gluing kernel. Explicit mutually inverse maps turn one half into a permutation with disjoint nested boundary restrictions. A six-term inverse-Cauchy formula counts this class; its proof includes the tableau-saturation argument needed to impose the empty corner. The resulting fixed-dimensional binomial sum proves that the ordinary generating function is a rational diagonal and that (U_n) is P-recursive. A uniform anisotropic boundary estimate and an endpoint convolution argument prove $U_n/A_n \rightarrow R_5$, where R_5 has an absolutely convergent series of nonnegative terms and is strictly positive. Consequently $U_n \sim (1536R_5/\pi^{3/2})16^n n^{-15/2}$. We also justify leading inverse thresholds with an $o(1)$ real error before integer rounding. No numerical value of R_5 , correction expansion, minimal recurrence, or worldwide novelty claim is made.

[write] **5 October 2026.** Part VI proves the correction expansion that this Part does not assert (Theorem 55.1); the “frozen leading-only baseline” of Part VI is this Part.

44 The sequence and the conclusions

An occurrence of the *classical* pattern 12345 is a choice of five positions whose values increase; the positions need not be adjacent. Let U_n be the number of permutations of $[n] = \{1, \dots, n\}$ with exactly one occurrence, and set $U_0 = \dots = U_4 = 0$. This is OEIS A224248 [6]. Its first nonzero terms are

$$1, 20, 270, 3142, 34291, 364462, 3844051, 40632886 \quad (n = 5, \dots, 12).$$

Let A_n count the avoiders, with $A_0 = 1$; these are OEIS A047889 [7]. Reversing positions interchanges increasing and decreasing patterns and does not change either count.

Nakamura and Zeilberger [11] gave functional-equation algorithms and many exact terms. Polynomial-time enumeration for a fixed pattern and occurrence count does not by itself imply a polynomial-coefficient recurrence. Bóna and Burstein [9], equation (6.1), already prove

$$U_n = \Theta(16^n n^{-15/2}). \quad (44.1)$$

Waite [10], Corollary 4.4, extends the corresponding order comparison to every fixed occurrence count. These cited order bounds are distinguished here from a leading equivalent: a two-sided Θ bound does not establish convergence of U_n/A_n . The known nonrationality result for this generating function is also consistent with rational diagonality. The cited even-length nonalgebraicity theorem does not decide algebraicity for length five.

[write] **Citations pinned, 5 October 2026.** Report 138 gives no citation for the two preceding sentences. They are [9, Theorem 5.3]: for every $k \geq 3$ the generating function of the permutations with exactly one copy of $k \cdots 21$ is not rational, which by reversal covers one 12345; and [9, Theorem 6.2], which proves nonalgebraicity for even $k > 2$ only. [10, Corollary 4.4] extends both to every fixed number of copies (not rational for odd k , not algebraic for even k). All three were checked against the arXiv texts (2101.00332v3 and 2506.05712v2). The obstruction of [9, Lemma 6.3] needs an integer exponent $m > 1$, so it does not apply to the exponent $15/2$ of (44.1); algebraicity for length five is open (question 4 of Section 54).

Here is an overview of the quantities that occur in our conclusions. Their combinatorial and algebraic definitions are proved below. A boundary vector is $v = (i_2, i_3; j_2, j_3) \in \mathbb{N}_0^4$, and

$$P = i_2 + i_3 + 1, \quad R = i_3 + 1, \quad Q = j_2 + j_3 + 1, \quad T = j_2 + 1. \quad (44.2)$$

The integer $H_s(v)$ counts northern halves with s extra points. For $w = (k_2, k_3; l_2, l_3)$ put

$$K(v, w) = \binom{i_2 + l_2}{i_2} \binom{i_3 + l_3}{i_3} \binom{j_2 + k_2}{j_2} \binom{j_3 + k_3}{j_3}, \quad \langle X, Y \rangle_K = \sum_{v, w \in \mathbb{N}_0^4} X(v) Y(w) K(v, w). \quad (44.3)$$

In particular $K(v, w) = K(w, v)$. The convention is that a southern half is obtained by *transposing* a northern half. It involves no reverse-complement or reversal of boundary indices.

For a partition α , s_α is its Schur function, and $f^{\alpha/\rho}$ counts standard tableaux of skew shape α/ρ . It is zero unless $\rho \subseteq \alpha$. Define

$$G_{P,R} = \sum_{\substack{\alpha \vdash P \\ \ell(\alpha) \leq 2}} f^{\alpha/(R)} s_\alpha, \quad \mathcal{T} = \{\emptyset, (1), (2), (1, 1), (2, 1), (2, 2)\}. \quad (44.4)$$

The adjoint $s_\tau^\perp$ and truncated inner product are defined in Section 47. Evaluation at $1^4 = (1, 1, 1, 1)$ is ordinary four-variable character dimension. Put

$$h(v) = 256 4^{-P-Q} \sum_{\tau \in \mathcal{T}} (-1)^{|\tau|} (s_\tau^\perp G_{P,R})(1^4) (s_\tau^\perp G_{Q,T})(1^4). \quad (44.5)$$

Although this finite formula has signs, the function h is nonnegative.

Theorem 44.1. *The sequence (U_n) is a multiple binomial sum over $\mathbb{Q}$. Its ordinary generating function is the diagonal of a rational power series, and (U_n) is P -recursive. Moreover*

$$R_5 = 2 \cdot 16^{-5} \sum_{t \geq 0} 16^{-t} \langle h, H_t \rangle_K \quad \text{converges absolutely,} \quad R_5 \geq \frac{15}{32768} > 0, \quad (44.6)$$

and

$$\frac{U_n}{A_n} \rightarrow R_5, \quad U_n \sim C_5 16^n n^{-15/2}, \quad C_5 = \frac{1536 R_5}{\pi^{3/2}}. \quad (44.7)$$

For $x > 0$ define the first integer threshold $n_*(x) = \min\{n \geq 0 : U_n \geq x\}$. With $L = \log x$ and $\lambda = \log 16$, let

$$F(x) = \frac{L + \frac{15}{2} \log L - \frac{15}{2} \log \lambda - \log C_5}{\lambda}. \quad (44.8)$$

There is a real function $\varepsilon(x) \rightarrow 0$ such that, for all sufficiently large x ,

$$n_*(x) = \lceil F(x) + \varepsilon(x) \rceil. \quad (44.9)$$

This is an error statement inside the ceiling; replacing it by $\lceil F(x) \rceil$ is not justified.

All combinatorial reductions and the estimates needed for the ratio limit are included in this article. Standard RSK, symmetric-function identities, Regev's avoidance asymptotic, and the binomial-sum theorem are the explicitly cited general inputs. The contribution is stated relative to the cited bounds and the derivation given here; an exhaustive literature-priority claim is not intended.

45 Cell geometry and exact gluing

Represent a permutation by its points, ordered left to right by position and bottom to top by value. In a permutation counted by U_n , mark its unique increasing chain $c_1, \dots, c_5$ in position order. For any extra point x , let a be the number of marked positions before x , and b the number of marked values below x . Thus $(a, b) \in \{0, \dots, 5\}^2$.

Lemma 45.1. *Every extra point satisfies $|a - b| \geq 2$. The ten cells $b \geq a + 2$ are called northern, and the ten cells $a \geq b + 2$ southern.*

Proof. Together with x , the first $\min(a, b)$ marked points and the last $5 - \max(a, b)$ marked points form an increasing chain of length $6 - |a - b|$. If $|a - b| \leq 1$, this chain contains a five-point subchain involving x , contrary to uniqueness. $\square$

A *northern half* is the standardized restriction to the marked points and all northern extras. Standardization replaces each value by its rank, preserving both orders. The half has exactly the marked five-chain. Define $H_s(i_2, i_3; j_2, j_3)$ by counting such halves with s extras, i_r extras in position gap r , and j_r extras in value gap r , for $r = 2, 3$. These four sets are pairwise disjoint: a northern point in position gap 2 or 3 has value gap at least 4, while a northern point in value gap 2 or 3 has position gap at most 1. Therefore

$$H_s(v) = 0 \quad \text{unless } i_2 + i_3 + j_2 + j_3 \leq s. \quad (45.1)$$

The all-marked half gives $H_0(0) = 1$.

Lemma 45.2. *Gluing any northern and southern halves along their marked chains, with any order-preserving shuffles in common position and value gaps, creates no additional increasing five-chain. In fact, an increasing chain using both kinds of extras has length at most three.*

Proof. First exclude three alternating runs. If a northern point in cell (a, b) precedes a comparable southern point in cell (c, d) , then $b \leq d$ and

$$c \geq d + 2 \geq b + 2 \geq a + 4 \geq 4.$$

A subsequent comparable northern point would have position gap at least 4, impossible because northern position gaps are at most 3. Thus an N, S, N triple of extras is impossible. Transposition excludes S, N, S . Any mixed chain has just one switch of type, ignoring marked points between its extras.

Suppose the switch is from north to south. Let x be the last northern extra, in value gap b , and y the first southern extra, in value gap d . Comparability gives $b \leq d$. Let L be the length of the chain up to and including x , and J its length from y onward. The first segment is entirely in the northern half. Appending $c_{b+1}, \dots, c_5$ gives a chain involving an extra, so $L + 5 - b \leq 4$. The last segment is entirely in the southern half; prepending $c_1, \dots, c_d$ similarly gives $d + J \leq 4$. There are at most $d - b$ marked points between x and y . The length of the whole chain is therefore at most

$$(b - 1) + (d - b) + (4 - d) = 3.$$

Transposition gives the other orientation. □

Theorem 45.3 (Four-binomial gluing). *For $n \geq 5$,*

$$U_n = \sum_{s+t=n-5} \langle H_s, H_t \rangle_K. \quad (45.2)$$

Proof. Take two northern halves with vectors v and w , and transpose the second to make a southern half. Northern and southern position gaps overlap precisely in 2 and 3. Their sizes there are respectively (i_2, i_3) and (l_2, l_3) . The value gaps overlap in 2 and 3 as well, with sizes (j_2, j_3) and (k_2, k_3) . Independently interleaving the two internal orders in each of these four gaps gives exactly $K(v, w)$ choices. All other cross-half comparisons are forced.

Two total orders on the same points, one for positions and one for values, specify one permutation. A point can participate in both a position shuffle and a value shuffle without any compatibility restriction: prescribing the two orders independently is precisely how a permutation diagram is specified. Lemma 45.2 proves that every such gluing is valid. Conversely, the unique marked chain in a resulting permutation identifies every northern and southern extra, and restricting the two total orders recovers both halves and all four shuffles. The construction is thus a bijection. There are $n - 5 = s + t$ extras in total. □

The exclusion of three runs uses the length-five geometry. It already fails in the analogous cell geometry for length six. No general arbitrary-length gluing theorem is inferred here.

46 Two explicit boundary bijections

The boundary locations are part of each object. Forgetting a boundary flag could identify different objects and would change the count. All positions, offsets, and ranks in this section are one-based.

46.1 A single boundary intersection

Fix s and v , and put

$$m = s + 3, \quad p = i_2 + i_3 + 2, \quad q = j_2 + j_3 + 2.$$

Delete the marked points c_2, c_4 from a northern half and standardize the result to a permutation $\beta \in S_m$.

Proposition 46.1. *This deletion is a bijection onto the boundary-marked permutations (β, p, q) satisfying the following conditions:*

1. β avoids 12345;
2. its final p entries avoid 123;
3. its q smallest values, in position order, avoid 123;
4. the two specified sets meet in exactly one point, the pivot;
5. the pivot has rank $j_2 + 2$ and offset $i_2 + 1$ in the final- p block.

In particular the pivot is neither the global minimum nor the final entry. The inverse inserts a new rank 2 immediately before the specified suffix, then a new rank $q + 2$ immediately before the final entry, shifting old ranks upward at each insertion as needed.

Proof. Northernness implies that c_1, c_2 are the two smallest values and c_4, c_5 are the final two entries. After deletion the specified suffix consists of the position-gap-2 extras, c_3 , the position-gap-3 extras, and c_5 . The specified low-value set consists of c_1 , the value-gap-2 extras, c_3 , and the value-gap-3 extras. Only c_3 is common to them, and the claimed pivot offset and rank follow.

A five-chain in β would already be a second five-chain in the original half. Any increasing triple in the suffix, preceded by c_1, c_2 , would likewise give a second five-chain; the original marked triple c_3, c_4, c_5 is unavailable because c_4 was deleted. The low-value condition follows in the same way by appending c_4, c_5 , since c_2 was deleted.

For the converse perform the two stated insertions, calling the new points c_2, c_4 . Let c_1 be the old minimum, c_3 the pivot, and c_5 the old final point. The minimum lies outside the suffix, since it is not the unique intersection; hence it precedes c_2 . Similarly c_5 lies outside the low-value set and is above c_4 . The five marked points form an increasing chain.

Any other five-chain must use at least one inserted point. If it uses c_2 but not c_4 , it has at most one smaller predecessor and would need three old suffix points in increasing order, a contradiction. If it uses c_4 but not c_2 , it has at most one successor and would need three increasing old low-value points, also a contradiction. If it uses both, it has at most one point before c_2 and at most one after c_4 . The only point lying between them in both coordinates is the pivot, by the unique-intersection condition. Thus its five points are exactly the marked chain.

It remains to check northernness. Every extra has value gap at least 2 and position gap at most 3. Extras in position gaps 2 and 3 are outside the low-value block, hence in value gaps at least 4. The only remaining potential violations are cells (1, 2) and (3, 4). The former would make c_1, x, c_3 an increasing triple in the old low-value block; the latter would make c_3, x, c_5 an increasing triple in the old suffix. Both are excluded. All extras are northern. Deletion and the prescribed insertions recover the same relative positions and ranks, proving both inverse identities. $\square$

46.2 Disjoint nested boundaries

Now delete the pivot from β , and standardize to $\gamma \in S_M$, where $M = s + 2$. Use the parameters in (44.2). Denote by $D_M(P, R; Q, T)$ the number of permutations γ with the following six properties:

1. γ avoids 12345;
2. its final P entries avoid 123;
3. its final R entries decrease;
4. its Q smallest values, in position order, avoid 123;
5. its T smallest values, in position order, decrease;
6. its final- P set and smallest- Q set are disjoint.

The natural range is

$$1 \leq R \leq P, \quad 1 \leq T \leq Q, \quad P + Q \leq M. \quad (46.1)$$

We define this count to be zero outside that range when using it for halves.

Theorem 46.2 (Nested boundary bijection). *For (44.2) and $M = s + 2$,*

$$H_s(v) = D_{s+2}(P, R; Q, T). \quad (46.2)$$

The inverse from γ to β inserts a new point of rank $T + 1$ immediately before the final R entries. Composing this with the two insertions of Proposition 46.1 recovers the northern half.

Proof. After pivot deletion, avoidance and disjointness are inherited. In β the pivot is smaller than every other suffix entry, so an increasing pair among the R entries following it would contradict suffix 123-avoidance. Those entries therefore decrease. Similarly, all T values smaller than the pivot occur before it and must decrease, or an increasing pair would extend to a triple ending at the pivot. This proves the forward conditions.

Conversely, insert rank $T + 1$ before the last R entries. The old suffix consists only of values above all old low-block values, by disjointness. Every old low-block entry is before the old suffix. The inserted point is thus the unique intersection of the enlarged suffix of size $P + 1$ and enlarged low-value block of size $Q + 1$. It is neither the minimum nor the final entry. In the enlarged suffix it is the minimum; any new triple through it would require an increasing pair in the final R entries. In the enlarged low block it is the last point; any new triple would require an increasing pair among the T smaller values. Neither is possible.

More generally, a chain through the pivot has at most one smaller predecessor *within that increasing chain*, and at most one larger successor, by the two decreasing conditions. Its length is at most three. Inserting the pivot cannot create a five-chain. Thus all of Proposition 46.1's conditions hold. Its offset is $P - R + 1 = i_2 + 1$, its rank is $T + 1 = j_2 + 2$, and the remaining statistics are $i_3 = R - 1$ and $j_3 = Q - T$. The operations are exact mutual inverses. $\square$

47 The six-term inverse-Cauchy formula

We give the coupling argument explicitly. Disjointness of the two boundaries cannot be imposed by simply multiplying independent RSK factors.

47.1 Symmetric functions and reversed positions

In the Hall inner product the Schur functions are orthonormal. The operator $s_\tau^\perp$ is adjoint to multiplication by s_τ ; equivalently,

$$s_\tau^\perp s_\alpha = \sum_\nu c_{\tau,\nu}^\alpha s_\nu,$$

where $c_{\tau,\nu}^\alpha$ are Littlewood–Richardson coefficients. Write $p_1 = s_{(1)}$, and define the truncated inner product by

$$\langle s_\rho, s_\eta \rangle_4 = \begin{cases} 1 & \rho = \eta, \ell(\rho) \leq 4, \\ 0 & \text{otherwise.} \end{cases}$$

Standard tableaux are equivalently saturated paths in Young’s lattice, so

$$[s_\rho] p_1^{|\rho|-|\nu|} s_\nu = f^{\rho/\nu}. \quad (47.1)$$

Reverse the positions of γ , writing $\sigma_i = \gamma_{M+1-i}$. Then σ avoids a decreasing subsequence of length five. Its initial P entries and its subsequence of the Q smallest values avoid 321; its initial R entries and subsequence of its T smallest values increase. Finally its first- P -position by lowest- Q -value rectangle is empty.

Under ordinary RSK, longest decreasing subsequence length is the number of rows of the common shape. The initial P recording-tableau labels have a shape α with at most two rows; the first R labels have single-row shape (R) . The number of their standard fillings is $f^{\alpha/(R)}$, since the initial row has one filling. The analogous insertion-tableau factor is $f^{\beta/(T)}$. The empty rectangle couples these two boundary data.

47.2 A literal matrix corner

Use ordered alphabets $A = (a_1, a_2)$ and $B = (b_1, b_2)$ of exactly two variables each, and arbitrary remaining alphabets C, D . Rows indexed by A precede rows indexed by C , and columns indexed by B precede columns indexed by D . For a nonnegative integer matrix, a decreasing chain uses strictly increasing row indices and strictly decreasing column indices, with positive entries in its cells. Matrix RSK gives the bounded-height series

$$\mathcal{K}_4(A + C, B + D) = \sum_{\ell(\rho) \leq 4} s_\rho(A + C) s_\rho(B + D). \quad (47.2)$$

Each matrix entry in row variable x and column variable y has weight $(xy)^m$ when it equals m . The equality follows from the RSK weight correspondence and the decreasing-chain theorem.

Let $\mathcal{E}_4(A, B; C, D)$ count the same matrices with the A -by- B corner forced to zero.

Lemma 47.1 (Harmless corner). *Restoring arbitrary entries in the two-by-two corner cannot create a decreasing chain of length five. Consequently*

$$\mathcal{E}_4(A, B; C, D) = \prod_{a \in A, b \in B} (1 - ab) \mathcal{K}_4(A + C, B + D). \quad (47.3)$$

Proof. A decreasing chain through a corner cell has at most two points up to and including that cell, because only two initial rows are available, and at most two points from that cell onward, because only two initial columns are available. Its total length is at most $2 + 2 - 1 = 3$. Any forbidden chain would therefore avoid the entire corner. Its four entries may be restored independently, contributing the geometric factors $\prod (1 - ab)^{-1}$. $\square$

This uses full rectangular matrix RSK. It does not invoke a Ferrers-board chain theorem for chains that may cross a missing corner. That distinction matters: on the board with row lengths $(2, 1)$, rooks at $(1, 2), (2, 1)$ form a global decreasing pair, but their bounding rectangle is not in the board. Ferrers chain statistics in the relevant growth-diagram theorem [22] require the bounding rectangle to be contained in the board.

47.3 Why the initial fillings can be summed independently

The next lemma supplies the saturation step behind coefficient extraction. It is not an assumption of independence between the two full tableaux.

Lemma 47.2 (Initial-filling saturation). *Fix the initial recording shape α for alphabet A , the initial insertion shape β for alphabet B , and the two outer skew tableaux filled with labels from C and D . Among the full RSK pairs with these data, emptiness of the A -by- B matrix block is independent of the fillings of the two initial shapes. The assertion holds for standard tableaux and for the semistandard tableaux of matrix RSK.*

Proof. Run inverse row insertion in the order specified by all recording labels in C , leaving those in A until later. The removable cells and their order are determined by the outer recording skew tableau; with repeated recording labels, the usual reverse-RSK tie order is used. The actual A filling is never consulted during this phase.

Call insertion entries from B special and those from D nonspecial. Every special entry is smaller than every nonspecial entry; in each row, the special cells form an initial segment. Track only the nonspecial entries and the shape occupied by special entries. At a reverse insertion, the carried entry and its next replacement follow the usual rule: in the preceding row replace the rightmost entry strictly smaller than the carried entry, and carry the displaced entry upward.

While the carried entry is nonspecial, if a smaller nonspecial entry exists in the next row, the rightmost such entry and its value depend only on the nonspecial data. If none exists, the selected entry is the rightmost special entry; its position is determined solely by the special-shape boundary. Replacing it by the nonspecial carried entry changes that boundary in a prescribed way and makes the next carried entry special. Once the carried entry is special, every subsequently displaced entry is special. Replacements then keep special cells special and do not change the nonspecial data or the special-shape boundary. Which special values are exchanged is irrelevant to the tracked data. The case where the first removed entry is already special has the same conclusion.

Induction through all reverse insertions with recording labels in C proves that the remaining nonspecial data and special shape, as well as the special or nonspecial type of every emitted value, are independent of the initial special filling. The remaining insertion tableau represents exactly the rows labeled by A . The A -by- B block is empty if and only if this tableau has no special entry. This proves independence from the B filling, and the first paragraph proved independence from the A filling. The argument uses only the order separation $B < D$ and the usual reverse-row-insertion rule, so it applies in both standard and semistandard cases. $\square$

It follows that the expansion

$$\mathcal{E}_4(A, B; C, D) = \sum_{\ell(\alpha), \ell(\beta) \leq 2} s_\alpha(A) s_\beta(B) E_{\alpha, \beta}(C, D) \quad (47.4)$$

has the following meaning: each outer pair allowed by the empty corner admits every initial filling of its specified shapes. Replacing an initial filling by any valid filling of the same shape preserves tableau validity across its interface: all initial labels are smaller than all outer labels. The two Schur

factors therefore count precisely those independent choices of fillings. Restricting a standard initial tableau of shape α to have first R labels in a row replaces its unrestricted filling count by $f^{\alpha/(R)}$, and similarly on the other side. This is also the empty-inner-corner instance of skew RSK [23]; the explicit reverse-insertion proof is included to identify the exact independence used here.

47.4 Coefficient extraction and the exact formula

Dual Cauchy in two variables on both sides gives

$$\prod_{a \in A, b \in B} (1 - ab) = \sum_{\tau \subseteq (2,2)} (-1)^{|\tau|} s_{\tau}(A) s_{\tau'}(B). \quad (47.5)$$

Both two-variable restrictions are essential: $s_{\tau}(A)$ requires at most two rows, and $s_{\tau'}(B)$ at most two columns. The six partitions are exactly $\mathcal{T}$ in (44.4).

Theorem 47.3 (Six-term formula). *For the range (46.1),*

$$D_M(P, R; Q, T) = \sum_{\tau \in \mathcal{T}} (-1)^{|\tau|} \langle p_1^{M-P} s_{\tau}^{\perp} G_{P,R}, p_1^{M-Q} s_{\tau'}^{\perp} G_{Q,T} \rangle_4. \quad (47.6)$$

Proof. Insert (47.2) and (47.5) into (47.3), and branch $s_{\rho}(A + C) = \sum_{\nu} s_{\nu}(A) s_{\rho/\nu}(C)$, with the analogous expansion in B, D . For $|\alpha| = P$, extracting the $s_{\alpha}(A)$ coefficient and weighting it by $f^{\alpha/(R)}$ moves multiplication by $s_{\tau}(A)$ to its adjoint $s_{\tau}^{\perp} G_{P,R}$. Initial-filling saturation, Lemma 47.2, proves that this weighted extraction imposes exactly the required initial R -label condition, even with the empty corner. The same extraction in B gives $s_{\tau'}^{\perp} G_{Q,T}$.

To standardize the remaining labels, use the Plancherel specializations $p_1(C) = z$, $p_j(C) = 0$ for $j > 1$, and $p_1(D) = w$, $p_j(D) = 0$ for $j > 1$. A skew Schur function of degree q becomes $f^{\rho/\nu} z^q / q!$. Extract $z^{M-P} w^{M-Q}$ and multiply by $(M-P)!(M-Q)!$. Equation (47.1) then gives exactly (47.6). In its τ term the common outer shape has size $M - |\tau|$: both homogeneous factors have that degree. The signs and size shifts are thus determined, rather than inferred from a factorization. $\square$

For $v = 0$, $P = R = Q = T = 1$ and $G_{1,1} = s_1$. Only $\tau = \emptyset, (1)$ survive, giving the useful exact check

$$H_s(0) = A_{s+2} - A_{s+1}. \quad (47.7)$$

The nonempty terms of (47.6) are the coupling correction. They cannot be suppressed merely because each boundary condition separately has a simple tableau interpretation.

48 A fixed-dimensional binomial sum

We explain why the exact expression belongs to the binomial-sum class, rather than just asserting that a finite tableau sum is holonomic.

For $\tau \in \mathcal{T}$ and a two-row partition ν , set

$$b_{P,R}^{\tau}(\nu) = \sum_{\substack{\alpha \vdash P \\ \ell(\alpha) \leq 2}} f^{\alpha/(R)} c_{\nu,\tau}^{\alpha}. \quad (48.1)$$

The six-term formula becomes the explicit fixed-height expansion

$$\sum_{\tau \in \mathcal{T}} (-1)^{|\tau|} \sum_{\substack{\rho \vdash M - |\tau| \\ \ell(\rho) \leq 4}} \left(\sum_{\substack{\nu \vdash P - |\tau| \\ \ell(\nu) \leq 2}} b_{P,R}^{\tau}(\nu) f^{\rho/\nu} \right) \left(\sum_{\substack{\eta \vdash Q - |\tau| \\ \ell(\eta) \leq 2}} b_{Q,T}^{\tau'}(\eta) f^{\rho/\eta} \right). \quad (48.2)$$

A sum over a negative size is empty. Partitions are padded with zeroes as necessary. For $\nu = (u, v)$, $\tau = (t_1, t_2)$, and $\alpha = (a, b)$, the two-row Littlewood–Richardson rule is

$$c_{\nu, \tau}^{\alpha} = 1 \iff \begin{aligned} |\alpha| &= |\nu| + |\tau|, & r &= u + t_1 - a = b - v - t_2, \\ 0 &\leq r \leq \min(u - v, t_1 - t_2). \end{aligned} \quad (48.3)$$

It is zero otherwise. This is the usual $GL(2)$ tensor-product rule; it also follows directly from two-row Littlewood–Richardson words. Thus it is an indicator of finitely many affine constraints.

For partitions ρ, ν with at most h rows, with h fixed, the skew-tableau determinant is

$$f^{\rho/\nu} = q! \det_{1 \leq i, j \leq h} \left(\frac{1}{(\rho_i - \nu_j - i + j)!} \right), \quad q = |\rho| - |\nu|. \quad (48.4)$$

This follows by applying the Plancherel specialization to the skew Jacobi–Trudi determinant: h_j becomes $z^j/j!$, while $s_{\rho/\nu}$ becomes $f^{\rho/\nu} z^q/q!$. Here a reciprocal factorial of a negative integer is zero, and a noncontained inner shape gives zero. In the determinant term indexed by $\pi \in S_h$, put $d_i = \rho_i - \nu_{\pi(i)} - i + \pi(i)$. Since $\sum_i d_i = q$, that term is

$$\text{sgn}(\pi) \binom{q}{d_1, \dots, d_h} = \text{sgn}(\pi) \prod_{j=1}^{h-1} \binom{q - \sum_{i < j} d_i}{d_j}, \quad (48.5)$$

when every $d_i \geq 0$, and zero otherwise. This cancellation of factorials is important: no arbitrary factorial quotient is being silently treated as a binomial sum.

Theorem 48.1. *The sequence $(U_n)_{n \geq 0}$ is a multiple binomial sum over $\mathbb{Q}$. Hence there exist an integer d and a rational power series $\mathcal{R}(x_1, \dots, x_d)$, regular at the origin, such that*

$$\sum_{n \geq 0} U_n z^n = \sum_{n \geq 0} [x_1^n \cdots x_d^n] \mathcal{R} z^n.$$

There are polynomials $q_0(n), \dots, q_r(n) \in \mathbb{Q}[n]$, not all zero, with $\sum_{j=0}^r q_j(n) U_{n+j} = 0$ for all sufficiently large n .

Proof. Use (48.3) and the determinant expansion (48.5) in (48.2). Every summand is a product of a fixed number of binomial coefficients with affine arguments and indicators of affine equalities and inequalities. The partition lengths are bounded by two or four, and the determinant permutations and τ range over fixed finite sets. Every coordinate can be bounded between 0 and $M + 4$; the number of coordinates is independent of M .

The gluing formula adds s and the eight boundary coordinates, with $t = n - 5 - s$, the affine support constraints, and the four binomial factors of K . All indices have affine upper bounds in n . Kronecker deltas and Heaviside functions imposing the affine constraints are binomial sums; products, affine substitutions, and bounded affine-index sums preserve that class [16, Definition 1.1 and Section 1]. This proves the first statement, including the finite zero initial segment. The equivalence between univariate binomial sums and rational diagonals is [16, Theorem 3.5]; P-recursiveness is its Corollary 3.6. $\square$

The proof is constructive in the sense of those closure operations. It does not supply a compact rational function or a minimal recurrence. The existence theorem is stronger than a numerical recurrence guess, and finite agreement with a guessed recurrence would not replace it.

[write] 5 October 2026. The check of Remark 13.2 applies, with one reading made explicit. In (48.5) the product runs over $j \leq h - 1$, so the condition $d_h \geq 0$ that the text requires (“when every $d_i \geq 0$, and zero otherwise”) must enter the summand as the indicator $h(d_h) = B(d_h, d_h)$ of Part II, just as Part II’s last factor $B(e_3, e_3)$ does. This Part describes its encoding in words; Parts II and IV display theirs.

49 Avoidance and a uniform boundary bound

Throughout the remaining proof, Haar measure on $U(4)$ has total mass one, and $p = 15/2$. RSK, transposition of shapes when necessary, and unitary character orthogonality give

$$A_m = \sum_{\substack{\rho \vdash m \\ \ell(\rho) \leq 4}} (f^\rho)^2 = \int_{U(4)} |\operatorname{tr} U|^{2m} dU. \quad (49.1)$$

Regev's bounded-height asymptotic [14], with its explicit normalization also recorded in [15, Section 3.1.4], specializes to

$$A_m \sim C_A 16^m m^{-p}, \quad C_A = \frac{1536}{\pi^{3/2}}. \quad (49.2)$$

For normalization, the standard h -row constant is

$$\frac{h^{h^2/2} \prod_{j=1}^{h-1} j!}{2^{(h^2+h-2)/2} \pi^{(h-1)/2}}; \quad h = 4: \quad \frac{4^8 (1!2!3!)}{2^9 \pi^{3/2}} = \frac{1536}{\pi^{3/2}}.$$

In particular there are global constants $0 < c < C < \infty$ with

$$c \frac{16^m}{(m+1)^p} \leq A_m \leq C \frac{16^m}{(m+1)^p} \quad (m \geq 0), \quad \frac{A_{m-d}}{A_m} \longrightarrow 16^{-d} \quad (49.3)$$

for every fixed integer shift d . The finite initial range is absorbed into c, C .

49.1 The aggregate two-row character bound

Dropping just disjointness from D_M leaves the two nested tableau restrictions, one in each RSK tableau. Thus the resulting count is the nonnegative empty- τ term of (47.6). It follows that

$$\begin{aligned} 0 \leq H_s(v) &\leq \langle p_1^{M-P} G_{P,R}, p_1^{M-Q} G_{Q,T} \rangle_4 \\ &\leq G_{P,R}(1^4) G_{Q,T}(1^4) \int_{U(4)} |\operatorname{tr} U|^{2M-P-Q} dU, \quad M = s + 2. \end{aligned} \quad (49.4)$$

The last inequality uses $|s_\alpha(U)| \leq s_\alpha(1^4)$ for a unitary character. No sign assumption on the five other correction terms is required.

For $\alpha = (a, b)$ with $a + b = P$, Weyl's formula gives

$$s_{(a,b)}(1^4) = \frac{(a-b+1)(a+2)(a+3)(b+1)(b+2)}{12} \leq \frac{(P+3)^5}{12}. \quad (49.5)$$

Moreover

$$\sum_{\substack{\alpha \vdash P \\ \ell(\alpha) \leq 2}} f^{\alpha/(R)} \leq 2^{P-R}. \quad (49.6)$$

Indeed a standard skew tableau specifies the successive rows used by its Young-lattice path from (R) ; there are at most two choices at each of the $P - R$ additions. This is a bound on the aggregate sum over endpoints, so no extra factor for the number of shapes is needed. Combining the last two displays bounds $G_{P,R}(1^4)$.

On the support (45.1), $P + Q \leq M$. For $E = 2M - P - Q$ we therefore have $M \leq E \leq 2M$. Let $r = \lfloor E/2 \rfloor$. The elementary bound $|\operatorname{tr} U| \leq 4$ and (49.3) give, including odd E ,

$$\int |\operatorname{tr} U|^E dU \leq 4^{E-2r} A_r \leq C \frac{4^E}{(M+1)^p}. \quad (49.7)$$

Proposition 49.1 (Anisotropic domination). *There is a constant C independent of s and v such that*

$$0 \leq H_s(v) \leq C A_s w(v), \quad w(v) = (P+3)^5 (Q+3)^5 2^{-i_2} 4^{-i_3} 4^{-j_2} 2^{-j_3}. \quad (49.8)$$

In addition

$$\sum_{v,w} w(v)w(w)K(v,w) < \infty, \quad \langle H_s, H_t \rangle_K \leq C A_s A_t. \quad (49.9)$$

Proof. Insert (49.5)–(49.7) into (49.4), and divide by the lower bound for A_s . Up to an absolute factor, the geometric dependence is

$$4^{-P-Q} 2^{P-R+Q-T} = 2^{-4} 2^{-i_2} 4^{-i_3} 4^{-j_2} 2^{-j_3}.$$

The fixed shift $M = s + 2$ is absorbed into C . Off support H_s is zero, giving the uniform assertion.

For kernel summability, each of its four factors pairs a coordinate with geometric weight $1/2$ and one with weight $1/4$. The nonnegative generating series

$$\sum_{a,b \geq 0} \binom{a+b}{a} x^a y^b = \frac{1}{1-x-y} \quad (x, y \geq 0, x+y < 1) \quad (49.10)$$

converges at $(x, y) = (1/2, 1/4)$ and at a slightly larger positive pair. Any fixed polynomial in a, b is absorbed by that enlargement. Each polynomial in sums of coordinates in $w(v)w(w)$ can also be bounded by a product of coordinate polynomials. Taking the product of the four convergent two-coordinate sums proves the first part of (49.9). The second follows from (49.8). $\square$

The unequal geometric weights are useful. A cruder symmetric replacement by $2^{-i_2-i_3-j_2-j_3}$ would put a paired kernel at $x+y=1$, losing this summability argument.

50 The positive half limit

We now identify the leading behavior for each fixed boundary vector. This requires concentration on the whole scalar circle, not just at the identity matrix.

Lemma 50.1 (Trace-weighted character limit). *Let F, E be fixed homogeneous symmetric polynomials of degrees d, e , evaluated on $U(4)$. As integers $r \rightarrow \infty$,*

$$\frac{1}{A_r} \int_{U(4)} (\text{tr } U)^{r-d} F(U) \overline{(\text{tr } U)^{r-e} E(U)} dU \longrightarrow \frac{F(1^4) \overline{E(1^4)}}{4^{d+e}}. \quad (50.1)$$

Only $r \geq \max(d, e)$ is used.

Proof. The maximum $|\text{tr } U| = 4$ occurs precisely at $\mathcal{S} = \{zI_4 : |z| = 1\}$. For any neighborhood of $\mathcal{S}$, compactness bounds the trace outside it by $4 - \eta$ for some $\eta > 0$. The lower bound in (49.3) implies that the probability measures

$$d\mu_r(U) = A_r^{-1} |\text{tr } U|^{2r} dU$$

concentrate on $\mathcal{S}$; the mass outside the neighborhood is bounded by a polynomial in r times $((4 - \eta)/4)^{2r}$.

On a sufficiently small neighborhood of $\mathcal{S}$, the function

$$\Psi(U) = \frac{F(U) \overline{E(U)}}{(\text{tr } U)^d (\text{tr } U)^e}$$

is continuous. Homogeneity gives $F(zI_4) = z^d F(1^4)$ and the analogous formula for E , so its value on all of $\mathcal{S}$ is $F(1^4)\overline{E(1^4)}4^{-d-e}$. This proves the desired limit inside the neighborhood. No division by zero is used outside it: bound the original numerator directly by $C(4-\eta)^{2r-d-e}$ and use the same polynomially corrected exponential suppression. The neighborhood can then shrink to $\mathcal{S}$. $\square$

Proposition 50.2. *For every fixed v ,*

$$\frac{H_s(v)}{A_s} \longrightarrow h(v), \quad 0 \leq h(v) \leq Cw(v), \quad h(0) = 240. \quad (50.2)$$

Proof. Fix a term τ in (47.6) and write $k = |\tau|$, $F = s_\tau^\perp G_{P,R}$, $E = s_\tau^\perp G_{Q,T}$. If either is zero there is nothing to prove. Otherwise their degrees are $d = P - k$, $e = Q - k$. The term is the Haar integral in (50.1) with $r = M - k$, since $r - d = M - P$ and $r - e = M - Q$. Here $M = s + 2$, so

$$\frac{A_{M-k}}{A_s} \longrightarrow 16^{2-k}, \quad 16^{2-k} 4^{-(P+Q-2k)} = 256 4^{-P-Q}.$$

The six terms therefore give exactly (44.5). Nonnegativity of H_s proves $h(v) \geq 0$ despite the alternating finite formula, and (49.8) passes to the limit. Finally (47.7) and (49.3) give $h(0) = 256 - 16 = 240$. $\square$

This proof identifies a nonnegative boundary limit but supplies no numerical summation of its infinite convolution amplitude. In particular it does not estimate R_5 by fitting large finite ratios.

51 Endpoint convolution and the leading equivalent

Set $N = n - 5$. For fixed t , dominated convergence in the boundary vectors, using (49.8) and (49.9), gives

$$\frac{\langle H_{N-t}, H_t \rangle_K}{A_{N+5}} \longrightarrow 16^{-t-5} \langle h, H_t \rangle_K. \quad (51.1)$$

Although H_t has finite support, the first vector ranges over an unbounded set as N grows; the weighted-kernel bound is what justifies that remaining limit interchange.

Theorem 51.1 (Positive ratio limit). *The series (44.6) converges absolutely and consists of nonnegative terms. It is strictly positive and equals $\lim_{n \rightarrow \infty} U_n/A_n$.*

Proof. The uniform estimate (49.9) and the two-sided avoidance bounds yield, for $0 \leq s \leq N$,

$$\frac{\langle H_s, H_{N-s} \rangle_K}{A_{N+5}} \leq C \frac{(N+6)^p}{(s+1)^p (N-s+1)^p}. \quad (51.2)$$

Fix an integer $L \geq 1$. On the middle range $L \leq s \leq N - L$, split at $N/2$. In the left part $(N+6)/(N-s+1)$ is bounded, and symmetry gives the same estimate for the right part. Hence

$$\sum_{s=L}^{N-L} \frac{\langle H_s, H_{N-s} \rangle_K}{A_{N+5}} \leq C \sum_{j \geq L} (j+1)^{-p}. \quad (51.3)$$

This tends to zero as $L \rightarrow \infty$, uniformly in all sufficiently large N , since $p = 15/2 > 1$.

For fixed L and $N > 2L$, the endpoint ranges $0 \leq s < L$ and $N - L < s \leq N$ are disjoint. Apply (51.1) to their finitely many terms. Symmetry of the kernel makes their limits equal, giving the

factor two. Let $N \rightarrow \infty$ first and then $L \rightarrow \infty$. Formula (45.2) and the vanishing middle bound prove (44.6) and the ratio limit.

Furthermore, $0 \leq h(v) \leq Cw(v)$ and the same kernel estimate give

$$0 \leq 16^{-t} \langle h, H_t \rangle_K \leq C 16^{-t} A_t \leq C(t+1)^{-p}.$$

Thus the series converges absolutely, with an analytical tail $O((L+1)^{1-p})$ after index L . The unspecified constant in this bound is not a numerical tail certificate. Finally H_0 is a unit mass at 0, $K(0,0) = 1$, and $h(0) = 240$. Retaining just $t = 0, v = w = 0$ gives

$$R_5 \geq 2 \cdot 16^{-5} \cdot 240 = \frac{15}{32768} > 0.$$

□

Combining this theorem with Regev's equivalent (49.2) proves (44.7). The exponent and exponential rate agree with the previously known Θ statement (44.1); the convergence to the specified positive amplitude is the additional conclusion of this argument. The small lower bound above is deliberately crude and is not presented as a useful decimal approximation.

[write] **5 October 2026.** Part VI proves the expansion to every order (Theorem 55.1). Numerical values, from the intake dossier and uncertified: floating-point partial sums of (44.6) through $t = 30$ reach 0.20407, a tail estimate suggests $R_5 \approx 0.2044 \pm 0.0003$, and Richardson extrapolation of exact ratios U_n/A_n for $28 \leq n \leq 40$ gives about 0.2040. The ratio itself is not monotone (0.166 at $n = 10$, a peak of about 0.3615 near $n = 30$, 0.3397 at $n = 40$). The two routes agree, but neither is an enclosure (Section 66.1, item 3).

52 Leading inversion with rounding justified

All logarithms are natural. We prove the last assertion of Theorem 44.1 without assuming any correction term or quantitative rate in (44.7).

Proposition 52.1. *Let $a_n > 0$ eventually and suppose $a_n \sim Ce^{\lambda n} n^{-p}$ with $C, \lambda > 0$ and fixed real p . If $n_a(x) = \min\{n \geq 0 : a_n \geq x\}$ is defined using a nonnegative finite initial segment as well, then, for large x ,*

$$n_a(x) = \lceil F_a(x) + o(1) \rceil, \quad F_a(x) = \frac{\log x + p \log \log x - p \log \lambda - \log C}{\lambda}.$$

The $o(1)$ is a real error before rounding. Equivalently, for every $\epsilon > 0$, all sufficiently large x satisfy

$$\lceil F_a(x) - \epsilon \rceil \leq n_a(x) \leq \lceil F_a(x) + \epsilon \rceil. \quad (52.1)$$

Proof. The assumed equivalent gives $a_{n+1}/a_n \rightarrow e^\lambda > 1$. Thus a_n is strictly increasing beyond some integer n_0 and tends to infinity. For x above the finite maximum of the earlier terms, the first threshold lies in this increasing tail.

Let $\ell(t)$ be the linear interpolation of $\log a_n$ between successive integers $n \geq n_0$. It is continuous and strictly increasing. Since

$$\log a_n = \lambda n - p \log n + \log C + r_n, \quad r_n \rightarrow 0,$$

interpolating gives

$$\ell(t) = \lambda t - p \log t + \log C + o(1) \quad (t \rightarrow \infty). \quad (52.2)$$

Indeed the interpolated r_n is bounded by the two endpoint absolute values, and the interpolation error for $\log t$ is $O(t^{-2})$ on each unit interval.

For large x there is a unique real $y(x)$ with $\ell(y(x)) = \log x$. Exact monotonicity of the interpolant implies

$$n_a(x) = \lceil y(x) \rceil, \quad (52.3)$$

including the case when x is exactly one of the a_n . Writing $L = \log x$, equation (52.2) first gives $y \sim L/\lambda$. It then yields $y = L/\lambda + O(\log L)$ and hence $\log y = \log L - \log \lambda + O(\log L/L)$. Substituting once more gives

$$y(x) = \frac{L + p \log L - p \log \lambda - \log C}{\lambda} + o(1).$$

Together with (52.3) this proves the real-error formulation and the two monotone-ceiling inequalities (52.1). $\square$

Apply the proposition to $a_n = U_n$, $C = C_5$, $p = 15/2$, and $\lambda = \log 16$. This gives (44.8)–(44.9). The existence of an $o(1)$ real error is stronger than merely stating $n_*(x) = F(x) + O(1)$, but it is not an exact rounding algorithm. If $F(x)$ is arbitrarily close to an integer, even an arbitrarily small unknown error can change its ceiling. If for some fixed $\delta > 0$ one knows $\text{dist}(F(x), \mathbb{Z}) > \delta$, then for sufficiently large such x the ceiling does agree with $\lceil F(x) \rceil$. Neither an effective onset nor such a separation condition for arbitrary inputs is supplied here.

[write] **5 October 2026.** Proposition 52.1 is part (1) of `p0:thm:staircase` (the front-matter section Relation to neighbouring reports) with the linear interpolation of $\log a_n$, an admissible interpolation in the sense of `p0:def:three-inverses`, followed by inversion of the leading model; the separation remark above is its part (2). Part VI's Theorem 63.1 sharpens it to every fixed order.

53 Exact checks and the reproducibility boundary

The proofs above establish identities and limits for all valid parameters. Finite checks are a separate safeguard against algebra, indexing, and implementation mistakes. They do not prove an infinite identity, Schur saturation, uniform domination, or an asymptotic theorem.

The companion's standard replay directly enumerates full permutations for $n = 5, \dots, 8$, and evaluates the six-term half formula and four-binomial gluing through $n = 24$. The latter values are compared with the published exact source data, including the overlapping OEIS terms. Representative full counts and total half counts are

n	U_n	$\sum_v H_{n-5}(v)$
5	1	1
6	20	10
7	270	81
8	3142	643
9	34291	5254
10	364462	44751

The optional C++ replay exhaustively checks the two boundary bijections through half size ten. It recovers the identical permutation, marked spine, boundaries, and all four statistics under both insertion/deletion round trips. The standard Python replay independently checks all 462 supported nested-boundary parameter tuples for $M = 2, \dots, 8$ against direct permutation

enumeration, including the full $(2, 2)$ inverse-Cauchy correction. It also checks 5,096 skew-dimension cases by comparing factorial determinants with removable-corner recursion, and 599,471 RSK tableau/boundary records for initial-filling saturation. The saturation check includes 131,854 groups with multiple initial fillings and verifies their complete filling cardinalities, as well as corner-emptiness invariance. These checks remain finite evidence for the corresponding proofs. The last published full count reproduced by gluing is

$$U_{24} = 181921327390401752024.$$

The companion archive contains the editable article, its PDF, exact source data with provenance, verification code, closed typed fixtures, and a complete file inventory. Its README specifies which checks are rerun by each command and distinguishes routine replay from optional deeper enumeration. No decimal value of R_5 is stored as a fixture. The code uses exact integer or rational arithmetic for the identities it checks. Fixtures are data, never executable input, and independently derived counts are compared rather than accepted solely because a stored hash matches.

The verifier requires isolated Python before its first shadowable import and uses explicit exceptions, including under optimized Python. Inventory, schema, mutation, output-path, and fresh-build checks concern reproducibility and guard behavior; they are not formal verification of the mathematical proofs. A manifest is an integrity record, not a digital signature. The PDF can be rebuilt from the included source, and the archive can be reconstructed deterministically in the stated toolchain. Exact command coverage and byte-comparison results are recorded with the release.

54 Questions suggested by the exact reduction

Several concrete questions remain open in this report.

1. Can the fixed-dimensional binomial sum be converted into a compact rational diagonal or a practical minimal recurrence for U_n ? The proof of existence does not predict a small order.
2. Can the nonnegative series for R_5 be turned into useful certified intervals? This requires effective constants and useful tail bounds, not just increasingly long finite ratio fits.
3. What corrections follow beyond the leading ratio? Uniform boundary estimates and endpoint interactions must be justified at the relevant orders before formal character expansions can be used. No correction or all-orders theorem is asserted here.
4. Is the length-five generating function algebraic or transcendental? Nonrationality and rational diagonality alone do not decide this question.
5. Which longer monotone patterns admit a different gluing or boundary description that controls alternating runs? The present three-run exclusion fails already at length six, so an extension needs an additional idea.
6. Does a similarly explicit positive ratio constant exist for more than one occurrence of 12345? The fixed-occurrence Θ results do not supply such a limit by themselves.

54.1 Further questions and research

[write] **Status of the questions above, 5 October 2026.** Question 1 is Part VI’s question 4: open (Section 66.1, item 5). Question 2 is Part VI’s question 1: open (item 3). Question 3: *answered* by Part VI, Theorem 55.1 (every fixed order, $r_1 = S_5$ explicit, no logarithms). Question 4: open (item 8); nonrationality

is known (note in Section 44). Questions 5 and 6: open (item 9). The two uncited claims of Section 44 are pinned in the note there. The identification of Lemma 47.2 with the empty-inner-corner case of skew RSK [23] is cited, not checked in the write; the Part's own proof does not depend on it. Nothing in this Part was found to be wrong.

Part VI

All orders asymptotics and inverse thresholds for unique 12345

Bundle Report 140, archive A224248_All_Orders_Asymptotics_and_Inverse_Thresholds_Source.zip, delivered as Report140.tex (18 pages), dated 2 October 2026; title block “All orders asymptotics and inverse thresholds for unique 12345”, author line “Report 140”. Printed as delivered, with [write] notes; a section number is Report 140’s plus 54, and equations are renumbered within sections. Its table of contents is replaced by the report’s. Section 65.1 is added in the write.

Reading conventions of Part VI	Elsewhere in the report
$U_n, A_n, H_s(v), P, R, Q, T, K$ as in Part V	
$p = 15/2, \alpha = 13/2, C_A, \lambda = \log 16$	Part II: $\alpha = -11/2$; Parts I–IV: $\lambda = \log 9$
$\mathbf{u}_j$ arrays in $E_{x,y}$; $a_s = 16^{-s} H_s$	Part III: U_j in E_r
$h(v), B(v)$ (59.5)–(59.6)	Parts II–III: $B(i, j)$
$q_j(n) = [z^n](1 - z)^{\alpha+j}$; $v_j, T_j, \mu_\ell, m_\ell$	Part III: $b_k(n), V_k, T_j$
r_j coefficients, $S_5 = r_1$	Part III: $r_j, S = r_1$ for 1234
$Q_k = \sum \theta_a^k$; Q a decay exponent; $D(\theta), D(F), J(F), V(F)$	Part I: Q_k polynomials
$d = P - t$; the constant d of Section 63;	
$M = C_A R_5$ there	
$N(X), L = \log X, n_J(X), P_j$	Part III: $N(T), Q_j$

Abstract of this Part (as delivered). Let U_n count permutations with exactly one classical increasing subsequence of length five, OEIS A224248, and let A_n count the avoiders. Building on the exact half decomposition in Report 138, we prove a full Poincaré expansion $U_n/A_n = \sum_{j=0}^J r_j n^{-j} + O_J(n^{-J-1})$ at every fixed order. The leading coefficient is the positive amplitude R_5 of that report, and the first correction has an explicit absolutely convergent series. The analytic proof has two parts: a character expansion uniform in an anisotropically weighted Banach space, and a coefficient-level convolution theorem for tails of exponent $15/2$. Ordinary endpoint moments converge through degree six; degree seven is the first requiring subtraction. Exact fractional-binomial models make all regularized moments absolutely convergent, and the product of any two models is a polynomial. This identity removes the entire candidate bulk sector and proves the absence of logarithms or an additional fractional relative power sector. We give finite coefficient recipes, a smooth-model inverse to all orders, and integer threshold brackets with the real error placed before the ceilings. No convergence of the infinite expansion, certified amplitude decimals, or effective numerical onset is claimed.

[write] **5 October 2026.** “Report 138” [3] is Part V of this report, and the “frozen baseline dependency” is Part V with the 138-leading- files.

55 The continuation and its scope

An occurrence of 12345 is a choice of five positions with increasing values, without an adjacency requirement. Write U_n for the number of permutations with exactly one such occurrence, with $U_0 = \dots = U_4 = 0$, and A_n for the number with no occurrence, with $A_0 = 1$. These are respectively A224248 and A047889. Put

$$p = \frac{15}{2}, \quad \alpha = p - 1 = \frac{13}{2}, \quad C_A = \frac{1536}{\pi^{3/2}}, \quad \lambda = \log 16. \quad (55.1)$$

All logarithms are natural.

Report 138 [3] supplies the exact combinatorial half bijections, the six-term inverse-Cauchy formula, the gluing identity, a global boundary estimate, and the positive leading ratio limit. It also proves rational diagonality and P-recursiveness. That report is a frozen leading-only baseline; its source, PDF, and reproducible companion are included unchanged as a dependency. The present report adds the analytic correction and all-orders theorems. It does not reinterpret the baseline's stated limits as stronger results already proved there. We restate exactly the baseline inputs needed below; all subsequent analytic steps are proved here.

Theorem 55.1 (All-orders ratio and first correction). *There is a unique sequence of real constants $r_0, r_1, \dots$ such that, for every fixed integer $J \geq 0$,*

$$\frac{U_n}{A_n} = \sum_{j=0}^J \frac{r_j}{n^j} + O_J(n^{-J-1}). \quad (55.2)$$

Here $r_0 = R_5 > 0$, and $r_1 = S_5$ is the absolutely convergent series in (60.2). In particular

$$U_n = C_A R_5 16^n n^{-p} \left(1 + \frac{S_5/R_5 - 135/8}{n} + O(n^{-2}) \right). \quad (55.3)$$

Every r_j has an explicit prescription from finitely many character and Gaussian calculations and absolutely convergent regularized Banach-valued sums. There are no logarithmic terms and no additional relative half-integer sector in the expansion.

The quantifier is “for every fixed J .” Constants in the remainder may depend on J , and the infinite formal series is not asserted to converge or to have any particular summability property. Nor do these formulas certify numerical values of R_5 , S_5 , or a usable numerical onset. The inverse theorem in Section 63 keeps this same fixed-order interpretation and explicitly preserves integer rounding.

56 Exact inputs and the weighted space

A boundary vector is $v = (i_2, i_3; j_2, j_3) \in \mathbb{N}_0^4$. Set

$$P = i_2 + i_3 + 1, \quad R = i_3 + 1, \quad Q = j_2 + j_3 + 1, \quad T = j_2 + 1. \quad (56.1)$$

The half count $H_s(v)$ of Report 138 counts the northern restriction of a permutation with its marked unique five-chain and s extra points. The four indices record the corresponding two internal position gaps and two internal value gaps. It is zero unless $i_2 + i_3 + j_2 + j_3 \leq s$, and H_0 is a unit mass at $v = 0$. For $w = (k_2, k_3; l_2, l_3)$ define

$$K(v, w) = \binom{i_2 + l_2}{i_2} \binom{i_3 + l_3}{i_3} \binom{j_2 + k_2}{j_2} \binom{j_3 + k_3}{j_3}, \quad (56.2)$$

$$\langle X, Y \rangle_K = \sum_{v, w \in \mathbb{N}_0^4} X(v) Y(w) K(v, w).$$

The kernel is symmetric. The exact gluing identity is

$$U_n = \sum_{s+t=n-5} \langle H_s, H_t \rangle_K \quad (n \geq 5). \quad (56.3)$$

This convention transposes a northern half to form a southern half; it does not reverse boundary coordinates.

Let s_ν be a Schur function, and let $f^{\nu/\rho}$ count standard tableaux of the skew shape, zero unless $\rho \subseteq \nu$. Set

$$G_{P,R} = \sum_{\substack{\nu \vdash P \\ \ell(\nu) \leq 2}} f^{\nu/(R)} s_\nu, \quad \mathcal{T} = \{\emptyset, (1), (2), (1,1), (2,1), (2,2)\}. \quad (56.4)$$

The operator $s_\tau^\perp$ is adjoint to multiplication by s_τ for the Hall inner product in which Schur functions are orthonormal; τ' denotes the transpose partition. For $\tau \in \mathcal{T}$ put

$$t = |\tau|, \quad F_\tau = s_\tau^\perp G_{P,R}, \quad G_\tau = s_{\tau'}^\perp G_{Q,T}, \quad d = P - t, \quad e = Q - t. \quad (56.5)$$

A zero character contributes zero, including when its indicated degree is negative. Haar measure on $U(4)$ has mass one. On the support $P + Q \leq s + 2$, the six-term formula becomes

$$H_s(v) = \sum_{\tau \in \mathcal{T}} (-1)^t \int_{U(4)} (\text{tr } U)^{s+2-P} F_\tau(U) \overline{(\text{tr } U)^{s+2-Q} G_\tau(U)} dU. \quad (56.6)$$

Both exponents outside the characters are nonnegative in this range. The character coefficients are real.

The other baseline inputs are

$$A_s = \int_{U(4)} |\text{tr } U|^{2s} dU \sim C_A 16^s s^{-p}, \quad (56.7)$$

$$\begin{aligned} c \frac{16^s}{(s+1)^p} &\leq A_s \leq C \frac{16^s}{(s+1)^p}, \\ 0 &\leq H_s(v) \leq C A_s (P+3)^5 (Q+3)^5 \omega(v), \\ \omega(v) &= 2^{-i_2} 4^{-i_3} 4^{-j_2} 2^{-j_3}. \end{aligned} \quad (56.8)$$

The constants in the last two lines are independent of s, v . The normalization in the first line is Regev's four-row constant; the derivation and references are in Report 138. The weighted estimate is an aggregate two-row tableau bound, rather than an estimate on one fixed boundary.

[write] The inputs, proved in Part V, 5 October 2026. (56.3) is Theorem 45.3; (56.6) is Theorem 47.3 written as Haar integrals, as in the proof of Proposition 50.2; (56.7) is (49.1) with (49.2) and (49.3); (56.8) is Proposition 49.1. The weighted space $E_{x,y}$ and everything after it are new in this Part.

Choose $x > 1/2$, $y > 1/4$ with $x + y < 1$. Let $E = E_{x,y}$ be the space of arrays with the norm

$$\|X\|_E = \sup_v |X(v)| x^{-i_2-j_3} y^{-i_3-j_2}. \quad (56.9)$$

Multiplication by the reciprocal weight identifies E with an ℓ^∞ space, so E is Banach. The kernel pairing extends to a continuous symmetric bilinear form, because

$$|\langle X, Y \rangle_K| \leq (1 - x - y)^{-4} \|X\|_E \|Y\|_E. \quad (56.10)$$

Indeed the four paired coordinate pairs each give $\sum_{a,b \geq 0} \binom{a+b}{a} x^a y^b = (1 - x - y)^{-1}$. The strict slack in $x > 1/2$, $y > 1/4$ absorbs every fixed polynomial in $P + Q$ multiplying $\omega(v)$. Consequently

$$\|H_s\|_E \leq C_E A_s, \quad a_s := 16^{-s} H_s = O_E((s+1)^{-p}). \quad (56.11)$$

57 A local character expansion with an actual remainder

This section proves a fixed-order Laplace expansion with sufficient uniformity to sum the boundary indices later. It is not an inference from a formal generating-function singularity.

57.1 The scalar maximum and the local measure

Write the eigenvalues as $e^{i\phi_a}$. Weyl integration gives a constant multiple of the eigenangle measure with density $\prod_{a<b} |e^{i\phi_a} - e^{i\phi_b}|^2$. The trace magnitude has maximum four precisely when all eigenvalues coincide. Homogeneity in (56.6) cancels the common phase between its two factors. We may therefore quotient by the scalar circle and use a centered local chart

$$\theta_1 + \theta_2 + \theta_3 + \theta_4 = 0, \quad \theta = (u, v, w, -u - v - w).$$

Any finite chart multiplicity and its constant Jacobian enter only the leading normalization already fixed by (56.7). Define $Q_k(\theta) = \sum_a \theta_a^k$, $\Delta(\theta) = \prod_{a<b} (\theta_a - \theta_b)$, $Z = \sum_a e^{i\theta_a}$, and $z = Z/4$. Near zero put

$$\Phi(\theta) = \log |z|^2, \quad D(\theta) = \prod_{a<b} \left(\frac{2 \sin((\theta_a - \theta_b)/2)}{\theta_a - \theta_b} \right)^2. \quad (57.1)$$

The removable factors in D have their continuous values on a diagonal. Both functions are real analytic and even under $\theta \mapsto -\theta$, with

$$\Phi = -\frac{Q_2}{4} + \frac{Q_4}{48} - \frac{Q_2^2}{64} + O(|\theta|^6), \quad (57.2)$$

$$D = 1 - \frac{Q_2}{3} + O(|\theta|^4). \quad (57.3)$$

For the first line, expand $Z\bar{Z}$ through degree four and then the logarithm, using $Q_1 = 0$. For the second, each sine quotient contributes $1 - (\theta_a - \theta_b)^2/12 + O(|\theta|^4)$ and $\sum_{a<b} (\theta_a - \theta_b)^2 = 4Q_2$. The full density is $\Delta^2 D$; Δ^2 has degree twelve.

On a sufficiently small fixed chart there are positive constants c, C for which $\Phi \leq -cQ_2$ and $|\Phi + Q_2/4| \leq CQ_2^2$. On its complement the original trace satisfies $|Z| \leq 4\rho$ for some fixed $\rho < 1$. Thus the local scaling $\theta = \xi/\sqrt{s}$ has degree $3 + 12 = 15$ and leads to $s^{-15/2}$.

57.2 Why finite expansion has a controlled error

For any fixed J , Taylor-expand the phase through degree $2J + 2$ and D through degree $2J$. After isolating the quadratic Gaussian, expand the remaining exponential only to the powers required for relative order J . The omitted phase begins at degree $2J + 4$ and is multiplied by s . Shrink the chart if necessary so every exponential remainder is bounded by a polynomial times e^{-csQ_2} . Taylor's integral remainder gives such a bound directly; it does not require the formal expansion to converge after integration. For each nonnegative integer b ,

$$\int Q_2^b \Delta^2 e^{-csQ_2} d\theta = O(s^{-p-b}), \quad (57.4)$$

by the change of variable $\theta = \xi/\sqrt{s}$. An omitted phase factor sQ_2^{J+2} is therefore of relative order s^{-J-1} . The same degree accounting applies to the finite exponential Taylor remainder and to products with the density. The part outside the chart is exponentially smaller. This proves an actual asymptotic expansion through order J with error $O(s^{-p-J-1})$ for every fixed-degree polynomial

insertion. The constants are uniform when the insertion coefficients range in the unit ball of that finite-dimensional polynomial space.

The same domination proves the moment estimate needed for parameter-dependent insertions. For every fixed $b \geq 0$ and $c_1 > 0$, there are $\epsilon > 0, C_b$ such that, for $0 \leq L \leq \epsilon s$,

$$\mathbb{E}_s[Q_2^b e^{c_1 L Q_2}; \text{local chart}] \leq C_b s^{-b}. \quad (57.5)$$

Here $\mathbb{E}_s$ uses the probability measure $A_s^{-1} |\text{tr } U|^{2s} dU$. Choose ϵ so that $c_1 \epsilon s Q_2$ can be absorbed into half the Gaussian decay; then use (57.4) and the lower bound in (56.7). This establishes the estimate uniformly in L , including when L grows linearly with s .

57.3 Gaussian coefficients and the avoidance correction

Let $\mathbb{E}_\gamma$ be normalized integration on the centered hyperplane with density $e^{-Q_2/4 \Delta^2}$. Its polynomial moments are finite rational numbers. One exact recipe uses the centered ordinary Gaussian $\mathbb{E}_0$ with covariance

$$\mathbb{E}_0[\theta_a \theta_b] = 2(\delta_{ab} - 1/4), \quad \mathbb{E}_\gamma P = \frac{\mathbb{E}_0[P \Delta^2]}{\mathbb{E}_0[\Delta^2]}. \quad (57.6)$$

Wick's rule evaluates the numerator and denominator by finite sums of products of these rational covariance entries. This recipe also proves rationality without leaving the Gaussian constants implicit. For example

$$\mathbb{E}_0 \Delta^2 = 18432, \quad \mathbb{E}_\gamma Q_2 = 30, \quad \mathbb{E}_\gamma Q_2^2 = 1020, \quad \mathbb{E}_\gamma Q_4 = 435. \quad (57.7)$$

The two radial values follow by scaling the normalizing integral: $Q_2/4$ has gamma moments with shape $p = 15/2$. The Q_4 value follows by expanding the twelve-degree Vandermonde square in (57.6); it is also an exact finite Wick calculation, replayed by the companion.

Applying the local expansion without a character insertion yields

$$A_s = C_A 16^s s^{-p} \left(\sum_{j=0}^J \frac{\alpha_j}{s^j} + O_J(s^{-J-1}) \right), \quad \alpha_0 = 1, \quad \alpha_j \in \mathbb{Q}. \quad (57.8)$$

The first correction is

$$\alpha_1 = \mathbb{E}_\gamma \left(\frac{Q_4}{48} - \frac{Q_2^2}{64} - \frac{Q_2}{3} \right) = \frac{435}{48} - \frac{1020}{64} - 10 = -\frac{135}{8}. \quad (57.9)$$

All higher coefficients are obtained by the same finite expansion and the moment rule (57.6).

58 Uniformity over the six boundary characters

Theorem 58.1 (Weighted all-orders half expansion). *For every admissible $E_{x,y}$ there are arrays $\mathbf{u}_j \in E_{x,y}$, independent of the choice of x, y , such that, for each fixed J ,*

$$\left\| a_s - s^{-p} \sum_{j=0}^J \mathbf{u}_j s^{-j} \right\|_E = O_{J,E}(s^{-p-J-1}). \quad (58.1)$$

Each entry of $\mathbf{u}_j$ lies in $C_A \mathbb{Q}$. For some constants C_j, D_j depending only on j ,

$$|\mathbf{u}_j(v)| \leq C_j (1 + P + Q)^{D_j} \omega(v). \quad (58.2)$$

The arrays are not asserted to be polynomials in the boundary indices: the two-row skew-tableau multiplicities in (56.4) remain present.

Proof. Every nonzero $F = F_\tau$ in (56.5) is Schur-positive and homogeneous of degree $d = P - t$. Write $D(F) = F(1, 1, 1, 1)$. The aggregate tableau-path bound and the four-variable two-row Weyl dimension formula give

$$D(F) \leq C(P+3)^5 2^{P-R}, \quad D(G) \leq C(Q+3)^5 2^{Q-T}. \quad (58.3)$$

For completeness, a standard skew tableau from (R) to a shape of size P with at most two rows is a sequence of $P - R$ row choices, so its aggregate count over final shapes is at most 2^{P-R} . The dimension of a two-row shape (a, b) is

$$s_{(a,b)}(1^4) = \frac{(a-b+1)(a+2)(a+3)(b+1)(b+2)}{12} \leq (P+3)^5/12.$$

Removing any fixed $\tau \subseteq (2, 2)$ has only a bounded number of possible two-row subshapes and bounded Littlewood–Richardson multiplicities, uniformly in P . This changes only the absolute constant. The second bound is identical.

A normalized positive character is a probability average of monomials:

$$\frac{F(e^{i\theta})}{D(F)} = \sum_{\eta} p_{\eta} e^{i\eta \cdot \theta}, \quad p_{\eta} \geq 0, \quad \sum_{\eta} p_{\eta} = 1, \quad \eta \in \mathbb{N}_0^4, \quad \sum_a \eta_a = d.$$

Its directional derivatives of fixed order h , in a fixed coordinate chart, have size at most $C_h(d+1)^h$. Since $|z|^{-d} = e^{-d\Phi/2} \leq e^{CdQ_2}$ locally, finite differentiation of z^{-d} gives

$$|D^h z^{-d}| \leq C_h(d+1)^h e^{C_h d Q_2}. \quad (58.4)$$

Consequently the real even part of

$$\Psi_{F,G}(\theta) = \frac{F(e^{i\theta}) \overline{G(e^{i\theta})}}{z(\theta)^d \overline{z(\theta)^e}} \quad (58.5)$$

has homogeneous Taylor terms of degree $2h$ bounded by $C_h D(F) D(G) (1 + P + Q)^{2h} Q_2^h$. After degree $2J$ its remainder is bounded by

$$C_J D(F) D(G) (1 + P + Q)^{2J+2} Q_2^{J+1} e^{C_J(P+Q)Q_2}. \quad (58.6)$$

One obtains an even expansion by averaging the Taylor formulas at θ and $-\theta$; conjugation of real character coefficients makes that average real. The odd part integrates to zero against the symmetric phase and density.

First consider $P + Q \leq \epsilon_J s$, for a sufficiently small $\epsilon_J > 0$. Combine (58.6) with (57.5) and the finite-order scalar expansion of Section 57. Every omitted term has the required relative factor s^{-J-1} times a fixed polynomial in $P+Q$ and $D(F)D(G)$. The finite shifts $r = s + 2 - t$ are uniformly harmless. Outside the local chart one must bound the original polynomial integrand in (56.6), rather than divide by the trace. It is at most

$$D(F)D(G)(4\rho)^{2s+4-P-Q}.$$

After division by A_s , its bound is a constant times

$$D(F)D(G)4^{4-P-Q} s^p \rho^{2s+4-P-Q}.$$

In the present range the exponent of ρ is at least $(2 - \epsilon_J)s + 4$, giving exponential suppression uniformly in the boundary. There is no inverse-trace singularity away from the chart. From (58.3), all coefficient and remainder bounds acquire the geometric factor

$$4^{-P-Q} 2^{P-R+Q-T} = 2^{-4} \omega(v), \quad (58.7)$$

times a fixed polynomial. Dividing by the Banach weights and taking the supremum therefore preserves the asserted remainder.

Now consider $P + Q > \epsilon_J s$. Set $\eta = \max\{(2x)^{-1}, (4y)^{-1}\} < 1$. The weight ratio obeys

$$\omega(v) x^{-i_2-j_3} y^{-i_3-j_2} \leq \eta^{P+Q-2}.$$

The global estimate (56.8) bounds a_s in this region by $C(s+1)^{-p}$ times a polynomial in $P+Q$ times that ratio, which is exponentially small in s . The coefficient arrays from the local calculation satisfy the same type of polynomial-geometric bound for every v , independently of whether v was in the small-boundary range used to estimate the remainder. Their tail in this complementary region is also exponentially small. The region outside the combinatorial support is covered at once: $H_s = 0$ there, while these coefficient tails are still exponentially small. This proves (58.1) globally.

Finally, all Taylor coefficients involved are rational after the common leading constant C_A is removed: the finite characters have integer coefficients, the phase and sine series have rational coefficients, and all Gaussian moments are rational by (57.6). This also yields the entrywise finite recipe and the bounds (58.2). Uniqueness of a Poincaré expansion coordinate by coordinate shows that the resulting arrays do not depend on the admissible weights. $\square$

58.1 A single finite recipe for every half coefficient

Here is a precise way to implement the coefficient construction, without relying on an unspecified Laplace algorithm. In the centered Gaussian variable ξ , introduce a formal variable w and define

$$\begin{aligned} \mathcal{L}_{\delta, F, G}(w, \xi) &= D(\sqrt{w} \xi) \Psi_{F, G}^{\text{ev}}(\sqrt{w} \xi) \\ &\times \exp \left(\delta \Phi(\sqrt{w} \xi) + \frac{\Phi(\sqrt{w} \xi) + w Q_2(\xi)/4}{w} \right). \end{aligned} \quad (58.8)$$

Here Ψ^{ev} denotes the real even part, so only integer powers of w occur. The numerator in the last quotient begins at w^2 , making the exponential a well-defined formal series with constant term one. Each fixed coefficient is a polynomial in ξ calculable by finite Taylor algebra. Then

$$\mathbf{u}_j(v) = C_A 256 4^{-P-Q} \sum_{\tau \in T} (-1)^t \mathbb{E}_\gamma [w^j] \mathcal{L}_{2-t, F_\tau, G_\tau}(w, \xi). \quad (58.9)$$

To check the prefactor, put $r = s + 2 - t$, $d = P - t$, $e = Q - t$ in the integral. The exponential normalization contributes $16^{2-t} 4^{-d-e} = 256 4^{-P-Q}$, and scaling by s gives $s^{-p} e^{-Q_2/4}$ with the remaining exponential exactly as in (58.8). The uniform proof above validates this coefficient recipe and its remainder. Similarly, the avoidance coefficients are

$$\alpha_j = \mathbb{E}_\gamma [w^j] \mathcal{L}_{0,1,1}(w, \xi). \quad (58.10)$$

No infinite-order Taylor interchange is made in either formula.

59 The explicit first half correction

For a homogeneous symmetric character F of degree d , define

$$\begin{aligned} V(F) &= \left[(x_1 \partial_{x_1})^2 - (x_1 \partial_{x_1})(x_2 \partial_{x_2}) \right] F|_{x_1=\dots=x_4=1}, \\ J(F) &= \frac{15}{4} [4V(F) - dD(F)], \quad D(F) = F(1^4). \end{aligned} \quad (59.1)$$

Set all three quantities to zero for the zero character. These are finite exact polynomial computations. By symmetry, the linear term of $F(e^{i\theta})$ vanishes on the centered hyperplane and its quadratic term is $-V(F)Q_2/2$. Also $z = 1 - Q_2/8 + O(|\theta|^3)$, so the quadratic part of F/z^d is

$$\left(-\frac{V(F)}{2} + \frac{dD(F)}{8} \right) Q_2. \quad (59.2)$$

Using $\mathbb{E}_r Q_2 = 30/r + O(r^{-2})$ gives

$$\begin{aligned} & \frac{1}{A_r} \int Z^{r-d} F \overline{Z^{r-e} G} dU \\ &= 4^{-d-e} \left[D(F)D(G) - \frac{J(F)D(G) + D(F)J(G)}{r} + O\left(\frac{\text{poly}(d+e)D(F)D(G)}{r^2} \right) \right]. \end{aligned} \quad (59.3)$$

The error is uniform in the small-boundary range of the preceding proof, and the weighted complementary bound extends the final half result to all boundaries.

A sometimes convenient equivalent expression for an irreducible Schur character of size d is

$$J(s_\nu) = s_\nu(1^4) \left(\kappa_\nu - \frac{d(d-1)}{4} \right), \quad \kappa_\nu = \sum_i \nu_i(\nu_i - 2i + 1). \quad (59.4)$$

This quadratic-character simplification is optional; the derivative formula (59.1) is the definition used in the proof and gives an independent finite way to compute every instance.

Define the arrays

$$h(v) = 256 4^{-P-Q} \sum_{\tau \in \mathcal{T}} (-1)^t D(F_\tau) D(G_\tau), \quad (59.5)$$

$$\begin{aligned} B(v) &= 256 4^{-P-Q} \sum_{\tau \in \mathcal{T}} (-1)^t [-p(2-t)D(F_\tau)D(G_\tau) \\ &\quad - J(F_\tau)D(G_\tau) - D(F_\tau)J(G_\tau)]. \end{aligned} \quad (59.6)$$

The finite-shift avoidance expansion is

$$\frac{A_{s+2-t}}{A_s} = 16^{2-t} \left(1 - \frac{p(2-t)}{s} + O(s^{-2}) \right). \quad (59.7)$$

The coefficient α_1 cancels in this ratio. Substituting (59.3) and (59.7) in the six-term formula therefore proves

$$\frac{H_s}{A_s} = h + \frac{B}{s} + O_E(s^{-2}), \quad \mathbf{u}_0 = C_A h, \quad \mathbf{u}_1 = C_A(\alpha_1 h + B). \quad (59.8)$$

Three useful exact checks are

v	$h(v)$	$B(v)$
$(0, 0; 0, 0)$	240	-3720
$(1, 0; 0, 0)$	224	-3600
$(0, 1; 0, 0)$	144	-2520

(59.9)

The first follows also from $H_s(0) = A_{s+2} - A_{s+1}$; the second from $H_s(1, 0; 0, 0) = A_{s+2} - 2A_{s+1}$. The third is obtained directly from the six characters. These are checks on exact formulas, not estimates of the convolution amplitude.

60 The first ratio correction as an ordinary convergent sum

The leading amplitude established in Report 138 is

$$R_5 = 2 \cdot 16^{-5} \sum_{t \geq 0} 16^{-t} \langle h, H_t \rangle_K, \quad R_5 \geq \frac{15}{32768} > 0. \quad (60.1)$$

The lower bound retains the $t = 0, v = w = 0$ summand, using $h(0) = 240$. Define

$$S_5 = 2 \cdot 16^{-5} \sum_{t \geq 0} 16^{-t} \langle B + p(t+5)h, H_t \rangle_K. \quad (60.2)$$

Both sums converge absolutely: the first summand is $O((t+1)^{-p})$ and the second is $O((t+1)^{1-p})$, by the Banach pairing bound.

Proposition 60.1. *One has $U_n/A_n = R_5 + S_5/n + O(n^{-2})$.*

Proof. For $0 \leq t \leq (n-5)/2$, put $s = n-5-t$. Uniformly on this range, (59.8), (57.8), and ordinary finite Taylor expansion at $(t+5)/n \leq 1/2 + O(n^{-1})$ give

$$\frac{H_{n-5-t}}{A_n} = 16^{-5-t} \left[h + \frac{B + p(t+5)h}{n} + O_E \left(\frac{(t+5)^2}{n^2} \right) \right]. \quad (60.3)$$

Pair this with H_t and sum. The error is $O(n^{-2})$, because

$$\sum_{t \geq 0} 16^{-t} A_t (t+5)^2 < \infty,$$

whose summand is $O((t+1)^{-11/2})$. Symmetry supplies the two endpoints. If the central index is integral, correcting its double count costs only $O(n^{-p})$ after division by A_n . Replacing the leading coefficient sum truncated at $t \leq (n-5)/2$ by the full sum costs $O(n^{1-p}) = O(n^{-13/2})$. The unweighted first-correction coefficient tail is $O(n^{2-p}) = O(n^{-11/2})$; its displayed factor n^{-1} makes its contribution again $O(n^{-13/2})$. All these errors are smaller than $O(n^{-2})$, proving the proposition. $\square$

This elementary argument continues while its required ordinary moments converge. Since $a_t = O_E(t^{-15/2})$, the moment of degree six is summable but the degree-seven estimate is not. One cannot complete an all-orders proof by replacing a divergent endpoint moment with a formally truncated sum. The next section gives an exact coefficient-level regularization and accounts for the entire potential interior contribution.

61 A coefficient-level half-integer convolution theorem

We state the analytic mechanism in a general Banach-space form. Let E, F be real or complex Banach spaces and let $\mathcal{K} : E \times E \rightarrow F$ be a continuous symmetric bilinear map. Assume $a_n = O_E((n+1)^{-p})$ and, for every fixed J ,

$$a_n = n^{-p} \sum_{j=0}^J \mathbf{u}_j n^{-j} + O_E(n^{-p-J-1}), \quad p = 15/2. \quad (61.1)$$

No generating-function continuation is among the hypotheses.

Theorem 61.1 (Regularized convolution). *The convolution $b_n = \sum_{s+t=n} \mathcal{K}(a_s, a_t)$ has*

$$b_n = n^{-p} \sum_{m=0}^J c_m n^{-m} + O_F(n^{-p-J-1}) \quad (61.2)$$

for every fixed J . The coefficients are the absolutely convergent regularized-moment prescriptions (61.19) below. Every model-model contribution is exactly a polynomial coefficient, so there are neither logarithms nor an extra sector starting at $n^{1-2p} = n^{-14}$.

61.1 Exact models and finite triangular matching

For $j \geq 0$ let

$$q_j(n) = [z^n](1-z)^{\alpha+j} = (-1)^n \binom{\alpha+j}{n} = \frac{\Gamma(n-\alpha-j)}{\Gamma(-\alpha-j)\Gamma(n+1)}. \quad (61.3)$$

The equality holds for every integer $n \geq 0$, since $\alpha+j$ is nonintegral. More generally write $q_x(n) = [z^n](1-z)^x$ when $x > 0$ is nonintegral, and set $q_x(n) = 0$ for negative integers n . The gamma-ratio expansion gives, to every fixed order H ,

$$q_j(n) = \frac{n^{-p-j}}{\Gamma(-\alpha-j)} \left(\sum_{h=0}^H g_h(\alpha+j) n^{-h} + O(n^{-H-1}) \right). \quad (61.4)$$

Here $g_0(x) = 1$ and the rational-polynomial coefficients are specified by the finite rule

$$g_h(x) = [w^h] \exp \left(\sum_{r=1}^h \frac{(-1)^{r+1} [B_{r+1}(-x) - B_{r+1}(1)]}{r(r+1)} w^r \right), \quad (61.5)$$

where B_k are Bernoulli polynomials. The fixed-order expansion follows by subtracting the Stirling expansions of $\log \Gamma(n-x)$ and $\log \Gamma(n+1)$, with their usual fixed-order real-positive-argument remainders; see [27]. In particular

$$g_1(x) = \frac{x(x+1)}{2}, \quad g_2(x) = \frac{x(x+1)(x+2)(3x+1)}{24}.$$

An alternative finite coefficient check uses the exact recurrence $q_x(n+1)/q_x(n) = (n-x)/(n+1)$.

Define arrays T_j, v_j triangularly:

$$\begin{aligned} T_0 &= \mathbf{u}_0, & T_j &= \mathbf{u}_j - \sum_{k=0}^{j-1} g_{j-k}(\alpha+k) T_k, \\ v_j &= \Gamma(-\alpha-j) T_j. \end{aligned} \quad (61.6)$$

Then, for every finite $L \geq 0$,

$$r_n^{[L]} := a_n - \sum_{j=0}^L v_j q_j(n) = O_E((n+1)^{-p-L-1}). \quad (61.7)$$

The expansion proves this for large n ; enlarging its constant includes all the finitely many small indices. For example

$$v_0 = \Gamma(-13/2) \mathbf{u}_0, \quad v_1 = \Gamma(-15/2) (\mathbf{u}_1 - 195 \mathbf{u}_0/8).$$

Only finitely many input coefficients are used at any stated order.

61.2 Canonical absolutely convergent moments

Write $n^\ell = n(n-1)\cdots(n-\ell+1)$ with $n^0 = 1$. Define

$$\mu_\ell = \sum_{n \geq 0} n^\ell \left[a_n - \sum_{j=0}^{\ell-7} v_j q_j(n) \right], \quad (61.8)$$

where the inner sum is empty for $\ell \leq 6$. For these first seven moments the norm of the summand is $O(n^{\ell-15/2})$, which is summable. For $\ell \geq 7$, the bracket is $r_n^{[\ell-7]}$ of order $n^{-\ell-3/2}$, so the weighted summand is $O_E(n^{-3/2})$. Thus every moment is an absolutely convergent Banach-valued series. The first subtraction is at degree seven, not degree six.

The prescription is invariant under any further model subtractions whose indices are larger than $\ell - 7$. To prove this, observe that for nonintegral $x > \ell$,

$$\sum_{n \geq 0} n^\ell q_x(n) = 0. \quad (61.9)$$

Indeed its absolute convergence follows from $q_x(n) = O(n^{-x-1})$. For $0 \leq z < 1$, differentiate the binomial series termwise:

$$\sum_{n \geq 0} n^\ell q_x(n) z^{n-\ell} = (-1)^\ell x^\ell (1-z)^{x-\ell}.$$

Take $z \uparrow 1$ using absolute convergence, and the right side tends to zero. Every additionally subtracted model satisfies $x = \alpha + j > \ell$. Consequently, whenever $L \geq \max\{0, \ell - 7\}$,

$$\mu_\ell = \sum_{n \geq 0} n^\ell r_n^{[L]}. \quad (61.10)$$

For $\ell \leq 6$ this says that even the unnecessary subtractions of $q_0, q_1, \dots$ do not change the ordinary moment.

For inverse-power formulas it is convenient to use ordinary powers: with $0^0 = 1$, define

$$m_\ell = \sum_{n \geq 0} n^\ell \left[a_n - \sum_{j=0}^{\ell-7} v_j q_j(n) \right]. \quad (61.11)$$

This also converges absolutely and obeys

$$m_\ell = \sum_{k=0}^{\ell} \left\{ \begin{matrix} \ell \\ k \end{matrix} \right\} \mu_k. \quad (61.12)$$

To verify it, use a common sufficiently large L in every term, apply the finite identity $n^\ell = \sum_k \left\{ \begin{matrix} \ell \\ k \end{matrix} \right\} n^k$, and then use (61.9). Thus (61.11) and (61.8) express the same regularization. Subtracting more models improves the summand decay by an arbitrary fixed polynomial power without changing the value; it is not a free choice of a finite-part constant.

61.3 An exact discrete Taylor estimate

Lemma 61.2. *Let $x > 0$ be nonintegral and let $c_n \in E$ satisfy $\|c_n\| = O((n+1)^{-Q})$. Fix $d \geq 0$ and suppose $Q > d + 2$. Then $\nu_\ell = \sum_{n \geq 0} n^\ell c_n$ converges absolutely for $0 \leq \ell \leq d + 1$, and*

$$\sum_{s=0}^n q_x(n-s) c_s = \sum_{\ell=0}^d \frac{(-1)^\ell}{\ell!} q_{x+\ell}(n) \nu_\ell + O_E(n^{-Q} + n^{-x-d-2}). \quad (61.13)$$

Proof. For $\nabla f(n) = f(n) - f(n-1)$, the binomial identity gives $\nabla q_x(n) = q_{x+1}(n)$. Repeated telescoping yields the exact identity

$$\begin{aligned} q_x(n-s) &= \sum_{\ell=0}^d (-1)^\ell \binom{s}{\ell} q_{x+\ell}(n) \\ &\quad + (-1)^{d+1} \sum_{k=0}^{s-d-1} \binom{s-1-k}{d} q_{x+d+1}(n-k). \end{aligned} \quad (61.14)$$

The last sum is empty when $s \leq d$; the identity can equivalently be proved by induction on s , using the first-difference relation. For $s \leq n/2$, its remainder is bounded by

$$C n^{-x-d-2} \sum_{k=0}^{s-d-1} \binom{s-1-k}{d} = C n^{-x-d-2} \binom{s}{d+1} \leq C(s+1)^{d+1} n^{-x-d-2}.$$

Summing this against $\|c_s\|$ gives $O(n^{-x-d-2})$ because $Q > d+2$. Extending each half-range moment to the whole series costs

$$O(n^{-x-\ell-1} n^{\ell+1-Q}) = O(n^{-x-Q}),$$

which is bounded by the displayed error. Finally $q_x \in \ell^1$ since $x > 0$, so the complementary range $s > n/2$ is bounded by

$$C n^{-Q} \sum_{t \geq 0} |q_x(t)| = O(n^{-Q}).$$

This proves the lemma using only coefficient estimates and finite identities. $\square$

61.4 Exact disappearance of the entire model product

Fix the desired order M , and put $S_n = \sum_{j=0}^M v_j q_j(n)$ and $r_n = a_n - S_n$. Then $r_n = O_E((n+1)^{-Q})$ with $Q = p + M + 1$. The convolution splits exactly into $S * S + 2S * r + r * r$, with $\mathcal{K}$ understood. For all $j, k \geq 0$,

$$\sum_{s+t=n} q_j(s) q_k(t) = [z^n](1-z)^{2\alpha+j+k} = [z^n](1-z)^{13+j+k}. \quad (61.15)$$

The identity is an equality of formal power series coefficients and requires no analytic continuation. Its right side is a polynomial coefficient, identically zero for $n > 13 + j + k$. Thus $(S * S)_n = 0$ exactly for $n > 13 + 2M$. This eliminates the full model-model contribution, including all possible higher algebraic orders of that contribution, rather than estimating only a leading interior integral.

For the residual convolution, a half-range split and $Q > 1$ give

$$\left\| \sum_{s+t=n} \mathcal{K}(r_s, r_t) \right\|_F \leq C n^{-Q} \sum_{s \geq 0} \|r_s\|_E = O_F(n^{-Q}). \quad (61.16)$$

For $q_j * r$, apply Lemma 61.2 with $x = \alpha + j$ and $d = M - j$. Its conditions hold since $Q - (d+2) = \alpha + j > 0$, and $x + d + 2 = Q$. The moments of r in the lemma are precisely μ_ℓ by (61.10). Hence

$$b_n = \sum_{m=0}^M W_m q_m(n) + O_F(n^{-p-M-1}), \quad W_m = 2 \sum_{j+\ell=m} \frac{(-1)^\ell}{\ell!} \mathcal{K}(v_j, \mu_\ell). \quad (61.17)$$

This proves the full expansion and its absolute-convergence claims. In fact the input expansion through M suffices for this output through M ; the all-orders hypothesis provides a single compatible family as M varies.

61.5 Two equivalent coefficient recipes

Expanding the finite sum in (61.17) gives

$$c_m = \sum_{h=0}^m \frac{g_{m-h}(\alpha + h)}{\Gamma(-\alpha - h)} W_h. \quad (61.18)$$

Equivalently, with the ordinary regularized moments of (61.11),

$$c_m = 2 \sum_{j+\ell=m} \frac{(p+j)_\ell}{\ell!} \mathcal{K}(\mathbf{u}_j, m_\ell). \quad (61.19)$$

Here $(a)_\ell = a(a+1) \cdots (a+\ell-1)$ is a rising factorial. To verify the equivalence by finite algebra, fix an integer $s \geq 0$. The coefficient at order n^{-p-m} in the shifted tail a_{n-s} is

$$\sum_{j+\ell=m} \frac{(p+j)_\ell}{\ell!} \mathbf{u}_j s^\ell.$$

Matching to the models and expanding the exact discrete Taylor identity gives the same polynomial, now in the basis s^ℓ . Equality for all nonnegative integers s implies equality of these finite polynomials. Apply the linear replacement $s^\ell \mapsto \mu_\ell$, equivalently $s^\ell \mapsto m_\ell$ by (61.12), and pair the resulting moment with the coefficient in the first factor. This gives (61.19), without any sum over unregularized divergent moments. The first examples are

$$\begin{aligned} c_0 &= 2\mathcal{K}(\mathbf{u}_0, m_0), \\ c_1 &= 2\mathcal{K}(\mathbf{u}_1, m_0) + 15\mathcal{K}(\mathbf{u}_0, m_1), \\ c_2 &= 2\mathcal{K}(\mathbf{u}_2, m_0) + 17\mathcal{K}(\mathbf{u}_1, m_1) + \frac{255}{4}\mathcal{K}(\mathbf{u}_0, m_2). \end{aligned} \quad (61.20)$$

Every exponent in (61.17) is p plus an integer. Together with the exact polynomial cancellation, this proves Theorem 61.1. In particular the putative absolute n^{-14} sector, of relative order $n^{-13/2}$, is absent. The absence of a logarithm here is a proved coefficient-level fact; no singularity-transfer assumption is imposed on the remainder.

[write] **5 October 2026.** For the integer exponent 4 of one 1234, Part III's Theorem 25.3 works with logarithmic models whose products are not polynomials, and logarithms survive; Remark 66.1 compares the two theorems.

62 All coefficients for the permutation ratio

Apply Theorem 61.1 to $a_s = 16^{-s} H_s$ in the weighted space and $\mathcal{K}(X, Y) = \langle X, Y \rangle_K$. Theorem 58.1 and (56.10) verify every hypothesis. The resulting constants c_m are given by (61.6), (61.11), and (61.19). With $N = n - 5$, the exact normalization is

$$U_n = 16^{n-5} \sum_{s+t=N} \langle a_s, a_t \rangle_K. \quad (62.1)$$

A finite shift expansion gives

$$\begin{aligned} U_n &= 16^n n^{-p} \left(\sum_{r=0}^J b_r n^{-r} + O_J(n^{-J-1}) \right), \\ b_r &= 16^{-5} \sum_{m=0}^r c_m \frac{(p+m)_{r-m} 5^{r-m}}{(r-m)!}. \end{aligned} \quad (62.2)$$

Divide by (57.8). Since $\alpha_0 = 1$, the ratio coefficients are defined recursively by

$$r_0 = b_0/C_A, \quad r_j = b_j/C_A - \sum_{k=1}^j \alpha_k r_{j-k} \quad (j \geq 1). \quad (62.3)$$

Division is a finite Taylor operation at any prescribed order, and its denominator tends to one. It preserves the stated error. This proves (55.2). Coefficient uniqueness follows successively by multiplying the difference of two expansions by n^j at the first possibly differing order and taking $n \rightarrow \infty$.

At orders zero and one, $m_0 = \sum_t 16^{-t} H_t$ and $m_1 = \sum_t t 16^{-t} H_t$ are ordinary convergent moments. Substituting $\mathbf{u}_0 = C_A h$ and $\mathbf{u}_1 = C_A(\alpha_1 h + B)$ gives

$$b_0 = C_A R_5, \quad b_1 = C_A(\alpha_1 R_5 + S_5).$$

Consequently $r_0 = R_5$ and $r_1 = S_5$, as also proved directly in Section 59 and Proposition 60.1. This checks the avoidance cancellation and the shift $n - 5$ independently in the all-orders formulas.

For clarity, the complete recipe at order J is finite except for explicit absolutely convergent moment sums:

1. Compute $\alpha_0, \dots, \alpha_J$ and the arrays $\mathbf{u}_0, \dots, \mathbf{u}_J$ by (58.9) and (58.10), evaluating polynomial moments by Wick's rule.
2. Match $v_0, \dots, v_J$ by (61.6) and form the regularized moments $m_0, \dots, m_J$ from (61.11). One may subtract all models through J in every moment without changing it.
3. Compute $c_0, \dots, c_J$ from (61.19), shift by (62.2), and divide by (62.3).

Each moment is Banach-valued and absolutely convergent, so continuity of the pairing justifies its use. This recipe does not claim that all desired numerical values can be enclosed to useful precision without additional effective error estimates.

63 Logarithms and smooth inverse thresholds

Set $M = C_A R_5 > 0$ and $\beta_j = b_j/M$ for $j \geq 1$. Equation (62.2) becomes

$$U_n = M 16^n n^{-p} \left(1 + \sum_{j=1}^J \beta_j n^{-j} + O_J(n^{-J-1}) \right). \quad (63.1)$$

The ordinary finite Taylor expansion of the logarithm yields

$$\log U_n = \lambda n - p \log n + \log M + \sum_{j=1}^J d_j n^{-j} + O_J(n^{-J-1}), \quad (63.2)$$

where

$$d_j = [w^j] \log \left(1 + \sum_{k=1}^j \beta_k w^k \right), \quad d_1 = \frac{S_5}{R_5} - \frac{135}{8}, \quad d_2 = \beta_2 - \frac{\beta_1^2}{2}. \quad (63.3)$$

The logarithmic leading factor $-p \log n$ comes from n^{-p} . There are no further $n^{-j} \log n$ terms.

For a large real target X , let

$$L = \log X, \quad z = \log L, \quad d = \log M + p \log \lambda, \quad e_j = d_j \lambda^j, \quad a = pz - d. \quad (63.4)$$

The letter d in this section is a scalar constant, unrelated to the earlier character degree. For $J \geq 0$ put

$$n_J(X) = \lambda^{-1} \left[L + p \log L - d + \sum_{j=1}^J \frac{P_j(\log L)}{L^j} \right]. \quad (63.5)$$

The polynomials P_j have degree at most j and are uniquely specified by finite reversion. Define $y_{j-1} = L + p \log L - d + \sum_{r=1}^{j-1} P_r(\log L) L^{-r}$ and set

$$P_j(z) = -[L^{-j}] \left(y_{j-1} - p \log y_{j-1} + d + \sum_{r=1}^j e_r y_{j-1}^{-r} - L \right), \quad (63.6)$$

expanding formally in L^{-1} with z held as a polynomial variable. The leading $\log L = z$ cancels, so this coefficient extraction is unambiguous. Adding $P_j L^{-j}$ to y_{j-1} changes that coefficient by exactly P_j , because the derivative of the expression in y is $1 + O(L^{-1})$. Induction shows both uniqueness and the degree bound. The first two polynomials are

$$\begin{aligned} P_1(z) &= pa - e_1, \\ P_2(z) &= -\frac{p}{2}a^2 + (p^2 + e_1)a - pe_1 - e_2. \end{aligned} \quad (63.7)$$

In particular

$$n_1(X) = \frac{1}{\lambda} \left[L + \frac{15}{2} \log L - d + \frac{\frac{225}{4} \log L - \frac{15}{2}d - \lambda(S_5/R_5 - 135/8)}{L} \right]. \quad (63.8)$$

Theorem 63.1 (Real error before integer ceilings). *Let $N(X) = \min\{n \geq 0 : U_n \geq X\}$. For each fixed $J \geq 0$, there exists a constant C_J such that, for all sufficiently large X ,*

$$\lceil n_J(X) - \epsilon_J(X) \rceil \leq N(X) \leq \lceil n_J(X) + \epsilon_J(X) \rceil, \quad \epsilon_J(X) = C_J \frac{(1 + \log \log X)^{J+1}}{(\log X)^{J+1}}. \quad (63.9)$$

The same error order bounds the difference between $n_J(X)$ and the large real root of the smooth truncated logarithmic model.

Proof. Define on large real x

$$f_J(x) = \lambda x - p \log x + \log M + \sum_{j=1}^J d_j x^{-j}.$$

Its derivative is $\lambda - p/x + O(x^{-2})$, bounded below by $\lambda/2$ for sufficiently large x . It has a unique sufficiently large root $\xi_J(X)$ of $f_J(\xi_J) = L$, with $\xi_J \sim L/\lambda$. Under $y = \lambda x$ its equation is

$$y - p \log y + d + \sum_{j=1}^J e_j y^{-j} = L.$$

The recursion cancels all powers through L^{-J} . Taylor's theorem in a fixed neighborhood of $y/L = 1$ bounds the residual at $y = \lambda n_J$ by $O_J((1 + \log L)^{J+1} L^{-J-1})$. The derivative is bounded away from zero, so the mean value theorem gives

$$|\xi_J(X) - n_J(X)| \leq C'_J (1 + \log L)^{J+1} L^{-J-1}. \quad (63.10)$$

This is the smooth-model inverse statement, justified by actual Taylor remainders rather than solely by formal reversion.

At integer k with $k \sim L/\lambda$, equation (63.2) gives $|\log U_k - f_J(k)| \leq D_J L^{-J-1}$. Choose C_J larger than C'_J by a sufficient fixed margin to absorb this error after using $f'_J \geq \lambda/2$. For

$$k_- = \lceil n_J - \epsilon_J \rceil - 1, \quad k_+ = \lceil n_J + \epsilon_J \rceil,$$

one has $k_- < n_J - \epsilon_J$ and $k_+ \geq n_J + \epsilon_J$. The derivative estimate and (63.10) then give $\log U_{k_-} < L$ and $\log U_{k_+} \geq L$ for all sufficiently large X .

Finally U_n is nondecreasing: append a new minimum at the final position of a permutation. It cannot lie in an increasing subsequence of length at least two, so this is an injection preserving exactly one five-pattern. Also $U_{n+1}/U_n \rightarrow 16$ by (63.1), so the positive tail is strictly increasing and unbounded. Large thresholds are past every exceptional initial index. The two integer witnesses therefore give exactly (63.9). $\square$

The inverse introduces polynomials in $\log \log X$ through inversion of $\lambda n - p \log n$; it does not introduce logarithms into the original coefficient expansion. Near an integer the two ceilings in (63.9) may differ. No eventual exact formula $N(X) = \lceil n_J(X) \rceil$ is claimed. If the whole error interval lies between two successive integers, its ceilings coincide and then determine the threshold. Linear interpolation of $\log U_n$ is sufficient for the baseline leading $o(1)$ inverse, but its $O(n^{-2})$ curvature error would obstruct arbitrary-order inverse claims. The proof above instead compares the smooth model directly with the integer counts.

[write] **5 October 2026.** Theorem 63.1 is an instance of parts (1)–(2) of `p0:thm:staircase` (the front-matter section Relation to neighbouring reports), with $n_J(X)$ as approximate inverse; its centre comes from a finite reversion, and no Lambert-function form is involved.

64 Exact checks and the reproducibility boundary

The companion separates the frozen baseline dependency from the new analytic statements. Its exact checks are finite independent calculations that supplement, rather than replace, the proofs. They cover rational Gaussian moments and the first avoidance correction; the finite boundary correction examples; gamma-ratio and triangular matching identities; the exact discrete Taylor formula; polynomial support of the fractional-binomial products; the change between falling and ordinary moment bases; the finite shift and ratio recursions; and finite inverse reversion. Checks of a model convolution are performed with rational binomial coefficients, without estimating a slowly convergent series.

The coefficient prescriptions contain genuine infinite sums and unquantified fixed-order remainder constants. A finite verifier does not certify those analytic theorems merely by testing examples. Nor does a file hash establish a mathematical proof or numerical amplitude certificate.

The package README documents isolated-mode execution, closed typed JSON fixtures, complete inventory checks, rejected malformed or unexpected inputs, independent exact replay, guard mutation tests, and clean PDF and deterministic archive reconstruction. Reproduction requires the stated local toolchain and does not fetch or execute remote dependencies. Integrity manifests are not digital signatures. The unchanged Report 138 companion remains a separate dependency; its theorem scope and its own checks are preserved. No public repository modification or publication is part of this release.

65 Further questions

The all-orders theorem closes the correction-expansion question left open by the frozen baseline, but leaves several useful next steps.

1. Can the weighted character and regularized-moment estimates be made effective enough to certify useful intervals for R_5 , S_5 , and subsequent coefficients? Increasing a numerical cutoff without an explicit tail constant does not answer this question.
2. How fast do the coefficient arrays and ratio coefficients grow with the order? The present fixed-order proof supplies no Gevrey bound, Borel summability result, optimal truncation rule, or controlled exponentially small sector.
3. Can the regularization be implemented with rigorous interval arithmetic and adaptive extra subtractions, while retaining a useful bound on cancellation error? The invariance under additional subtractions is exact, but numerical stability is a separate issue.
4. Can the baseline binomial sum be turned into a compact rational diagonal or a practical recurrence that independently exposes these all-orders coefficients? P-recursiveness by itself does not supply a small or computationally convenient recurrence.
5. For which other half decompositions does the exponent sum make every binomial-model product a polynomial? When it does not, a surviving interior sector or resonant logarithms must be analyzed rather than suppressed by analogy.
6. Can one extend the positive-amplitude and regularized-convolution method to several occurrences, or to longer monotone patterns with a new boundary construction? The length-five mixed-chain exclusion used in the baseline does not automatically extend to length six.

65.1 Further questions and research

[write] **Status of the questions above, 5 October 2026.** Questions 1–4: open (Section 66.1, items 3, 4, 6 and 5; question 4 is Part V’s question 1). Question 5: *answered in the negative for one* 1234 by Part III, as Remark 66.1 explains: that half decomposition has the integer exponent 4, its model products $(1 - z)^m \log^2(1 - z)$ are not polynomials, and the resonant logarithms survive with $\kappa_3 > 0$ and $\kappa_4 \neq 0$ (Theorem 23.1). For other decompositions it is open. Question 6: open (item 9). The two delivered receipts of separate checks (`data/140-allorders-sources-independent_extension_checks.json` and `data/140-allorders-sources-check_avoidance_recurrence.json`) record runs whose scripts were not delivered; the Part says that its replay does not rerun them, and nothing here rests on them. Nothing in this Part was found to be wrong.

Synthesis

66 The six Parts together

This section is added in the write (5 October 2026). Parts I–III and Parts V–VI apply one method to two patterns: an exact decomposition of a permutation with a unique occurrence into two halves glued by binomial shuffles; tableau or character formulas for the halves; a uniform expansion of the normalized halves; and a convolution theorem. Part IV applies the first two steps to 1243 with a different gluing (a shared decreasing spine) and stops after the leading term and a vanishing square-root correction.

Remark 66.1 ([**write**] One method, opposite outcomes). Parts III and VI end with the same step: an expansion of the normalized half arrays a_n in a weighted Banach space (Proposition 24.1, Theorem 58.1), followed by a coefficient-level convolution theorem (Theorems 25.3 and 61.1). They differ in the exponent p of the half tails, $\|a_n\| \asymp n^{-p}$: $p = 4$ for one 1234 and $p = 15/2$ for one 12345. In both cases p is Regev’s exponent $(h^2 - 1)/2$ for strips of height h , with $h = 3$ and $h = 4$.

For $p = 4$ the exact tail models are $b_k(n) = [z^n](1 - z)^k \log(1 - z)$, $k \geq 3$. A product of two models is $(1 - z)^m \log^2(1 - z)$, whose coefficients (25.9) carry a harmonic number and hence a $\log n$, and products of a model with the regular part give further single-logarithm models. The first logarithm comes at relative order three, with coefficient $\ell_3 = 40\langle U_0, U_0 \rangle_{\mathcal{K}}$. This is strictly positive because $U_0 = C_0 C$ has positive entries by (5.6) and the kernel is positive, and it gives $\kappa_3 = 43020\sqrt{3}/\pi$; Part III computes $\kappa_4 = -5041845\sqrt{3}/\pi \neq 0$. For $p = 15/2$ the models are $q_j(n) = [z^n](1 - z)^{13/2+j}$, the product of two models is the polynomial $(1 - z)^{13+j+k}$ (61.15), all model–model terms vanish for large n , and no logarithm and no extra sector at n^{-14} arises.

This answers Part VI’s question 5 in the negative for one 1234: that half decomposition does not make every model product a polynomial, and the resonant logarithms survive, with nonzero coefficients. For one 1243 the spine convolution of Part IV has a different endpoint law, $e_t \sim Dt^{-3/2}$ (Theorem 41.1), and its expansion beyond the vanishing square-root term is open.

A heuristic, not a theorem: if a two-half decomposition of this kind existed for one $12 \cdots k$ with $k \geq 6$ (Part V notes that its mixed-chain exclusion already fails at length six), its half exponent would presumably be $((k - 1)^2 - 1)/2$, an integer for even k and a half-integer for odd k , and the parity of k would then decide between the two outcomes.

66.1 Further questions and research across the report

Under Vladimir’s standing rule of 4 October 2026, here are the open problems of all six Parts, with their sources, what is known and what is missing. Each Part’s subsection “Further questions and research” says which of its own questions a later Part answers.

1. *Certified values of the 1234 constants* (Report 134, Section 10; Report 135, question 1; Report 137, question 1). Proved: $R > 0.50009$ (Part I) and $R > 0.50153$ (Proposition 6.1); no upper bound and no closed form. S , S_2 and r_3 are exact convergent series (Parts II–III) whose values and signs are uncertified. The intake’s exact partial sums of Part II’s signed series for S are positive and increasing, 7.16 at $T = 44$; since the series is signed, this is not a bound. Missing: effective tail constants for the domination and convolution steps.
2. *Is $R_{1243} = 149/160$?* (Conway and Guttmann, numerically; Report 136, question 1.) Proved: $R_{1243} > L_{35} \approx 0.6214$. Numerical evidence: the note after (37.2), consistent with $149/160$. Missing: effective control of the endpoint tails, which decay only like $T^{-1/2}$ (Report 136, question 2).

3. *Certified values of R_5 and S_5* (Report 138, question 2; Report 140, question 1). Proved: $R_5 \geq 15/32768$. Uncertified estimates: $R_5 \approx 0.2044$ (note after Theorem 51.1).
4. *Effective constants and onsets* for the direct and inverse remainders in every Part (Report 134, Corollary 8; Report 135, question 1; Report 137, question 2; Report 136, question 2; Report 138, Section 9; Report 140, question 1). Without them no statement here gives an exact threshold algorithm; a shrinking error interval does not decide the ceiling when it crosses an integer.
5. *Usable annihilating equations* (Report 135, question 2; Report 137, question 4; Report 136, question 4; Report 138, question 1 = Report 140, question 4): explicit differential equations or recurrences for the three generating functions, whose existence Parts II, IV and V prove. Also open: whether Part IV's 40-variable constant term is a rational diagonal of the kind whose existence is proved.
6. *Growth and summability of the full expansions* (Report 137, question 3; Report 140, questions 2–3): coefficient growth, Gevrey order, Borel summability, optimal truncation, exponentially small terms, and rigorous interval implementations of the regularized sums.
7. *One 1243 beyond the square-root cancellation* (Report 136, question 3): the first nonzero correction, an $O(n^{-1})$ rate, every order, and whether logarithms occur. Parts III and VI suggest a route through a Banach expansion of spine arrays; the spine convolution has three indices and the endpoint law $e_t \sim Dt^{-3/2}$, so their theorems do not apply as stated.
8. *Algebraicity for one 12345* (Report 138, question 4). Known: D-finite (Part V) and not rational ([9, Theorem 5.3]); the obstruction of [9, Lemma 6.3] needs an integer exponent and does not apply to $15/2$.
9. *Longer patterns, other patterns, several occurrences* (Report 134, Section 10; Report 137, question 5; Report 138, questions 5–6; Report 140, questions 5–6). Part V's mixed-chain exclusion fails at length six. The parity observation in Remark 66.1 is a heuristic.

Answered inside the report (dated notes at the questions): Report 134's first and higher corrections (Parts II and III) and its all-orders inverse (Part III, without effective constants); Report 135's question 3 (Part III) and, in part, its question 4; Report 137's question 5, in part (Parts IV–VI); Report 138's question 3 (Part VI); Report 140's question 5, in the negative for one 1234 (Part III, Remark 66.1).

Wrong claims. The one wrong claim on record is external: Conway and Guttmann's conjecture $R = 1/2$, refuted in Part I (Sections 2–6, Proposition 6.1). Nothing in the six manuscripts was found to be wrong. Report 138's two uncited claims are pinned in the note in Section 44.

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- [2] *Rational structure and logarithmic asymptotics for A217057*, Report 135, October 2, 2026. **In this report: Part II.** As cited by Report 137: final version with the second correction and first logarithmic term; included unchanged with its full reproducibility companion. As cited by Report 136: Sections 6–7, a companion research report; the weaker boundary and avoidance estimates needed there are proved in full in Section 40.
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